

[MS-MWBE]:

Microsoft Web Browser Federated Sign-On Protocol Extensions

Intellectual Property Rights Notice for Open Specifications Documentation

- **Technical Documentation.** Microsoft publishes Open Specifications documentation ("this documentation") for protocols, file formats, data portability, computer languages, and standards support. Additionally, overview documents cover inter-protocol relationships and interactions.
- **Copyrights.** This documentation is covered by Microsoft copyrights. Regardless of any other terms that are contained in the terms of use for the Microsoft website that hosts this documentation, you can make copies of it in order to develop implementations of the technologies that are described in this documentation and can distribute portions of it in your implementations that use these technologies or in your documentation as necessary to properly document the implementation. You can also distribute in your implementation, with or without modification, any schemas, IDLs, or code samples that are included in the documentation. This permission also applies to any documents that are referenced in the Open Specifications documentation.
- **No Trade Secrets.** Microsoft does not claim any trade secret rights in this documentation.
- **Patents.** Microsoft has patents that might cover your implementations of the technologies described in the Open Specifications documentation. Neither this notice nor Microsoft's delivery of this documentation grants any licenses under those patents or any other Microsoft patents. However, a given Open Specifications document might be covered by the Microsoft [Open Specifications Promise](#) or the [Microsoft Community Promise](#). If you would prefer a written license, or if the technologies described in this documentation are not covered by the Open Specifications Promise or Community Promise, as applicable, patent licenses are available by contacting iplg@microsoft.com.
- **License Programs.** To see all of the protocols in scope under a specific license program and the associated patents, visit the [Patent Map](#).
- **Trademarks.** The names of companies and products contained in this documentation might be covered by trademarks or similar intellectual property rights. This notice does not grant any licenses under those rights. For a list of Microsoft trademarks, visit www.microsoft.com/trademarks.
- **Fictitious Names.** The example companies, organizations, products, domain names, email addresses, logos, people, places, and events that are depicted in this documentation are fictitious. No association with any real company, organization, product, domain name, email address, logo, person, place, or event is intended or should be inferred.

Reservation of Rights. All other rights are reserved, and this notice does not grant any rights other than as specifically described above, whether by implication, estoppel, or otherwise.

Tools. The Open Specifications documentation does not require the use of Microsoft programming tools or programming environments in order for you to develop an implementation. If you have access to Microsoft programming tools and environments, you are free to take advantage of them. Certain Open Specifications documents are intended for use in conjunction with publicly available standards specifications and network programming art and, as such, assume that the reader either is familiar with the aforementioned material or has immediate access to it.

Support. For questions and support, please contact dochelp@microsoft.com.

Revision Summary

Date	Revision History	Revision Class	Comments
10/22/2006	0.01	New	Version 0.01 release
1/19/2007	1.0	Major	Version 1.0 release
3/2/2007	1.1	Minor	Version 1.1 release
4/3/2007	1.2	Minor	Version 1.2 release
5/11/2007	1.3	Minor	Version 1.3 release
6/1/2007	1.3.1	Editorial	Changed language and formatting in the technical content.
7/3/2007	1.3.2	Editorial	Changed language and formatting in the technical content.
7/20/2007	1.3.3	Editorial	Changed language and formatting in the technical content.
8/10/2007	1.4	Minor	Clarified the meaning of the technical content.
9/28/2007	1.4.1	Editorial	Changed language and formatting in the technical content.
10/23/2007	1.5	Minor	Clarified the meaning of the technical content.
11/30/2007	1.6	Minor	Clarified the meaning of the technical content.
1/25/2008	1.6.1	Editorial	Changed language and formatting in the technical content.
3/14/2008	1.6.2	Editorial	Changed language and formatting in the technical content.
5/16/2008	1.6.3	Editorial	Changed language and formatting in the technical content.
6/20/2008	1.6.4	Editorial	Changed language and formatting in the technical content.
7/25/2008	1.6.5	Editorial	Changed language and formatting in the technical content.
8/29/2008	1.6.6	Editorial	Changed language and formatting in the technical content.
10/24/2008	2.0	Major	Updated and revised the technical content.
12/5/2008	3.0	Major	Updated and revised the technical content.
1/16/2009	3.0.1	Editorial	Changed language and formatting in the technical content.
2/27/2009	3.0.2	Editorial	Changed language and formatting in the technical content.
4/10/2009	3.0.3	Editorial	Changed language and formatting in the technical content.
5/22/2009	3.1	Minor	Clarified the meaning of the technical content.
7/2/2009	4.0	Major	Updated and revised the technical content.
8/14/2009	5.0	Major	Updated and revised the technical content.
9/25/2009	5.1	Minor	Clarified the meaning of the technical content.
11/6/2009	5.1.1	Editorial	Changed language and formatting in the technical content.
12/18/2009	5.1.2	Editorial	Changed language and formatting in the technical content.
1/29/2010	5.2	Minor	Clarified the meaning of the technical content.

Date	Revision History	Revision Class	Comments
3/12/2010	5.2.1	Editorial	Changed language and formatting in the technical content.
4/23/2010	5.2.2	Editorial	Changed language and formatting in the technical content.
6/4/2010	5.2.3	Editorial	Changed language and formatting in the technical content.
7/16/2010	6.0	Major	Updated and revised the technical content.
8/27/2010	6.0	None	No changes to the meaning, language, or formatting of the technical content.
10/8/2010	6.0	None	No changes to the meaning, language, or formatting of the technical content.
11/19/2010	6.0	None	No changes to the meaning, language, or formatting of the technical content.
1/7/2011	7.0	Major	Updated and revised the technical content.
2/11/2011	7.0	None	No changes to the meaning, language, or formatting of the technical content.
3/25/2011	7.0	None	No changes to the meaning, language, or formatting of the technical content.
5/6/2011	7.0	None	No changes to the meaning, language, or formatting of the technical content.
6/17/2011	7.1	Minor	Clarified the meaning of the technical content.
9/23/2011	7.1	None	No changes to the meaning, language, or formatting of the technical content.
12/16/2011	8.0	Major	Updated and revised the technical content.
3/30/2012	8.0	None	No changes to the meaning, language, or formatting of the technical content.
7/12/2012	8.0	None	No changes to the meaning, language, or formatting of the technical content.
10/25/2012	8.0	None	No changes to the meaning, language, or formatting of the technical content.
1/31/2013	8.0	None	No changes to the meaning, language, or formatting of the technical content.
8/8/2013	9.0	Major	Updated and revised the technical content.
11/14/2013	9.0	None	No changes to the meaning, language, or formatting of the technical content.
2/13/2014	9.0	None	No changes to the meaning, language, or formatting of the technical content.
5/15/2014	9.0	None	No changes to the meaning, language, or formatting of the technical content.
6/30/2015	10.0	Major	Significantly changed the technical content.
7/14/2016	10.0	None	No changes to the meaning, language, or formatting of the technical content.

Date	Revision History	Revision Class	Comments
6/1/2017	11.0	Major	Significantly changed the technical content.
12/1/2017	11.0	None	No changes to the meaning, language, or formatting of the technical content.

Table of Contents

1	Introduction	7
1.1	Glossary	7
1.2	References	10
1.2.1	Normative References	10
1.2.2	Informative References	11
1.3	Overview	11
1.3.1	Query String Response Transfer Protocol	11
1.3.2	SAML 1.1 Assertion Extension	12
1.4	Relationship to Other Protocols	12
1.5	Prerequisites/Preconditions	12
1.6	Applicability Statement	12
1.7	Versioning and Capability Negotiation	12
1.8	Vendor-Extensible Fields	13
1.9	Standards Assignments.....	13
2	Messages.....	14
2.1	Transport	14
2.1.1	Query String Response Transfer Protocol	14
2.2	Message Syntax.....	14
2.2.1	XML Namespace References	14
2.2.2	Query String Response Transfer Protocol	14
2.2.2.1	wsignin1.0 Message	14
2.2.2.1.1	Common Parameters	14
2.2.2.1.2	wsignin1.0 Response	14
2.2.3	SAML 1.1 Assertion Extension	15
2.2.3.1	SAML Advice Elements.....	15
2.2.3.2	WindowsIdentifiers Structure.....	15
2.2.3.2.1	WindowsIdentifierFlags Structure	16
2.2.3.2.2	PACKED_SIDs Structure	16
2.3	Directory Service Schema Elements	17
3	Protocol Details.....	18
3.1	IP/STS Details	18
3.1.1	Abstract Data Model.....	18
3.1.1.1	Query String Response Transfer Protocol	18
3.1.1.1.1	Pending Result	18
3.1.1.1.2	Maximum Query String Response Message Length.....	18
3.1.2	Timers	19
3.1.3	Initialization.....	19
3.1.4	Higher-Layer Triggered Events	19
3.1.5	Processing Events and Sequencing Rules	19
3.1.5.1	Query String Response Transfer Protocol	19
3.1.5.1.1	Receiving a wsignin1.0 Request That Does Not Specify a ttpindex	19
3.1.5.1.2	Receiving a wsignin1.0 Request That Specifies a ttpindex of 0	19
3.1.5.1.3	Receiving a wsignin1.0 Request That Specifies a ttpindex Other Than 0 ..	19
3.1.5.1.4	Responding to a wsignin1.0 Request That Specifies a ttpindex	20
3.1.5.2	SAML 1.1 Assertion Extension.....	20
3.1.5.2.1	Responding to a wsignin1.0 Request	20
3.1.5.2.1.1	ClaimSource Element	20
3.1.5.2.1.2	CookieInfoHash Element	20
3.1.5.2.1.3	WindowsUserIdentifier Element	21
3.1.5.2.1.4	WindowsUserName Element	21
3.1.5.2.1.5	WindowsIdentifiers Element	21
3.1.6	Timer Events.....	21
3.1.7	Other Local Events.....	21

3.2	Relying Party Details	21
3.2.1	Abstract Data Model.....	21
3.2.1.1	Query String Response Transfer Protocol	21
3.2.1.1.1	Aggregated Result	21
3.2.2	Timers	21
3.2.3	Initialization.....	22
3.2.4	Higher-Layer Triggered Events	22
3.2.5	Processing Events and Sequencing Rules	22
3.2.5.1	Query String Response Transfer Protocol	22
3.2.5.1.1	Sending a wsignin1.0 Request.....	22
3.2.5.1.2	Receiving a wsignin1.0 Response That Does Not Specify a ttpindex	22
3.2.5.1.3	Receiving a wsignin1.0 Response That Specifies a ttpindex	22
3.2.5.1.4	Processing the Complete Aggregated Result.....	23
3.2.5.2	SAML 1.1 Assertion Extension.....	23
3.2.6	Timer Events.....	23
3.2.7	Other Local Events.....	23
3.3	Web Browser Requestor Details	23
3.3.1	Abstract Data Model.....	23
3.3.2	Timers	24
3.3.3	Initialization.....	24
3.3.4	Higher Layer Triggered Events	24
3.3.5	Processing Events and Sequencing Rules	24
3.3.6	Timer Events.....	24
3.3.7	Other Local Events.....	24
4	Protocol Examples	25
4.1	Query String Response Transfer Protocol.....	25
4.1.1	Annotated Example.....	25
4.1.2	Full Network Trace	28
4.2	SAML 1.1 Assertion Extension.....	42
5	Security	44
5.1	Security Considerations for Implementers	44
5.1.1	Data Integrity	44
5.1.2	Privacy	44
5.1.3	Authorization Validation and Filtering	44
5.2	Index of Security Parameters	44
6	Appendix A: Product Behavior	45
7	Change Tracking.....	49
8	Index.....	50

1 Introduction

This specification extends the Microsoft Web Browser Federated Sign-On Protocol described in [\[MS-MWBF\]](#). It is assumed that the reader is familiar with its terms, concepts, and protocols.

The extensions defined in this specification enable **web browser requestors** that do not support scripting (to create POST messages) and enable passing **security identifiers (SIDs)** in Security Assertion Markup Language (SAML) 1.1 assertions. These extensions are referred to, respectively, as the Query String Response Transfer Protocol and the SAML 1.1 Assertion Extension.

The Microsoft Web Browser Federated Sign-On Protocol specifies the use of HTTP POST to transmit the **wsignin1.0** result. The use of HTTP POST requires web browser requestors to support scripting for automated form submittal, but web browser requestors do not always have scripting support. The Query String Response Transfer Protocol provides a method for using a series of HTTP GET messages instead of a single HTTP POST to transmit the result of a wsignin1.0 action. This eliminates the scripting requirement for the web browser requestor. That is, the extension increases the number of messages needed to perform a wsignin1.0 action to avoid the POST message.

The SAML 1.1 Assertion Extension is an extension of the Microsoft Web Browser Federated Sign-On Protocol that specifies a method for transmitting SIDs as elements in **SAML advice**.

Sections 1.5, 1.8, 1.9, 2, and 3 of this specification are normative. All other sections and examples in this specification are informative.

1.1 Glossary

This document uses the following terms:

account: A **user** (including machine account), group, or alias object. Also a synonym for security principal or principal.

Active Directory: The Windows implementation of a general-purpose directory service, which uses **LDAP** as its primary access protocol. Active Directory stores information about a variety of objects in the network such as user accounts, computer accounts, groups, and all related credential information used by Kerberos [\[MS-KILE\]](#). Active Directory is either deployed as Active Directory Domain Services (AD DS) or Active Directory Lightweight Directory Services (AD LDS), which are both described in [\[MS-ADOD\]](#): Active Directory Protocols Overview.

aggregated result: The assembly of received parts transferred using the Query String Response Transfer Protocol. The **aggregated result** is assembled at a **relying party** and might not represent the complete result if all parts have not been received. Once complete, the **relying party** extracts a **RequestSecurityTokenResponse (RSTR)** from the **aggregated result**. For more information, see section 3.2.1.1.1.

base64 encoding: A binary-to-text encoding scheme whereby an arbitrary sequence of bytes is converted to a sequence of printable ASCII characters, as described in [\[RFC4648\]](#).

claim: A declaration made by an entity (for example, name, identity, key, group, privilege, and capability). For more information, see [\[WSFederation1.2\]](#).

domain: A set of users and computers sharing a common namespace and management infrastructure. At least one computer member of the set has to act as a domain controller (DC) and host a member list that identifies all members of the domain, as well as optionally hosting the **Active Directory** service. The domain controller provides authentication of members, creating a unit of trust for its members. Each domain has an identifier that is shared among its members. For more information, see [\[MS-AUTHSOD\]](#) section 1.1.1.5 and [\[MS-ADTS\]](#).

forest: One or more **domains** that share a common schema and trust each other transitively. An organization can have multiple **forests**. A **forest** establishes the security and administrative

boundary for all the objects that reside within the **domains** that belong to the **forest**. In contrast, a **domain** establishes the administrative boundary for managing objects, such as users, groups, and computers. In addition, each **domain** has individual security policies and trust relationships with other **domains**.

global group: An **Active Directory** group that allows user objects from its own **domain** and **global groups** from its own **domain** as members. Also called domain global group. **Universal groups** can contain **global groups**. A group object *g* is a **global group** if and only if `GROUP_TYPE_ACCOUNT_GROUP` is present in *g*! `groupType`; see [MS-ADTS] section 2.2.12, "Group Type Flags". A **global group** that is also a security-enabled group is valid for inclusion within ACLs anywhere in the **forest**. If a **domain** is in mixed mode, then a **global group** in that **domain** that is also a security-enabled group allows only user object as members. See also domain local group, security-enabled group.

identity provider/security token service (IP/STS): An STS that might also be an identity provider (IP). This term is used as shorthand to see both identity that verifies token services and general token services that do not verify identity. Note that the "/" symbol implies an "or" relationship.

Lightweight Directory Access Protocol (LDAP): The primary access protocol for **Active Directory**. Lightweight Directory Access Protocol (LDAP) is an industry-standard protocol, established by the Internet Engineering Task Force (IETF), which allows users to query and update information in a directory service (DS), as described in [MS-ADTS]. The Lightweight Directory Access Protocol can be either version 2 [\[RFC1777\]](#) or version 3 [\[RFC3377\]](#).

little-endian: Multiple-byte values that are byte-ordered with the least significant byte stored in the memory location with the lowest address.

NetBIOS: A particular network transport that is part of the LAN Manager protocol suite. **NetBIOS** uses a broadcast communication style that was applicable to early segmented local area networks. A protocol family including name resolution, datagram, and connection services. For more information, see [\[RFC1001\]](#) and [\[RFC1002\]](#).

pending result: The transformed **RequestSecurityTokenResponse (RSTR)** that an **identity provider/security token service (IP/STS)** maintains for the duration of a Query String Response Transfer Protocol message series. Each message in the Query String Response Transfer Protocol transfers a portion of the **pending result** to the **relying party**, where the portions are assembled into the **aggregated result**. For more information, see section 3.1.1.1.1.

relative identifier (RID): The last item in the series of SubAuthority values in a **security identifier (SID)** [\[SID\]](#). It distinguishes one account or group from all other accounts and groups in the domain. No two accounts or groups in any domain share the same RID.

relying party (RP): A web application or service that consumes **security tokens** issued by a security token service (STS).

requestor IP/STS: An **IP/STS** in the same **security realms** as the **web browser requestor**. The **requestor IP/STS** has an existing relationship with the **user** that enables it to issue **security tokens** containing **user** information.

RequestSecurityTokenResponse (RSTR): An XML element used to return an issued **security token** and associated metadata. An **RSTR** element is the result of the **wsignin1.0** action in the Web Browser Federated Sign-On Protocol. For more information, see [\[MS-MWBF\]](#) section 2.2.4.1.

resource IP/STS: An **IP/STS** in the same **security realm** as the **web service (WS) resource**. The **resource IP/STS** has an existing relationship with the **WS resource** that enables it to issue **security tokens** that are trusted by the **WS resource**.

SAML advice: The advice element of a **SAML assertion**. The data in the advice element is advisory and can be ignored without affecting the validity of the assertion. See [\[SAMLCore\]](#) section 2.3.2.2. The SAML 1.1 Assertion Extension includes **security identifiers (SIDs)** and related data in the **SAML advice** element.

SAML assertion: The Security Assertion Markup Language (SAML) 1.1 assertion is a standard XML format for representing a **security token**. For more information, see [\[SAMLCore\]](#) section 2.

security identifier (SID): An identifier for security principals that is used to identify an account or a group. Conceptually, the **SID** is composed of an account authority portion (typically a **domain**) and a smaller integer representing an identity relative to the account authority, termed the **relative identifier (RID)**. The **SID** format is specified in [\[MS-DTYP\]](#) section 2.4.2; a string representation of **SIDs** is specified in [\[MS-DTYP\]](#) section 2.4.2 and [\[MS-AZOD\]](#) section 1.1.1.2.

security realm or security domain: Represents a single unit of security administration or trust, for example, a Kerberos realm (for more information, see [\[RFC4120\]](#)) or a Windows Domain (for more information, see [\[MSFT-ADC\]](#)).

security token: A collection of one or more **claims**. Specifically in the case of mobile devices, a **security token** represents a previously authenticated user as defined in the Mobile Device Enrollment Protocol [\[MS-MDE\]](#).

subject: The entity to which the **claims** and other data in a **SAML assertion** apply. For more information, see [\[SAMLCore\]](#) section 1.3.1.

trusted forest: A forest that is trusted to make authentication statements for security principals in that forest. Assuming forest A trusts forest B, all domains belonging to forest A will trust all domains in forest B, subject to policy configuration.

universal group: An **Active Directory** group that allows user objects, **global groups**, and **universal groups** from anywhere in the **forest** as members. A group object g is a **universal group** if and only if GROUP_TYPE_UNIVERSAL_GROUP is present in g! groupType. A security-enabled universal group is valid for inclusion within ACLs anywhere in the **forest**. If a **domain** is in mixed mode, then a **universal group** cannot be created in that **domain**. See also domain local group, security-enabled group.

user: A person who employs a **web browser requestor** to access a **WS resource**.

user agent: An HTTP user agent, as specified in [\[RFC2616\]](#).

web browser requestor: An HTTP 1.1 web browser client that transmits protocol messages between an **IP/STS** and a **relying party**.

web service (WS) resource: A destination HTTP 1.1 web application or an HTTP 1.1 resource serviced by the application. In the context of this protocol, it refers to the application or manager of the resource that receives identity information and assertions issued by an **IP/STS** using this protocol. The **WS resource** is a **relying party** in the context of this protocol. For more information, see [\[WSFederation1.2\]](#).

wsignin1.0: A protocol message exchange defined in [\[WSFederation1.2\]](#) sections 2.1 and 3.1. The **wsignin1.0** request and response are the HTTP binding for the WS-Trust Issue action and response; as such, the WS-Trust **RSTR** element is used to return the issued **security token** in the **wsignin1.0** response ([\[WSTrust\]](#) section 3.2). For more information, see [\[MS-MWBF\]](#) section 2.2.

MAY, SHOULD, MUST, SHOULD NOT, MUST NOT: These terms (in all caps) are used as defined in [\[RFC2119\]](#). All statements of optional behavior use either MAY, SHOULD, or SHOULD NOT.

1.2 References

Links to a document in the Microsoft Open Specifications library point to the correct section in the most recently published version of the referenced document. However, because individual documents in the library are not updated at the same time, the section numbers in the documents may not match. You can confirm the correct section numbering by checking the [Errata](#).

1.2.1 Normative References

We conduct frequent surveys of the normative references to assure their continued availability. If you have any issue with finding a normative reference, please contact dochelp@microsoft.com. We will assist you in finding the relevant information.

[MS-ADA1] Microsoft Corporation, "[Active Directory Schema Attributes A-L](#)".

[MS-ADA3] Microsoft Corporation, "[Active Directory Schema Attributes N-Z](#)".

[MS-DTYP] Microsoft Corporation, "[Windows Data Types](#)".

[MS-MWBF] Microsoft Corporation, "[Microsoft Web Browser Federated Sign-On Protocol](#)".

[RFC1950] Deutsch, P., and Gailly, J-L., "ZLIB Compressed Data Format Specification version 3.3", RFC 1950, May 1996, <http://www.ietf.org/rfc/rfc1950.txt>

[RFC1951] Deutsch, P., "DEFLATE Compressed Data Format Specification version 1.3", RFC 1951, May 1996, <https://www.rfc-editor.org/info/rfc1951>

[RFC2045] Freed, N., and Borenstein, N., "Multipurpose Internet Mail Extensions (MIME) Part One: Format of Internet Message Bodies", RFC 2045, November 1996, <https://www.rfc-editor.org/info/rfc2045>

[RFC2119] Bradner, S., "Key words for use in RFCs to Indicate Requirement Levels", BCP 14, RFC 2119, March 1997, <https://www.rfc-editor.org/info/rfc2119>

[RFC2279] Yergeau, F., "UTF-8, a transformation format of ISO 10646", RFC 2279, January 1998, <http://www.rfc-editor.org/rfc/rfc2279>

[RFC2396] Berners-Lee, T., Fielding, R., and Masinter, L., "Uniform Resource Identifiers (URI): Generic Syntax", RFC 2396, August 1998, <https://www.rfc-editor.org/info/rfc2396>

[RFC2616] Fielding, R., Gettys, J., Mogul, J., et al., "Hypertext Transfer Protocol -- HTTP/1.1", RFC 2616, June 1999, <https://www.rfc-editor.org/info/rfc2616>

[RFC2965] Kristol, D. and Montulli, L., "HTTP State Management Mechanism", RFC 2965, October 2000, <https://www.rfc-editor.org/info/rfc2965>

[RFC4395] Hansen, T., et al., "Guidelines and Registration Procedures for New URI Schemes", BCP 115, RFC 4395, February 2006, <http://www.ietf.org/rfc/rfc4395.txt>

[SAMLCore] Maler, E., Mishra, P., Philpott, R., et al., "Assertions and Protocol for the OASIS Security Assertion Markup Language (SAML) V1.1", September 2003, <http://www.oasis-open.org/committees/download.php/3406/oasis-sstc-saml-core-1.1.pdf>

[WSFederation1.2] Kaler, C., McIntosh, M., "Web Services Federation Language (WS-Federation)", Version 1.2, May 2009, <http://docs.oasis-open.org/wsfed/federation/v1.2/os/ws-federation-1.2-spec-os.html>

[WSFederation] Kaler, C., Nadalin, A., Bajaj, S., et al., "Web Services Federation Language (WS-Federation)", Version 1.1, December 2006, <http://specs.xmlsoap.org/ws/2006/12/federation/ws-federation.pdf>

[XMLSCHEMA2] Biron, P.V., Ed. and Malhotra, A., Ed., "XML Schema Part 2: Datatypes", W3C Recommendation, May 2001, <https://www.w3.org/TR/2001/REC-xmlschema-2-20010502/>

[XML] World Wide Web Consortium, "Extensible Markup Language (XML) 1.0 (Fourth Edition)", W3C Recommendation 16 August 2006, edited in place 29 September 2006, <http://www.w3.org/TR/2006/REC-xml-20060816/>

1.2.2 Informative References

[FIPS180] FIPS PUBS, "Secure Hash Standard", FIPS PUB 180-1, April 1995, <https://www.nist.gov/ir/ia/GetFile.aspx?pid=63>

[IANASCHEME] IANA, "Uniform Resource Identifier (URI) Schemes per RFC4395", November 2006, <http://www.iana.org/assignments/uri-schemes/uri-schemes.xhtml>

[MAXURL] Microsoft Corporation, "Maximum URL Length Is 2,083 Characters in Internet Explorer", September 2009, <https://support.microsoft.com/en-us/help/208427/maximum-url-length-is-2,083-characters-in-internet-explorer>

[MS-ADTS] Microsoft Corporation, "[Active Directory Technical Specification](#)".

[MS-PAC] Microsoft Corporation, "[Privilege Attribute Certificate Data Structure](#)".

[MS-WPO] Microsoft Corporation, "[Windows Protocols Overview](#)".

[SIDD] Microsoft Corporation, "How Security Identifiers Work", March 2003, <http://technet.microsoft.com/en-us/library/cc778824.aspx>

1.3 Overview

This document specifies the Query String Response Transfer Protocol and the SAML 1.1 Assertion Extension. These extensions are based on the federated sign-on protocol described in [\[MS-MWBF\]](#). The extensions specified in this document broaden the applicability of the protocol to simpler **web browser requestors** and a wider range of protected applications. The extensions in this specification do not change the services of authentication, identity federation, or single sign-on provided by [MS-MWBF].

1.3.1 Query String Response Transfer Protocol

The scripting capability for forms submittal is not specified as part of HTTP [\[RFC2616\]](#); consequently, not all **web browser requestor** implementations support forms submittal as recommended for **wsignin1.0** responses in [\[MS-MWBF\]](#) section 2.1. In addition, the **RequestSecurityTokenResponse (RSTR)** can be too large to be transferred in a single HTTP GET message. The Query String Response Transfer Protocol addresses these issues by eliminating the web browser requestor requirement for scripting support.

When using the Query String Response Transfer Protocol, the wsignin1.0 message exchange in [MS-MWBF] is replaced by a series of wsignin1.0 message exchanges. In [MS-MWBF], the RSTR is transmitted using a single HTTP POST message. The Query String Response Transfer Protocol transmits the RSTR in pieces in query string parameters using multiple HTTP GET messages. The **relying party** accumulates the pieces from the responses in the series to produce an **aggregated result**. When the relying party has accumulated all the pieces, it extracts the RSTR from the aggregated result.

1.3.2 SAML 1.1 Assertion Extension

The Microsoft Web Browser Federated Sign-On Protocol described in [\[MS-MWBF\]](#) does not specify a method for including **SIDs** in a **security token**. For applications requiring SIDs, **claims** are not sufficient for authorization.

The SAML 1.1 Assertion Extension provides a method for including SIDs in a **SAML assertion**. How an **identity provider/security token service (IP/STS)** obtains the security identifiers and how a **relying party** interprets them is implementation-specific.

1.4 Relationship to Other Protocols

The Web Browser Federated Sign-On Protocol and the extensions specified in this document use standard web protocols, XML (as specified in [\[XML\]](#)), WS-Federation Passive Requestor Profile (as specified in [\[WSFederation1.2\]](#) section 13), and SAML 1.1 (as specified in [\[SAMLCore\]](#)). The reader has to be familiar with the specifications listed in [\[MS-MWBF\]](#) section 1.4.

A **relying party** uses the Query String Response Transfer Protocol instead of the **wsignin1.0** messages from Web Browser Federated Sign-On Protocol to avoid the use of HTTP POST messages.

The SAML 1.1 Assertion Extension provides a facility for including **security identifiers** in **SAML assertions**. In order to understand the Windows behavior relating to this extension, the reader has to be familiar with the Active Directory Technical Specification [\[MS-ADTS\]](#), security concepts in [\[MS-WPO\]](#) section 9, and security identifiers [\[SID\]](#).

The Web Browser Federated Sign-On Protocol and the extensions specified in this document can be applicable where other web-based authentication protocols are used. For more information, see [\[MS-MWBF\]](#) section 1.4.

1.5 Prerequisites/Preconditions

The SAML 1.1 Assertion Extension requires that an **IP/STS** have a source of **security identifiers** and that the **relying party** have an authorization framework in which to interpret them. The exact methods by which the IP/STS obtains the SIDs, and the methods by which the relying party interprets them, are implementation-specific. [<1><2>](#)

1.6 Applicability Statement

The Query String Response Transfer Protocol is applicable where the Web Browser Federated Sign-On Protocol described in [\[MS-MWBF\]](#) is applicable. [<3>](#) The Query String Response Transfer Protocol widens the applicability of the Microsoft Web Browser Federated Sign-On Protocol to include **web browser requestors** that do not implement scripting or form submittal via scripting.

The SAML 1.1 Assertion Extension is applicable when the protected HTTP web application requires **SIDs** to perform authorization. [<4>](#)

1.7 Versioning and Capability Negotiation

The Web Browser Federated Sign-On Protocol as described in [\[MS-MWBF\]](#) section 1.7 defers all versioning and capability negotiation to [\[WSFederation\]](#), [\[WSFederation1.2\]](#), and [\[RFC2616\]](#).

When using the Query String Response Transfer Protocol, an **IP/STS** uses the presence of the *ttpindex* parameter (as specified in section [2.2.2.1.1](#)) in the request to determine whether the Query String Response Transfer Protocol is used for the response.

The SAML 1.1 Assertion Extension uses **SIDs**. SIDs have a revision mechanism. For more information, see [\[MS-DTYP\]](#), as specified in section 2.1.

1.8 Vendor-Extensible Fields

The extensions specified in this document make use of vendor-extensible fields that are specified as part of [\[MS-MWBF\]](#) section 1.8. Specifically, the Query String Response Transfer Protocol specifies new message parameters as allowed by [\[WSFederation1.2\]](#) section 13.2.1, and the SAML 1.1 Assertion Extension specifies new elements in the **SAML advice** as allowed by [\[SAMLCore\]](#) section 2.3.2.2.

The Query String Response Transfer Protocol does not introduce any vendor-extensible fields that are not present in [\[MS-MWBF\]](#) section 1.8.

The SAML 1.1 Assertion Extension introduces the [ClaimSource \(section 3.1.5.2.1.1\)](#) element whose value is a URI that can be extended by vendors. Uniqueness of URIs is scheme-dependent. For more information, see [\[IANAScheme\]](#) and [\[RFC4395\]](#).

1.9 Standards Assignments

There are no standards assignments beyond those for XML namespaces and standard ports specified in [\[MS-MWBF\]](#) section 1.9. [<5>](#)

2 Messages

The Query String Response Transfer Protocol and SAML 1.1 Assertion Extension extend the messages specified in [\[MS-MWBF\]](#) section 2 as described in this section.

2.1 Transport

No additional transport is required other than that provided for in [\[MS-MWBF\]](#) section 2.1.

2.1.1 Query String Response Transfer Protocol

In the Query String Response Transfer Protocol, all **wsignin1.0** messages MUST use HTTP GET.

2.2 Message Syntax

The Query String Response Transfer Protocol extends the **wsignin1.0** message specified in [\[MS-MWBF\]](#) section 2.2. The SAML 1.1 Assertion Extension extends the **SAML assertion** specified in [\[MS-MWBF\]](#) section 2.2.4.2.

2.2.1 XML Namespace References

Prefixes and XML namespaces used in this specification include the following:

Prefix	Namespace URI	Reference
saml	"urn:oasis:names:tc:SAML:1.0:assertion"	[SAMLCore]
adfs	"urn:Microsoft:federation"	This document

2.2.2 Query String Response Transfer Protocol

The Query String Response Transfer Protocol extends the **wsignin1.0** message from [\[MS-MWBF\]](#) section 2.2 to enable passing results in pieces rather than using POST. The protocol does not extend any other message types.

2.2.2.1 wsignin1.0 Message

[\[MS-MWBF\]](#) section 2.2 specifies how parameters are encoded in messages, including new parameters added by extensions. Specifically, it describes that in HTTP GET messages, the parameters are encoded as query string parameters for transmission in the URL. It also specifies how invalid values are handled.

Section [2.2.2.1.1](#) specifies parameters included in both **wsignin1.0** requests and wsignin1.0 responses that use this protocol.

Section [2.2.2.1.2](#) specifies parameters included only in wsignin1.0 responses that use this protocol.

2.2.2.1.1 Common Parameters

ttpindex: The length, in characters, of the **aggregated result** as a 32-bit unsigned integer in decimal notation.

2.2.2.1.2 wsignin1.0 Response

ttpsize: The length in characters of the **pending result** as a 32-bit unsigned integer in decimal notation.

wresult: The current part of the result being transferred, as a string.

2.2.3 SAML 1.1 Assertion Extension

The SAML 1.1 Assertion Extension extends the **SAML assertion** subset as specified in [MS-MWBF] section 2.2.4.2. This extension uses the **SAML advice** element specified in [SAMLCore] section 2.3.2.2 as an extensibility point.

The sections that follow define new elements to convey the source of **claims**, hash data, **user** name, and **SIDs**. These new elements are included as child elements of the advice element in a SAML assertion. These elements use the XML namespace "urn:microsoft:federation". The element content is described using XML schema data types, as specified in [XMLSCHEMA2] section 3.

2.2.3.1 SAML Advice Elements

ClaimSource (optional): A Uniform Resource Identifier (URI) identifying a **requestor IP/STS** or other authentication service (such as a local **account** store) that is the source of the **claims** in the **security token**. The content is of type any URI (as specified in [XMLSCHEMA2] section 3.2.17).

CookieInfoHash (optional): A **base64-encoded** implementation-specific hash value. The content is of type base64Binary (as specified in [XMLSCHEMA2] section 3.2.16).[<6>](#)

WindowsUserIdentifier (optional): A **SID** identifying the **subject** of the **SAML assertion**. The content is of type string (as specified in [XMLSCHEMA2] section 3.2.1) and MUST follow the restrictions for the string representation of a SID, as specified in [MS-DTYP] section 2.4.2).

WindowsUserName (optional): A **user** name associated with the subject of the SAML assertion. The content is of type string (as specified in [XMLSCHEMA2] section 3.2.1) and MUST be of the form "DOMAIN\user name".[<7>](#)

WindowsIdentifiers (optional): A base64-encoded binary structure that defines a set of SIDs that identify the subject of the SAML assertion and a set of flags that specify the use of the SIDs. The content is of type base64Binary (as specified in [XMLSCHEMA2] section 3.2.16), and the binary data MUST be structured as specified in [WindowsIdentifiers Binary Structure \(section 2.2.3.2\)](#).

2.2.3.2 WindowsIdentifiers Structure

The WindowsIdentifiers structure has variable length. It defines a set of **SIDs** and flags. To reduce the overall data size, the SIDs are not included in full binary expansion. Rather, [PACKED SIDS](#) structures are created for each group of SIDs that are identical except for the last subauthority.

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
WindowsIdentifierFlags																															
PackedSidsCount																															
PackedSids (variable)																															
...																															

WindowsIdentifierFlags (4 bytes): A 32-bit **WindowsIdentifierFlags** structure (see [2.2.3.2.1](#)).

PackedSidsCount (4 bytes): A 4-byte, **little-endian**, unsigned integer that defines the number of **PackedSids** fields in this structure. This field **MUST NOT** be 0.

PackedSids (variable): A sequence of PACKED_SIDS structures of variable size, each of which defines a set of SIDs. The sequence defines a set of SIDs, which is the union of the sets of SIDs defined by all the elements.

2.2.3.2.1 WindowsIdentifierFlags Structure

The WindowsIdentifierFlags structure is a field of 32 bits.

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
WindowsIdentifierFlags																															

WindowsIdentifierFlags (4 bytes): Bits marked 0 **MUST** be 0. The T and N bits operate independently.

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
0	T	N	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

T - TryLocalAccount (1 bit): A value of 1 indicates that the SAML NameIdentifier (as specified in [\[SAMLCore\]](#) section 2.4.2.2) takes precedence over the [WindowsIdentifiers structure \(section 2.2.3.2\)](#) element. A value of 0 indicates that the WindowsIdentifiers structure (section 2.2.3.2) element takes precedence over the SAML NameIdentifier (as specified in [\[SAMLCore\]](#) section 2.4.2.2).

N - NoUserSid (1 bit): A value of 1 indicates that a **user SID** is not encoded in the WindowsIdentifiers structure. A value of 0 indicates that a user SID is encoded in the WindowsIdentifiers structure.

2.2.3.2.2 PACKED_SIDs Structure

The PACKED_SIDs structure encapsulates a set of **SIDs** that are identical except for the value of the final subauthority, which is called the **relative identifier (RID)**. The identical portion of the SIDs is included in the **DomainSid** field, and each RID is included separately. The PACKED_SIDs structure has a variable length.

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
DomainSid (variable)																															
...																															
RidCount																															
RID (variable)																															
...																															

DomainSid (variable): A SID structure of variable size that defines the identical portion of the SIDs encoded in this structure. For details on the SIDs structure, see [\[MS-DTYP\]](#) section 2.4.2.

RidCount (4 bytes): A 4-byte, **little-endian**, unsigned integer that defines the number of RID fields in this structure. This field MUST NOT be zero.

RID (variable): A sequence of 4-byte, little-endian, unsigned integers that define the RIDs for the SIDs encoded in this structure.

2.3 Directory Service Schema Elements

This protocol accesses the following Directory Service schema classes and attributes listed in the following table.

For the syntactic specifications of the following **<Class>** or **<Class><Attribute>** pairs, refer [\[MS-ADTS\]](#), [\[MS-ADA1\]](#), [\[MS-ADA3\]](#).

Class	Attribute
User	All

3 Protocol Details

The following sections specify the **IP/STS** and **relying party** protocol details. Each section details role-specific behavior for the extensions specified in this document. There is not a section for the **web browser requestor** role because additional protocol details for the web browser requestor other than those specified in [\[MS-MWBF\]](#) section 3.4 do not exist.

The IP/STS details (see section [3.1](#)) apply to both the **requestor IP/STS** and **resource IP/STS** roles. The relying party details (see section [3.2](#)) apply to both the resource IP/STS and **Web service (WS) resource** roles.

Because the behavior for issuance and consumption of the **SIDs** is implementation-specific, an abstract data model is not introduced for the SAML 1.1 Assertion Extension. Hence, sections [3.1.1](#) and [3.2.1](#) do not have subsections for that extension.

3.1 IP/STS Details

The following sections specify the protocol details of the **IP/STS**, including details for the Query String Response Transfer Protocol and SAML 1.1 Assertion Extension. These details apply to both the **resource IP/STS** and the **requestor IP/STS** roles.

3.1.1 Abstract Data Model

This section describes a conceptual organization of data that an implementation maintains to participate in this protocol. The described organization is provided to facilitate the explanation of how the protocol behaves. This document does not mandate that implementations adhere to this model as long as their external behavior is consistent with that described in this document.

3.1.1.1 Query String Response Transfer Protocol

The following sections specify the abstract data model for transmitting the **pending result** as a series of parts in separate messages that are assembled by a **relying party**.

3.1.1.1.1 Pending Result

When the Query String Response Transfer Protocol is used, the result of a **wsignin1.0** action is transmitted by a series of HTTP 302 responses.

The first message in the exchange establishes the **pending result**. Each message exchange in the series transmits a portion of the pending result to the **relying party** until all parts have been delivered. Consequently, the pending result must be available to the **IP/STS** when processing each message exchange until the series is complete.

The message exchange also includes parameters for the current position and the total length of the result (see section [2.2.2.1](#)) to ensure proper construction and error/completion detection. The method used to maintain the availability of the pending result during the series of exchanges is implementation-specific. The IP/STS MUST discard the pending result when cleaning up local state. See [\[MS-MWBF\]](#) section 3.2.5.3.3 for **requestor IP/STS**. See [\[MS-MWBF\]](#) section 3.3.5.4.2 for **resource IP/STS**.[<8>](#)

3.1.1.1.2 Maximum Query String Response Message Length

Although the HTTP protocol (as specified in [\[RFC2616\]](#)) does not place a limit on the length of a URL, some **web browser requestor** implementations have such a limit. **IP/STS** implementations SHOULD use a maximum query string response message length that will allow a broad range of web browser implementations to act as the web browser requestor. This is a limit on the length of the URL after

escaping, including the scheme, authority, path, and query components (as specified in [\[RFC2396\]](#) section 3). The recommended value is 2,083 octets for all messages. For more information, see [\[MAXURL\].<9>](#)

3.1.2 Timers

There are no timers required other than any specified in [\[MS-MWBF\]](#) section 3.1.2; however, a timer MAY [<10>](#) be used to manage state associated with the **pending result**.

3.1.3 Initialization

There are no new initializations beyond any described in [\[MS-MWBF\]](#) section 3.1.3; however, the **pending result** is initialized after the protocol has been initiated (see section [3.1.5.1.2](#)).

3.1.4 Higher-Layer Triggered Events

There are no higher-layer triggered events other than any described in [\[MS-MWBF\]](#) section 3.1.4.

3.1.5 Processing Events and Sequencing Rules

3.1.5.1 Query String Response Transfer Protocol

The following sections specify protocol details for the Query String Response Transfer Protocol when receiving **wsignin1.0** requests.

Sections [3.1.5.1.1](#), [3.1.5.1.2](#), and [3.1.5.1.3](#) specify how **wsignin1.0** requests are processed given the value of the *ttpindex* parameter. Section [3.1.5.1.4](#) specifies common behavior for responding to such requests when the Query String Response Protocol is used.

3.1.5.1.1 Receiving a **wsignin1.0** Request That Does Not Specify a *ttpindex*

When the **IP/STS** receives a **wsignin1.0** request that does not specify the *ttpindex* parameter, the response MUST NOT use the Query String Response Transfer Protocol. The IP/STS MUST discard any **pending result**, and the IP/STS MUST process the message as specified in [\[MS-MWBF\]](#) section 3.1.5.4.

3.1.5.1.2 Receiving a **wsignin1.0** Request That Specifies a *ttpindex* of 0

When the **IP/STS** receives a **wsignin1.0** request that specifies a *ttpindex* parameter value of 0, the IP/STS MUST process the **wsignin1.0** request (as specified in [\[MS-MWBF\]](#) section 3.1.5.4) up to the point where the IP/STS has constructed the **RSTR** element (as specified in [\[MS-MWBF\]](#) section 3.1.5.4.6). The IP/STS MUST apply the following transforms to the RSTR element to produce the **pending result**:

1. Convert the XML string to binary data by applying UTF-8 encoding, as specified in [\[RFC2279\]](#).
2. Compress the result from step 1 to the zlib format (as specified in [\[RFC1950\]](#)) by using the deflate algorithm (as specified in [\[RFC1951\]](#)).
3. Base64-encode the result from step 2, as specified in [\[RFC2045\]](#) section 6.8.

The IP/STS MUST respond as specified in section [3.1.5.1.4](#), by using the result from step 3 as the pending result.

3.1.5.1.3 Receiving a **wsignin1.0** Request That Specifies a *ttpindex* Other Than 0

When the **IP/STS** receives a **wsignin1.0** request that specifies a *ttpindex* parameter value other than 0, the following conditions MUST be evaluated in order:

1. The *ttpindex* value does not conform to message syntax rules (for example, if it is not a number, as specified in section [2.2.2.1.1](#)).
2. The corresponding **pending result** is not available.
3. The value of *ttpindex* is greater than or equal to the length in characters of the pending result.

If the message meets any of the conditions, the IP/STS MUST reject the message and return an HTTP 500 response. Otherwise, the IP/STS MUST respond as specified in section [3.1.5.1.4](#).

3.1.5.1.4 Responding to a wsignin1.0 Request That Specifies a ttpindex

The **IP/STS** MUST construct a **wsignin1.0** response for the **relying party** by using the following values as properly escaped query string parameters in the returned status URL, as specified in [\[WSFederation1.2\]](#) section 13.2.3.

- *ttpindex* MUST be the same value of *ttpindex* as requested.
- *ttpsize* MUST be the same length in characters as the **pending result**.
- *wctx* MUST be the same value of *wctx* as requested.
- *wresult* MUST contain a portion of the pending result, beginning at the character of index *ttpindex* (zero-based), and including as many characters as possible without exceeding the maximum query string response message length. Note that the maximum query string response message length applies after the escaping rules for URL query string parameters.

The IP/STS MUST transmit the message to the **web browser requestor** using HTTP 302 as specified in [\[WSFederation1.2\]](#) section 13.2.3.

3.1.5.2 SAML 1.1 Assertion Extension

The **IP/STS** uses the SAML 1.1 Assertion Extension when responding to **wsignin1.0** requests. Although the protocol details are largely similar, it is necessary in this section to distinguish between the **requestor IP/STS** and **resource IP/STS** roles.

The IP/STS is a requestor IP/STS when issuing a token to a **relying party** that is in a different **security realm** than the IP/STS. Conversely, the IP/STS is a resource IP/STS when issuing a token to a relying party that is in the same security realm as the IP/STS.

3.1.5.2.1 Responding to a wsignin1.0 Request

When responding to a **wsignin1.0** request, the **IP/STS** MAY include any of the [ClaimSource](#) (section [3.1.5.2.1.1](#)), [CookieInfoHash](#) (section [3.1.5.2.1.2](#)), [WindowsUserIdentifier](#) (section [3.1.5.2.1.3](#)), [WindowsUserName](#) (section [3.1.5.2.1.4](#)), and [WindowsIdentifiers](#) (section [3.1.5.2.1.5](#)) elements in the issued **SAML assertion**. For syntax details, see section [2.2.3](#).

The following sections describe the processing semantics for each of these optional SAML assertion elements.

3.1.5.2.1.1 ClaimSource Element

The **IP/STS** MAY [<11>](#) include the ClaimSource element in the issued **SAML assertion**.

3.1.5.2.1.2 CookieInfoHash Element

The **IP/STS** MAY<12> include the CookieInfoHash element in the issued **SAML assertion**.

3.1.5.2.1.3 WindowsUserIdentifier Element

The **IP/STS** MAY<13> include the WindowsUserIdentifier element in the issued **SAML assertion**.

3.1.5.2.1.4 WindowsUserName Element

The **IP/STS** MAY<14> include the WindowsUserName element in the issued **SAML assertion**.

3.1.5.2.1.5 WindowsIdentifiers Element

The **IP/STS** MAY<15> include the WindowsIdentifiers element in the issued **SAML assertion**.

3.1.6 Timer Events

Timer events are not required other than any specified in [MS-MWBF] section 3.1.6; however, a timer event MAY<16> be used to manage the state associated with the **pending result**.

3.1.7 Other Local Events

There are no other local events that impact the operation of this protocol.

3.2 Relying Party Details

These details apply to both the **resource IP/STS** and **WS resource** roles (when acting as a **relying party** for a received **security token**).

3.2.1 Abstract Data Model

This section describes a conceptual model of a possible data organization that an implementation maintains to participate in this protocol. The described organization is provided to facilitate the explanation of how the protocol behaves. This document does not mandate that implementations adhere to this model as long as their external behavior is consistent with that described in this document.

3.2.1.1 Query String Response Transfer Protocol

The following sections specify the abstract data model for receiving and concatenating the **aggregated result** as a series of messages.

3.2.1.1.1 Aggregated Result

When the Query String Response Transfer Protocol is used, the result of a **wsignin1.0** action is transmitted by a series of message exchanges. When an **aggregated result** is not available, it is considered to be empty (with size 0). The **relying party** concatenates each portion of the result to the aggregated result when it is received, so the aggregated result must be available to the relying party when processing each message exchange. The method used to maintain the availability of the aggregated result is implementation-specific. The relying party **MUST** discard the aggregated result when it receives a **wsignoutcleanup1.0** message (as specified in [MS-MWBF] section 3.3.5.4.2).<17>

3.2.2 Timers

There are no timers required other than any specified in [MS-MWBF] section 3.3.2; however, a timer MAY<18> be used to manage a state associated with the **aggregated result**.

3.2.3 Initialization

There are no new initializations other than any described in [\[MS-MWBF\]](#) section 3.3.3; however, the **aggregated result** is initialized after the protocol has been initiated (as specified in section [3.2.5.1.3](#)).

3.2.4 Higher-Layer Triggered Events

There are no new higher-layer triggered events other than any described in [\[MS-MWBF\]](#) section 3.3.4.

3.2.5 Processing Events and Sequencing Rules

The Query String Response Transfer Protocol and the SAML 1.1 Assertion Extension introduce new message processing rules.

3.2.5.1 Query String Response Transfer Protocol

The following sections specify protocol details for the Query String Response Transfer Protocol when processing **wsignin1.0** messages.

Section [3.2.5.1.1](#) specifies how to send a **wsignin1.0** request by using this protocol. Sections [3.2.5.1.2](#) and [3.2.5.1.3](#) specify how to process a **wsignin1.0** response given the presence or absence of the *ttpindex* parameter. Section [3.2.5.1.4](#) specifies how the completed **aggregated result** is processed.

3.2.5.1.1 Sending a **wsignin1.0** Request

When using the Query String Response Transfer Protocol, the **relying party** sends a **wsignin1.0** request to the **IP/STS** as specified in [\[MS-MWBF\]](#) section 3.3.5.1 and MUST add the *ttpindex* parameter. When included, the value of the *ttpindex* parameter MUST be the length in characters of the **aggregated result**.

Note If the aggregated result is not available, it is considered to have length 0 (see section [3.2.1.1.1](#)). Therefore, in the first request, the value of *ttpindex* MUST be 0. [<19>](#)

3.2.5.1.2 Receiving a **wsignin1.0** Response That Does Not Specify a *ttpindex*

When the **relying party** receives a **wsignin1.0** response that does not specify the *ttpindex* parameter, the message does not use the Query String Response Transfer Protocol, and the relying party MUST process the message as specified in [\[MS-MWBF\]](#) section 3.3.5.2.

3.2.5.1.3 Receiving a **wsignin1.0** Response That Specifies a *ttpindex*

When the **relying party** receives a **wsignin1.0** response that specifies the *ttpindex* parameter, it MUST evaluate the following conditions in order:

1. The *ttpindex* value does not conform to message syntax rules (for example, it is not a number; as specified in section [2.2.2.1.1](#)).
2. The *ttpindex* parameter is not equal to the length, in characters, of the **aggregated result**. Note that if the aggregated result is not available, it is considered to have length 0 (as specified in section [3.2.1.1.1](#)).

If the message meets one of these conditions, the relying party MUST reject the message and return an HTTP 500 response.

The relying party MUST construct a new aggregated result. If *ttpindex* is 0, the aggregated result MUST be the value of the *wresult* parameter; otherwise, the aggregated result MUST be constructed by appending the *wresult* parameter to the current aggregated result.

The relying party MUST evaluate the *ttpsize* parameter as follows:

- If the length in characters of the new aggregated result is greater than the value of the *ttpsize* parameter, the relying party MUST reject the message and return an HTTP 500 response.
- If the length, in characters, of the new aggregated result is equal to the value of the *ttpsize* parameter, the relying party MUST process the completed result as specified in section [3.2.5.1.4](#).
- If the length, in characters, of the new aggregated result is less than the value of the *ttpsize* parameter, the relying party MUST request the next portion of the result as specified in section [3.2.5.1.1](#).

3.2.5.1.4 Processing the Complete Aggregated Result

When the **aggregated result** is complete, the **relying party** MUST apply the following transforms to the aggregated result to produce an XML string:

1. Base64-decode the aggregated result to binary data (as specified in [\[RFC2045\]](#) section 6.8).
2. Decompress the result from step 1 from the zlib format (as specified in [\[RFC1950\]](#)) by using the inflate algorithm (as specified in [\[RFC1951\]](#)).
3. Interpret the binary data from step 2 as a UTF-8-encoded string (as specified in [\[RFC2279\]](#)).

If any error occurs when applying the transforms, the relying party MUST reject the message and return an HTTP 500 response.

The relying party MUST process the **wsignin1.0** response (as specified in [\[MS-MWBF\]](#) section 3.3.5.2), substituting the string output from step 3 for the *wresult* parameter.

3.2.5.2 SAML 1.1 Assertion Extension

The method of evaluating **SIDs** transmitted by using the SAML 1.1 Assertion Extension is implementation-specific. For specifications about SIDs and their semantics in Windows, see [\[MS-DTYP\].<20>](#)

3.2.6 Timer Events

There are no timer events required other than any events specified in [\[MS-MWBF\]](#) section 3.3.6; however, a timer event MAY [<21>](#) be used to manage a state associated with the **aggregated result**.

3.2.7 Other Local Events

There are no other local events that impact the operation of this protocol.

3.3 Web Browser Requestor Details

This section specifies the **web browser requestor** role in transporting protocol messages.

3.3.1 Abstract Data Model

A **web browser requestor** does not need to understand any protocol-specific data for the correct operation of the protocol. It MUST be able to support HTTP query string and POST body parameterization. To provide the best end-user experience, it SHOULD be able to support HTTP cookies (for more information, see [\[RFC2965\]](#)).<22>

3.3.2 Timers

A **web browser requestor** does not depend on timers beyond those that are used by the underlying transport to transmit and receive messages over HTTP and SSL/TLS, as specified in section [3.1.2](#).

A web browser requestor does not need to be aware of an implementation's use of timers to determine when the validity intervals of **security tokens** and authentication contexts expire.

3.3.3 Initialization

There is no protocol-specific initialization for a **web browser requestor**. It only needs to be ready to perform the standard HTTP 1.1 methods required for accessing **WS resources**. Specifically, it **MUST** support HTTP GET and POST methods and properly respond to HTTP 1.1 redirection and error responses.

3.3.4 Higher Layer Triggered Events

Protocol messages are exchanged between a **requestor IP/STS** and a **relying party**. The only function of the **web browser requestor** with respect to the protocol is to transport these messages. The web browser requestor can be triggered to begin protocol message exchange by receipt of an HTTP/1.1 302 found that includes a location directive, or an HTTP/1.1 200 OK that includes a form with method set to POST.

3.3.5 Processing Events and Sequencing Rules

A **web browser requestor** plays a passive role in the operation of the protocol. Its only function is to transport protocol message requests and responses between a **requestor IP/STS** and one or more relying parties. It is not required to understand the types or content of these protocol messages. The web browser requestor **SHOULD** transport all protocol message requests and responses between a requestor IP/STS and a **relying party** without changing the messages at all. [<23>](#)

3.3.6 Timer Events

A **web browser requestor** does not need to interact with any timers, or service any timer events, beyond those that might be used by the underlying transport to transmit and receive messages over HTTP and SSL/TLS, or those specified in section [3.1.6](#).

3.3.7 Other Local Events

A **web browser requestor** does not have dependencies on local events beyond those specified in section [3.1.7](#).

4 Protocol Examples

4.1 Query String Response Transfer Protocol

4.1.1 Annotated Example

The following is a protocol example for the Query String Response Transfer Protocol.

The Query String Response Transfer Protocol is best understood as occurring abstractly between an **IP/STS** and a **relying party**, because the changes to the Web Browser Federated Sign-On Protocol [MS-MWBF] are applied consistently whether between **requestor IP/STS** and **resource IP/STS** or between resource IP/STS and **WS resource**.

This annotated example shows a Query String Response Transfer Protocol exchange between a requestor IP/STS and a resource IP/STS. It is part of a larger network trace (see section 4.2) that also uses the Query String Response Transfer Protocol between the resource IP/STS and the WS resource.

The following table specifies the protocol roles of the hosts.

Protocol role	Host name
IP/STS	adatumsts-7
Relying party	treysts-7

Each HTTP message is prefaced by an annotation that describes its recipient and purpose. This annotated example omits many elements of the HTTP messages. For example, implementation-specific cookies and superfluous HTTP headers are not included. The full messages are specified in section 4.1.2. The following parameters are specified in this document and appear in the HTTP messages that follow:

- *ttpindex*
- *ttpsize*
- *wresult*

The following are the processing steps of this annotated example:

1. Just prior to the example, the **web browser requestor** made a GET request to the relying party (treysts-7).
2. The relying party returns an HTTP 302 message that specifies a **wsignin1.0** request for the IP/STS. This wsignin1.0 request includes the *ttpindex=0* parameter, which initiates the Query String Response Transfer Protocol (see section 3.2.5.1). At this time, the relying party's **aggregated result** is empty, as specified in section 3.2.5.1.1.

```
HTTP/1.1 302 Found
Location:
https://adatumsts-7/adfs/ls/?wa=wsignin1.0&
wtrealm=urn%3afederation%3atreys+research&
wct=2006-07-13T07%3a32%3a21Z&
wctx=https%3a%2f%2ftreys-test%2fclaims%2f%5chttps%3a%2f%
2ftreys-test%2fclaims%2fDefault.aspx&ttpindex=0
```

3. The web browser requestor relays the wsignin1.0 request to the IP/STS (adatumsts-7) in an HTTP GET message.

```
GET /adfs/ls/?wa=wsignin1.0&wtrealm=urn%3afederation
%3atrey+research&wct=2006-07-13T07%3a32%3a21Z&wctx=https%
3a%2f%2ftreyws-test%2fclaims%2f%5chttps%3a%2f%2ftreyws-test%
2fclaims%2fDefault.aspx&ttpindex=0 HTTP/1.1
```

4. The IP/STS engages in a series of messages outside the scope of the protocol whereby it ascertains the **user** identity. These messages are omitted, as they have no bearing on the Query String Response Transfer Protocol. For more details, see [MS-MWBF] section 3.1.5.4.3.
5. Once the user's identity has been determined, the IP/STS creates an **RSTR**, as specified in [MS-MWBF] section 3.1.5.4.6. The RSTR is transformed into the **pending result**, and the first portion is returned in a wsignin1.0 response (as specified in section 3.1.5.1.4). In this message, the **wresult** parameter is the first 1,727 characters of the pending result. The **ttpsize** parameter indicates the length of the full pending result, 2,652.

```
HTTP/1.1 302 Found
Location: https://treysts-7/adfs/ls/?wa=wsignin1.0&
ttpsize=2652&ttpindex=0&wctx=https%3a%2f%
2ftreyws-test%2fclaims%2f%5chttps%3a%2f%2ftreyws-test%2fclaims%
2fDefault.aspx&wresult=eNrNWftzqsos%
2fUW6zGVBYg3rCRlhquAaAlyssphOEShVEGRpz1e9BoTNYt59R62D4NPW3311%
2f3dDM8lDgFWHBXQJzb0CuyOK%2bmaA1TC%2bItSjFshJJNigde7bEZ5f12QFHYi2Di
4u9kByN3%2bx1lIVViqkXTHYpuUXlW4Lz59HBjGfofbD89YDfZDADGMMtjlDauK1V8
bP436NAMGzD%2bvddps%2fftFuzd91ewc79iu5y7Yv1%2b4HeadRXjAqopzt2UQCPOu
%2fd0755hp3RvwLYGrZ7zppM9NossHQTQh51bOxm4vpsXSbNhuK8om8MME
%2bFjkyGCOP0oOAdfoz0bQS6O8SB1E4gHuTEwTEaMN
%2fpGxsJoPkWm4BSP64FuDFGOQ8D1MFfoSQKk3SSgSCvsd7q9C86F8YKP4apB0ly8i
z2avtXR59Unj7FnGewamQQzFzozgfo%2b6n55%2bbpz4FdnHo7
%2bPahOsHmOyidUyyEqChi6M3%2buqdM31J7GUIoyC%2fQdZ8UqK5vj2Kulay0tQy
KoWL1HhnM8%2fS%2bvGB%2bondK94312%2fg8wimeeydrNq5m8OEPDc%2byv9QMB%
2bvDZhhYp%2bxfD4oDco49VGJLzmyi9Ur9PK3pzEpFdWvtYMYZg0ZZYn7
%2b5Pkbdw4wdTsedx8An4SpzFJhpuj7D%2fnsV3uoeQt%2fo%2fGL6xcAVC
%2f4%2bXCWk4yvSpy%2bHnj3xPGR5yN66r%2b22NTStx4A3yflDZuftzEW9eDXwDZ
%2f0xj7m4K%2bOT%2bHvUn7c%2fSP%2bEWUJKgtF7%2fXdQGUwydKL87yBVMLRs
%2fy5Iod4kCU7lID9d8ME222G919HSf3yXNlXmJiiyt5m6hVXWZbfs
%2fYeh3QlqmI5iij4OA6%2fNc%2f%2fgn7d%2bp4eBDdFKTnJm
%2fh406AaYBMiMlyj5BcmGYgha5P38ODde0w7%2fdZsUDd4vmjma7IMu
%2fc4cpmTJQsGMKsnR2NmQY%2fNb18b4U8P08xNMRmNCb5Z%2f284YLqHG7SF
%2fj2%2bhHOC9HVZv2CHugUnxiGzif8PTleOzibOVTPhN2GVHqWsG5S6BHgV4f3E
0eycpnEu9V8k07ckvVtLVyzdlYUYTg1clGn%2fJ5011Fmx1mmaTOBae%2bMEojVHb
%2bd96NDIj09jLvrlopNJXwsPNN3kWWtuE2Vj4t90Zram4XtYg1Y2jTyLGrUFmYHmd
V4WW87i6VIm9VO3UNABoiR5XrV0emvSA5kpm9SXZK7rXmHZ
%2fGWXdBd5mpuJX07ZiQ4ru01X%2fmi6gf4De4zmQG
%2fAPOqz0US06Nce6uXteCFVLVVCPLtLYVfVkreEAWRWCuuVBqOoKGyayCeqfdJiCMR
%2bud9E6Vris5oE5k4EIHMPyStlcinPTFKWYm3OVTeXiFOS1VmeWcHtFBFAu6XL8C
lhjqlbGFJSGOHZPsnNFJplkBr88yCKw%2bXA854ExFVryermwmNViTHsVzRqQehiA
VgR7p9jqihVnQcYBQFSZA1HgY1PnQ10017ue4uB4smhLvZZth3SxY7QX1thxeQF0p
%2b9W%2bNmJNkfjZb%2fTga7DDMWjyOeTmeZ07vb8EogLjTrK%2bZRq6weJ1WNn02
k9F6RVHa3%2bWqcZXRkto8gHGZR5oZfYq4g72IsKhFxFvFycmf0Eas4dKvg9xCec
SI0h1rSDYWDpzG7tcUxxz7Q5WOpisAEPGqrPD0S7p1%2bKYaES4t%2bBuaQIsGJIJ
yWhmTexqyJpcka4qzWVXX
```

6. The web browser requestor relays the wsignin1.0 response to the relying party (treysts-7) in an HTTP GET message.

```
GET /adfs/ls/?wa=wsignin1.0&
ttpsize=2652&ttpindex=0&wctx=https%3a%2f%2ft
reyws-test%2fclaims%2f%5chttps%3a%2f%2ftreyws-test%2fclaims
%2fDefault.aspx&wresult=eNrNWftzqsos%2fUW6zGVBYg3rCRlh
quAaAlyssphOEShVEGRpz1e9BoTNYt59R62D4NPW3311%2f3dDM8lDgFWHBXQJzb0
CuyOK%2bmaA1TC%2bItSjFshJJNigde7bEZ5f12QFHYi2Di4u9kByN3%2bx1lIVViq
kXTHYpuUXlW4Lz59HBjGfofbD89YDfZDADGMMtjlDauK1V8bP436NAMGzD%2bvddps
%2fftFuzd91ewc79iu5y7Yv1%2b4HeadRXjAqopzt2UQCPOu%2fd0755hp3RvwLYGr
```

Z7zppM9NossHQTh51bOxm4vpsXSbNhuK8om8MME%2bFjkyGCOP0oOAdfoz0bQS608
 SB1E4gHuTewgTEaMN%2fpgXsJoPkWm4BSP64FuDFGOQ8D1MFfoSQKk3SSgSCvsd7q9
 C86F8YKP4apB01y8iz2avtXR59Unj7FnGewamQQQzfzofqo%2b6n55%2bbpz4FdnH
 o7%2bPahOsHmOyidUyyEqChi6M3%2buqdm31J7GUIOyC%2fQdZ8UqK5vj2Kulay0tQ
 yKoWL1HhnM8%2fS%2bvGB%2bondK94312%2fg8wimeeydrNq5m8OEPDc%2byv9QMB%
 2bVDZhHyP%2bxfD4oDco49VGJLzmyi9Ur9PK3pzEpFdWvtYMYZg0ZZYn7%2b5Pkbaw
 4wdTsedx8An4SpzFJhpuj7D%2fnsV3uoEQ%2fo%2fGL6xcAVC%2f4
 %2bXCWk4yvSpy%2bHnj3xPGR5yN66r%2b22NTStx4A3yflDZuftzEW9eDXwDZ
 %2fOxj7m4K%2bOT%2bHvUn7c%2fSP%2bEWUJKgtF7%2fXdQGQUwydKL87yBVM1Rs
 %2fy5Iod4kCU7inID9d8ME22G919HSf3yXN1xmJiIyt5m6hVXWZbfs
 %2fYeh3QlmgI5iij4OA6%2fNc%2f%2fgn7d%2bp4eBdDFKtNjM%2fh406AaYBMiM1
 yj5BcmGXgha5P38ODde0w7%2fdZsUDd4vmjma7IMu%2fc4cpmTJQsGMKsnR2NmQY
 %2fNb18b4U8P08xNMRmNCb5Z%2f284YLqHG7SF%2fj2%2bhHOC9HVzv2CHugUnxi
 GZiF8PT1eOzibOVTPHN2GVHqWsg5S6BHgV4f3E0eycpnEu9V8oK7ckvVtLVyzd1
 YUYTg1c1Gn%2fJ5011Fmx1mmaTOBae%2bMEojVHb%2bd96NDIj09jLvrlopNjXws
 PNN3kWwtUE2Vj4t90Zram4XtYglY2jTyLGrUFmYHmdV4WW87i6VIm9VO3UNaBPOi
 R5XrV0emvSA5kpm9SXZK7rXmHZ%2fGWXdBd5mpuJX07ZIQA4ru01X%2fmI6gf4De
 4zmQG%2fAPQz0US06Nce6uXteCFVLVVCPLtLYVfVkrEEAWRWCuVBqOoKGyayCe
 qfdJiCMR%2bud9E6Vris5oE5k4EIHMPyStlcinPTFKWYm3OVTeXiFOSlVmeWcHtf
 BFAu6XL8ClhjqlbGfJSGOHZPsnFJplkBr88yCKw%2bXA854ExFVryermwNViTH
 sVzRqQehiAVGr7p9jqihVNqcYBQFsZA1HgY1PnQ10017uE4uB4smhLvZZth3SxY7
 QX1thxeQF0p%2b9W%2bNmJNkfjZb%2fTGa7DDMWjyOeTmeZO7vb8EogLjTrK%2bZ
 Rq6weJlWNn02k9F6RVHa3%2bWqcZXRkto8gHGZR5oZfyq4g72IsKhFxVavFycmf0E
 as4dKvg9cUEcSIOh1rSDYWPzG7tcUxxz7Q5WOpisAEPGqrPDos7p1%2bKYaES4t%
 2bBuaQISGJIJyWhmTexqyJpcka4qzWVXX HTTP/1.1

7. Because the relying party's aggregated result is empty, the *wresult* parameter becomes the aggregated result. Because the length of the *wresult* parameter was 1,727 characters, which is less than the length of the *ttpsize* parameter of 2,652 characters, the relying party needs to request more data from the IP/STS. It does this by sending an HTTP 302 with a *wsignin1.0* request that includes the *ttindex=1727* parameter. For further specifications, see section [3.2.5.1.3](#).

HTTP/1.1 302 Found
 Location: https://adatumsts-7/adfs/ls/?wa=wsignin1.0&wtrealm=urn%3afederation%3atrety+research&wct=2006-07-13T07%3a32%3a27Z&wctx=https%3a%2f%2ftreyws-test%2fclaims%2f%5chttps%3a%2f%2ftreyws-test%2fclaims%2fDefault.aspx&ttindex=1727

8. The web browser requestor relays the *wsignin1.0* request to the IP/STS (adatumsts-7) in an HTTP GET message.

GET /adfs/ls/?wa=wsignin1.0&wtrealm=urn%3afederation%3atrety+research&wct=2006-07-13T07%3a32%3a27Z&wctx=https%3a%2f%2ftreyws-test%2fclaims%2f%5chttps%3a%2f%2ftreyws-test%2fclaims%2fDefault.aspx&ttindex=1727 HTTP/1.1

9. The IP/STS returns the remaining 925 characters of the pending result, beginning with character 1,727. For further specifications, see section [3.1.5.1.3](#).

HTTP/1.1 302 Found
 Location: https://treysts-7/adfs/ls/?wa=wsignin1.0&ttpsize=2652&ttindex=1727&wctx=https%3a%2f%2ftreyws-test%2fclaims%2f%5chttps%3a%2f%2ftreyws-test%2fclaims%2fDefault.aspx&wresult=kqNHeG5OcCSAUgEHIEISLriGRPIQJBEM%2bnGviaAj8ERCSwkseRfLs13wPzbYkh%2bZs6Vn0YZP76zacMJFGrZbrVe%2fYXvUjtZTKE7YND8pSqH2XS543Z0Piuxye9155PixlBGZyO6WHThaJc9Xt2NPJctVWbL2tsZNZOTdPugEvtcupyOtIikMLOZI7tGhPRPtRa7wh9RP5CwuNEqvwqs7ritXKUcJsl6y291pc4gtElqL3Zmte%2bdK88BIOrwTux%2fojel5SlqGDCC9YgeuWLyWeV5UxqsfdkNR3%2breIOauD%2fKFVtUL%2bflnbVmTGV8LPf16dhUYvF3PalX%2fhMzVLC7xzZABdmSjOdqXMuqOK8xbKe4wL6bSWV6m2Xykh791vFMvVaOs1lrkk

```
%2fuCc7LnmQTigP58D%2fnQOyLk1eXYX3x3GU8rrUsn84LO%2bcGRAtt4eNHYY1VD
NEDcGh1%2fYOCujud%2fup3%2bZeFDpVb3s91IYB7%2b3VXuK%2bwLyUR23Le07Y
WTLPlDl1p62FeOiHAQuF53jLdrGtc%2b3jPFPkwJFdrk%2b9HlC7ZNDdiyLPcivf
SV2bCmN6ZEmxMRJ0z3KS6G426mldQl0MYcV1WJ3zzJHR8pWVozlral0wbkuYqT0l
eD60fT1hBCVDicNLlmwCRiKRIAu9QjsGoiff4T63Xg3zIFiq6rSQ%2fRvr29y6LF
oyBn2NHQ6T1oTaG%2f1lJe46h3nqJM8dbElFMpNTRKW7SYduLSjlaHXZKO2m0
%2fTykrVn3ny93nGygPpVKClJt3qFQfFAfe6YZ8m5m1LXDvvee6%2fvKpfLKxH8
7tJe4u2AtJhNDPEUXb8LbL%2fwXaBdT8Qt2sRedfow4A6k1N%2biOM3f3xwu9ty
v2euTi%2bHppT9Owzebb7eAP9x8bzXPTz9gOclvYv3Iy08%2fkzz9A6cx%2fkW
%3d
```

10. The web browser requestor relays the `wsignin1.0` response to the relying party (treysts-7) in an HTTP GET message.

```
GET /ads/ls/?wa=wsignin1.0&
ttptime=2652&ttptimeindex=1727
&wctx=https%3a%2f%2ftreyws-test%2fclaims%2f%5chttps%3a
%2f%2ftreyws-test%2fclaims%2fDefault.aspx&
wresult=kqNHeG5OcCSAUgEHIEISLriGRPIJQBEM
%2bnGViaAj8ERCSwkseRfLs13wPzbYkh%2bZs6Vn0YZP76zacMJFGrZbrVe
%2fYXUvJtZTKE7YND8pSqH2XS543Z0Piuxye9155PixlBGZy06WHThaJc9Xt2NP
JctVWbL2TsZNZOTdPugEvtcupyOtIikMLOZI7tGhPRPtRa7wh9RP5CwuNEqvwqs
7ritXKUcJsl6y291pc4gtElqL3Zmte%2bdK88BIOrwTux%2fojel5SlqGDCC9Y
geuWLyWeV5UxqSfdkNR3%2bREI0auD%2fKFVTuL%2bflnbVmtGV8LPfl6dhUYvF
3PalX%2fHmzVLC7xzZABdmSjOdqXMuqOK8xbKe4wL6bSWV6m2XyKH791vFMvVaO
s1lrkk%2fuCc7LnmQTigP58D%2fnQOyLk1eXYX3x3GU8rrUsn84LO%2bcGRAtt4
eNHYY1VDNEDcGh1%2fYOCujud%2fup3%2bZeFDpVb3s91IYB7%2b3VXuK%2bwLyU
R23Le07YWTLPlDl1p62FeOiHAQuF53jLdrGtc%2b3jPFPkwJFdrk%2b9HlC7ZND
diyLPcivfSV2bCmN6ZEmxMRJ0z3KS6G426mldQl0MYcV1WJ3zzJHR8pWVozlral
0wbkuYqT0leD60fT1hBCVDicNLlmwCRiKRIAu9QjsGoiff4T63Xg3zIFiq6rSQ
%2fRvr29y6LFoyBn2NHQ6T1oTaG%2f1lJe46h3nqJM8dbElFMpNTRKW7SYduLS
jlaHXZKO2m0%2fTykrVn3ny93nGygPpVKClJt3qFQfFAfe6YZ8m5m1LXDvvee6
%2fvKpfLKxH87tJe4u2AtJhNDPEUXb8LbL%2fwXaBdT8Qt2sRedfow4A6k1N
%2biOM3f3xwu9tyv2euTi%2bHppT9Owzebb7eAP9x8bzXPTz9gOclvYv3Iy08
%2fkzz9A6cx%2fkW%3d HTTP/1.1
```

11. The relying party appends the `wresult` parameter to the aggregated result. The new aggregated result is 2,652 characters long, which indicates the completion of the Query String Response Transfer Protocol (see section 4.1). The relying party extracts the RSTR from the aggregated result (as specified in section 3.2.5.1.4) and processes it as specified in [MS-MWBF] section 3.3.5.2. The relying party's next action is outside the scope of the protocol, though in this case the relying party, which was a resource IP/STS, issued a new **SAML assertion** and used the Query String Response Transfer Protocol to transmit it to a WS resource.

4.1.2 Full Network Trace

The following is the full network trace for the protocol example in [Annotated Example \(section 4.1.1\)](#).

The following table specifies the protocol roles of the hosts.

Protocol role	Host name
Requestor IP/STS	adatumsts-7
Resource IP/STS	treysts-7
WS resource	treyws-test

HTTP requests are prefaced by a line with three right-angle brackets ("**>>>**") and the name of the host to which the request was sent. Requests are followed by the HTTP response, which is prefaced by

a line with three left-angle brackets ("<<<"). The final three HTTP messages (HTTP 302, HTTP GET, and HTTP 200 OK) are an internal implementation detail and shown for completeness only. These messages are not necessary for interoperability.

```
>>> treyws-test (WS Resource)
GET /claims/ HTTP/1.1

<<<
HTTP/1.1 302 Found
Location: https://treysts-7/adfs/ls/?wa=wsignin1.0&wreply=https%3a%2f%2ftreyws-test%2fclaims%2f&wct=2006-07-13T07%3a32%3a21Z&wctx=https%3a%2f%2ftreyws-test%2fclaims%2fDefault.aspx&ttpindex=0

>>> treysts-7 (Resource IP/STS)
GET /adfs/ls/?wa=wsignin1.0&wreply=https%3a%2f%2ftreyws-test%2fclaims%2f&wct=2006-07-13T07%3a32%3a21Z&wctx=https%3a%2f%2ftreyws-test%2fclaims%2fDefault.aspx&ttpindex=0 HTTP/1.1

<<<
HTTP/1.1 302 Found
Location: https://adatumsts-7/adfs/ls/?wa=wsignin1.0&wtrealm=urn%3afederation%3atreys-research&wct=2006-07-13T07%3a32%3a21Z&wctx=https%3a%2f%2ftreyws-test%2fclaims%2f%5chttps%3a%2f%2ftreyws-test%2fclaims%2fDefault.aspx&ttpindex=0
Set-Cookie: _TTPRealm=urn:federation:adatum; path=/adfs/ls; secure; HttpOnly

>>> adatumsts-7 (Requestor IP/STS)
GET /adfs/ls/?wa=wsignin1.0&wtrealm=urn%3afederation%3atreys-research&wct=2006-07-13T07%3a32%3a21Z&wctx=https%3a%2f%2ftreyws-test%2fclaims%2f%5chttps%3a%2f%2ftreyws-test%2fclaims%2fDefault.aspx&ttpindex=0 HTTP/1.1

<<<
HTTP/1.1 302 Found
Location: /adfs/ls/auth/integrated/?wa=wsignin1.0&wtrealm=urn%3afederation%3atreys-research&wct=2006-07-13T07%3a32%3a21Z&wctx=https%3a%2f%2ftreyws-test%2fclaims%2f%5chttps%3a%2f%2ftreyws-test%2fclaims%2fDefault.aspx&ttpindex=0

>>> adatumsts-7 (Requestor IP/STS)
GET /adfs/ls/auth/integrated/?wa=wsignin1.0&wtrealm=urn%3afederation%3atreys-research&wct=2006-07-13T07%3a32%3a21Z&wctx=https%3a%2f%2ftreyws-test%2fclaims%2f%5chttps%3a%2f%2ftreyws-test%2fclaims%2fDefault.aspx&ttpindex=0 HTTP/1.1

<<<
HTTP/1.1 401 Unauthorized
WWW-Authenticate: Negotiate
WWW-Authenticate: NTLM

>>> adatumsts-7 (Requestor IP/STS)
GET /adfs/ls/auth/integrated/?wa=wsignin1.0&wtrealm=urn%3afederation%3atreys-research&wct=2006-07-13T07%3a32%3a21Z&wctx=https%3a%2f%2ftreyws-test%2fclaims%2f%5chttps%3a%2f%2ftreyws-test%2fclaims%2fDefault.aspx&ttpindex=0 HTTP/1.1
Authorization: Negotiate TlRMTVNTUAABAAAAB4IIogAAAAAAAAAAAAAAAAAAAA
FAs4OAAAADw==

<<<
HTTP/1.1 401 Unauthorized
WWW-Authenticate: Negotiate TlRMTVNTUAACAAAEAAQAdgAAAAFgomi1Z0tC7za4
J0AAAAAAAAAAAAQBBAFIAAAABQL0DgAAAA9BAEQARgBTAFYATQataEEAAgAQAEARABGAF
MAVgBNAC0AQQABABYAQQBEAEAVABVAE0AUwBUAFMALQA3AAQAQgBhAGQAZgBzAHYAbQA
tAGEALgBuAHQAdABlAHMAdAAuAG0AaQBjAHIAbwBzAG8AZgB0AC4AYwBvAG0AAwBSAGEA
ZABhAHQAdQBtAHMAdABzAC0ANwAuAGEAZABmAHMAdgBtAC0AYQAuAG4AdAB0AGUAcwB0A
C4AbQBpAGMAcgBvAHMAbwBmAHQALgBjAG8AbQAFADoAYQBkAGYAcwB2AG0ALQBhAC4Abg
B0AHQAZQBzAHQALgBtAGkAYwByAG8AcwBvAGYAdAAuAGMAbwBtAAAAA=
```

```
>>> adatumsts-7 (Requestor IP/STS)
GET /adfs/ls/auth/integrated/?wa=wsignin1.0&wtrealm=urn%3afederation%3atreys+research&wct=2006-07-13T07%3a32%3a21Z&wctx=https%3a%2f%2ftreys-test%2fclaims%2f5chttps%3a%2f%2ftreys-test%2fclaims%2fDefault.aspx&ttindex=0 HTTP/1.1
Authorization: Negotiate TlRMTVNTUAADAAAAGAAAYIAAAAAAYABgAmAAAAABAAEABI
AAAAAGgAaAFgAAAAOAA4AcgAAAAAAACwAAAAABYKIOgUCzg4AAAAFYQBkAGYAcwB2AG0AL
QBhAGEAZABtAGkAbgBpAHMAdABYAGEAdABvAHIASgBTAETALQBEAEUAVgBRfVjBrWkBSQ
AAAAAAAAAAAAAAAAAADZmf/wdoShwGwc7CBCpFNGdUHCLsDZPUU=
```

```
<<<
HTTP/1.1 401 Unauthorized
WWW-Authenticate: Negotiate
WWW-Authenticate: NTLM
```

```
>>> adatumsts-7 (Requestor IP/STS)
GET /adfs/ls/auth/integrated/?wa=wsignin1.0&wtrealm=urn%3afederation%3atreys+research&wct=2006-07-13T07%3a32%3a21Z&wctx=https%3a%2f%2ftreys-test%2fclaims%2f5chttps%3a%2f%2ftreys-test%2fclaims%2fDefault.aspx&ttindex=0 HTTP/1.1
Authorization: Negotiate YIIFRwYgKwYBBQUCoIIFOzCCBTegJDAiBgkqhkiC9xI
BAGIGCSqSgIb3EgECAGYKKwYBBAGCNwICCqKCBQ0EggUJYIIIFBQYJKoZIhvcSAQICAQB
uggtOMIIE8KADAgEFOQMAQ6iBwMFACAAACjggQDYYID/zCCA/ugAwIBBAEfGx1BREZ
TVk0tQS5OVFRFUF1QuTU1DUk9TT0ZULkNPTaIeMBygAwIBAqEVMBMBBehUVFABc2FkYXR
lbXN0cy03o4IDstCCA62gAwIBF6EDAgECooIDnwSCA5syDvJHfsRntPF0oUy00/THHoA
eX2fGt0ND8DenGhkYzdIHU4r98+vIgvn+8io6qyZR8r8ZPpqIhEltafOXWnBCckq8XES
ARwj4oX1LlNja0i+zoylvQ1voj8FRVxwiE0Jta5bLnQA+luMhrRot2F3VAYeZK5kXQFDz
7G9PuBSyhrk6nwd9gORXk5AiHlX7Hog03RHe4hYoJ7jTkT9g7lm0wq7RvtsnlvvzW6E1
an3UZBjak61OlhlkH/zC4YYFKIRtnL2ESlQ2teVeRSolMzyRJu46n4SEY2I5X3J+5Svq
+oAwkWUq9B41xqBFEEfIl/4Vc9jwgmmDg8UUVdjiCa4qaAWrsv2JuxiDxHYWYQyEiuTC
AmZx+AniZ4IAAnQATA/iQV0YEkMcUkwBH6YVz3sz/Sw7Kowmuq8lzhbmgpQdU7GtBkkG7
Dj3e3uQffYfLuw5AFkxDJlby+q/TBaOtwq/kUfWKTcg1TziWl3wNyj3CmShctPXlxsJU
SUMJj5uVYm9gl7cv0CnaCRM4TJVlQd+eWcuXlwcMrujg2ktZZHAgmk8VxfqjrUyWJbw
OGqalZ60uxTId9k3Rxt1kwBICtWgPgeHsDQ3nMOPpHXFlTNqPQvYmKOnCNeJAsqAJ34m
OHw4D/Q3qfj9PI6DWlfxolDBM6AlUJ/rpXHGyV3z5EJKLPYqRYPyAm9hrWmmu4/SIA1r
+5rZHWb5vTuh4jLiHDGb12ReqIPhMvP6CUVJGggbIjyvrX5VoYCIIm+uRPP3GFej6C4eX9
fMrwqN3ktk76bRWZf9HKb9VHuYX/k+tMlktf2VrYAifyQf8oJxMq9XON+Iej3KvqyP3d
ZgZfaw9389iITrdX4YnzFwKAf/ot/dLyGrqgWxR/xMTYzmMgPjGjGm8ZVHj6M/CZugh7e
mVzS2WzVFWzk2fufFovcgxHc5hzuWav4VtxD0KWp1UDnr8DmAK3T65dJ8AsgKpfKaxn
vbZ0+0jDWFZVErgGqgnJCefuOU9QDm/oAftGgaruQrAUdeAJVUrfNE6rott6jMXECVIO
IxtDbAtugbtNhw/9f61Tc0xY+7y/tmY+N5MK1Y/my8a/jJ0+DjFR1UJXUWJMDKun0ITt
lxaAmjJAmGsgkmkhV6MoAd7JDq3je3oogKcz2rfAavRZt/Gy8Zhvb+QKNFeMWpKq7wXm
fUa9w/z6V6q3VHomzipcSGUkr2RqrRpnTcIBopGZcy/kbVlG9RqSB0zCB0KADAgEXooH
IBIHFGfGfTuh/3yxOaGfythDxT33t4KCBiNzQ/SLH5M/iH4oxfgtJNbJ5tNM3rXlRkY+
36tO0GgLfJUTaaBS5P0Fms5lHir9PbK4nY7n335HcDVJQXBfiudOkqGU3U80Kuh6UPrS+
ihacMygHCR5E0lDgl1tqvEObraoRR0ct1EvnMBQG6sajBRt800sqEjuldl77U7W0Aa9I
jDWGreZ+nTzu/rhy+Ab0e2tjjgQ8+T+77FUHeYy4B61w7+y5QgnHBBuSP/KAIXAa=
```

```
<<<
HTTP/1.1 302 Found
Date: Thu, 13 Jul 2006 07:32:27 GMT
Server: Microsoft-IIS/6.0
X-Powered-By: ASP.NET
WWW-Authenticate: Negotiate oYGgMIGdoAMKAQChCwYJKoZIgvcSAQICooGIBIGFYI
GCBgkqhkiG9xIBAGICAG9zMHGgAwIBBAEDAgEPomUwY6ADAgEXolwEWjj9oljcrPxx9ipQ
kjQo69bf5SYXFD7mxzAlpl8q5jKcV4ETZcXVawsYxvnHGV/wTsl/yg8CHMZ8D07cNqsp
IEUkG6joNvB9NQAga4q5441JbEvaBLzPZ/9w==
Location: https://treys-7/adfs/ls/?wa=wsignin1.0&ttsize=2652&ttind
ex=0&wctx=https%3a%2f%2ftreys-test%2fclaims%2f5chttps%3a%2f%2ftreys
-test%2fclaims%2fDefault.aspx&wresult=eNrNWFTzqsoS%2fIUW6zGVBYg3rCRlhq
uAaAAlYssphOESHVEGRpZ1e9BoTNYt59R62D4NPW3311%2f3dDM8lDgFWHBXQJzb0CuyOK
%2bmaAlTC%2bItSjFsHJJNigDE7bEZ5f12QFHYi2Di4u9kByN3%2bx11IVViqkXTHYpuUX
lW4Lz59HBjGfofbd89YdfZDADGMMtj1DauK1V8bP436NAMGzD%2bvddps%2fftFuzd9lew
c79iu5y7Yv1%2b4HeaDRXjAqopzt2UQCPOu%2fd0755hp3RvwLYGrZ7zppM9NossHQTQh5
lbOxm4vpsXSbNhuK8om8MME%2bFjkyGCOP0oOAdfoz0bQS6O8SB1E4gHuTewgTEAMN%2fp
gXsJoPkWm4BSP64FuDFGOQ8DlMFfoSQKk3SSgSCvsd7q9C86F8YKP4apB0ly8iz2avtXR5
9Unj7FnGewamQQZfzfoqf%2b6n55%2bbpz4FdnHo7%2bPahOsHmOyidUyyEqChi6M3%2
buqdm31J7GUIoyC%2fQdZ8UqK5vj2Kulay0tQyKoWl1HhnM8%2fs%2bvGB%2bondK94312
```

```
path=/adfs/ls/; secure; HttpOnly
Set-Cookie: _LSCleanup=2006-07-13:07:32:27Zr0urn:federation:trey
research; path=/adfs/ls/; secure; HttpOnly
Set-Cookie: TTPDest=urn:federation:trey research; path=/adfs/ls/;
secure; HttpOnly
Set-Cookie: TTPData=eNrNWftzqsoS/iuW6zGVBYg3rCr1hquAaAA1yssphOESHVEGR
Pz1e9BoTNYt59R62D4NPW3311/3dDM81DgfWHBXQJzb0CuyOK+maA1TC+ItSjFshJUNigd
E7bEZ5f12QFHYi2Di4u9kByN3+xl1IVViqkXTHYpUx1WL4Lz59HBjGfofD89YdFZDADGM
Mtlj1DauK1V8p436NAMGzd+vdpps/ftFuzd91ewc79iu5y7Yv1+4HeaDrXbJAqopzt2UQCP
Ou/0d75sh3RvwLYGRZ7zppm9NossHOTQh5lboxm4vpsXSbNhuK8om8MME+FjkvGCOP0oo
```

```
>>> treysts-7 (Resource IP/STS)
GET /ads/ls/?a=wsignin1.0&ttspize=2652&ttindex=0&wctx=https%3a%2f%
2ftreyws-test%2f2fcliams%2f2f5chttps%3a%2f%2ftreyws-test%2f2fcliams%2f
Default.aspx&result=eNrNWFtZqsoS%2fuiuW6zGVBYg3rCRlhuAaaA1yssphOESHv
EGRPzle9BoTNYt59R62D4NPW311%2f2f3dM8lDgFWhBXQJzb0CuyOK%2bmaA1TC%2btTsJ
F5HJUNjId67bEZSf12QrHYi2Di4u9kBYN3%2b1x1IVViqKXTHyzpuUXlW4Lz59HBjGfofb
89YdZ2DADGMMTj1DauKlV8bP436NMGZD%2bvddps%2fftFuzD1ewc79i5uY7v1%2b4
HeadRXjAqoppz2UQCPOu%2fd0755hp3RwLYgrZ7zppM9NossHQTQh31u0xm4ypsXSNhu
K8om8MME%2bfjkyGCOP0o0Adfoz0bQS608SB1E4gHuTewgTEaMN%2fpqXsJoPkWm4BSF64
FuDFGQG8D1MFfo5qSKk3SSgScvsd7q9C86f8YKP4apB0ly8iezavtXR59UNj7fNgewamQQQ
zfzofqo%2b6n5%2bbp2z4FdnHo7%2bPahOsHmOyidUyyEgChi6M3%2buqdm31J7GUioyC
%2fQd28Uqf5vJ2KulayOtQyKoWl1HhnM8%2f2bVbG%2bondK94312%2fg8w1meeydrNq
5m8OEPDc%2byv9QMB%2bVDZzhHyP%2bxfD4oDco49VGJLzmyi9Ur9PK3pzEpFdwvtYMYZg0
ZZYN7%2b5Pkbdw4dTxsedx8An4SpzFJbpuj7D%2fnsfv3uoefz%2f2fGL6xcAVC%2f4%2
bXCWk4yvSpY%2bHnJ3pPGR5Yn66r%2b22NTSt4A3yflDZuftEw9eDXdw2%2foxj7m4K%
2bOT%2bHvUn7c%2fSP%2b2EWWJkgTgr7%2fxdQGQWYfLdK87yVBMLR%2fyf5Iod4KU7iNID
9d8ME222G9l9HSf3yXNlxmJIIyt5m6hVXWZbfS%2fyEh3QlmgI5iij4OA6%2fNc%2f%2fg
n7d%2bp4eBdFFKTnJm%2f2h406AaYBMiMly5YBcmGyqha5P380bDdeow7%2fdZsUGd4vmnjMa
7IMu%2fh0c4cpmTJQsGMXknR2NmqqY%2fnb18b4U8P08xNMRMCb5Z%2f2f84YLHd7SF%2fj2
%2bhHOC9HVz2vC2HugUniGziI8FTPEoZibOVTPhN2GVHqWsgS5S6BhgV4f3E0eycpncEu9V
8oK7ckvVtLVyzdlYUYTglc1Gn%2fJ5011FmxlmmaTOBae%2bMEojVhB%2bd96NDIjO9jLv
rlopNjXrVsPNN3kWwtue2Vj4t90Zram4XtYg1Y2jTyLGRUFmYHmdv4W87i6vIN9VO3UNaB
POiR5XrV0emvSa5kpm9SXKZrXmHXZ%2f6WXdBd5mpuJX07YIqA4ru0lX%2fmI6gf4De4zm
QG%2fAPOqzOUS06Nce6uXteCFVLVVCPLTLVYVfKREEAWRWCuVbQoKGyagCeqdfJiCMR%
2bud9E6Vris5oE5k4EIHMpYstlclnPTFKWym3OVTeXiFoSlVmeWcHtFBFAu6XL8Clhjqlb
GFJSGOHZPsuNFjp1kBr88yCkUw%2bXA854ExFVryermwmNViThSvZRqgEhiAVgr7p9jqihV
NqcYQGFsZa1HgYlPnK10017uE4uB4smhLzZzTh3sY7QX1thxeQZ0p%2b9W%2bNmJNkfjz
b%2ftGa7DDMWjyoeTmeZ02v7b8EogUjTrK%2b2Z8q6weJ1WNn02k9f6RVhA3%2bwqcZXRkto
8gHGZR5oZfyYq4g72IsKhFxVavFycmf0Eas4dKvg9xCEcSiOhlrSDYWDpzG7tcUxxz7Q5WO
pisAEPGqrPDoS7p1%2bKyAEs4t%2bBuaQtsGJIjYWhmTexqYJpcka4qzWVXX HTTP/1.1
Accept: image/gif, image/x-bitmap, image/jpeg, image/pjpeg,
application/vnd.ms-excel, application/vnd.ms-powerpoint,
```



```

application/msword, application/x-shockwave-flash, */*
Accept-Language: en-us
UA-CPU: x86
Accept-Encoding: gzip, deflate
User-Agent: Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.2;
SV1; .NET CLR 1.1.4322; .NET CLR 2.0.50727; InfoPath.1)
Host: treysts-7
Connection: Keep-Alive
Cookie: _TTPRealm=urn:federation:adatum

<<<
HTTP/1.1 302 Found
Date: Thu, 13 Jul 2006 07:32:27 GMT
Server: Microsoft-IIS/6.0
X-Powered-By: ASP.NET
X-AspNet-Version: 2.0.50727
Location: https://adatumsts-7/adfs/ls/?wa=wsignin1.0&wtrealm=urn%
3afederation%3atrey+research&wct=2006-07-13T07%3a32%3a27Z&wctx=
https%3a%2f%2ftreyws-test%2fclaims%2f%5chttps%3a%2f%2ftreyws-test
%2fclaims%2fDefault.aspx&ttindex=1727
Set-Cookie: _TTPData=eNrNWftzqsoS/iuW6zGVBYg3rCRlhquAaAA1yssphOEShVEG
RPz1e9BoTNYt59R62D4NPW3311/3dDM81DgfWHBXQJzb0CuyOK+maA1TC+ItSjFsHJJNi
gdE7bEZ5f12QFHYi2Di4u9kByN3+x1lIVViqkXTHYpuUXlW4Lz59HBjGfobfD89YDfZDA
DGMmtj1lDauK1V8bP43NAMGzD+vddps/ftFuzd91ewc79iu5y7Yv1+4HeaDRXjAqopzt2
UQCPOu/d0755hp3RvwLYGrZ7zppM9NossHQTQh51bOxm4vpsXSbNhuK8om8MME+FjkyGC
OP0oAdfoz0bQS608SB1E4gHuTewgTEaMN/pgXsJoPkWm4BSP64FuDFGOQ8DlMFfoSQKk
3SSgSCvsd7q9C86F8YKp4apB0ly8iz2avtXR59Unj7FnGewamQQQzfzofqo+6n55+bpz
4FdnHo7+PahOsHmOyidUyyEqChi6M3+uqdm31J7GUIoYc/QdZ8UqK5vj2Kulay0tQyKoW
L1HhnM8/S+vGB+ondK94312/g8wimeeydrNq5m8OEPdc+yv9QMB+VDZhHyP+xfD4oDco4
9VGJLzmyi9Ur9PK3pzEpFdWvtYMYZg0ZZYn7+5Pkbdw4wdTsedx8An4SpzFJhpuj7D/ns
v3uoEQt/o/GL6xcAVC/4+XCWk4yvSpy+Hnj3xPGR5yN66r+22NTstx4A3yflDZuftzEW9
eDXwDZ/Oxj7m4K+OT+HvUn7c/SP+EWUJKgtF7/XdQGQUwydKL87yBVMlRs/y5Iod4kCU7
inID9d8ME222G919HSf3yXNlxmJIiYt5m6hVXWZbfs/YEh3Qlmi5iij4OA6/Nc//gn7d
+p4eBDdFKTnJm/h406AaYBMiM1y5BcmGYqha5P38ODde0w7/dZsUdd4vmjma7IMu/c4c
pmTJQsGMKsnR2NmQy/Nb18b4U8P08xNMRmNCb5Z/284YLqHG7SF/j2+hHOC9HVzv2CHug
UnxiGzif8PT1eOzibOVTPPhN2GVHqWsg5S6BHgV4f3E0eycpnEu9V8oK7ckvVtLVyZdlY
UYTg1clGn/J5011FmxlmmaTOBae+MEoJvHb+d96NDIj09jLvrlonNjXwsPNN3kWwtuE2V
j4t90Zram4XtYg1Y2jTyLGrUFmYHmdV4WW87i6VIm9V03UNaBPOiR5XrV0emvSA5kpm9S
XZK7rXmHZ/GWXdBd5mpuJX07ZiQ4aru01X/mi6gf4De4zmQG/AP0qz0US06Nce6uXteCf
VLVVCPLtLYVfVhKREAEAWRWCuVBqOoKGyayCeqfdJiCMR+ud9E6VriS5oE5k4EIHMPyStl
cinPTFKWYm3OVTeXiFOS1VmeWcHtFBFAu6XL8ClhjqlbGfJSGOHZPsnNFJplkBr88yCKw
+XA854ExFVryermwMNViTHsVzRqQEhiAVGR7p9jqihVNqcYBQFsZA1HgY1PnQ10017uE4
uB4smhLvZzth3SxY7QX1thxeQF0p+9W+NmJNkfjZb/TGa7DDMWjyOeTmeZ07vb8EogLjT
rK+ZRq6weJ1WNn02k9F6RVHa3+WqcZXRkto8gHGZR5oZfYq4g72IsKhFxVavFycmf0Eas
4dKvg9xCcSIOh1rSDYWDpzG7tcUxxz7Q5WOpisAEPGqrPD0S7p1+KYAES4t+BuaQISGJ
IJyWhmTexqyJpcka4qzWVXX; path=/adfs/ls/; secure; HttpOnly

>>> adatumsts-7 (Requestor IP/STS)
GET /adfs/ls/?wa=wsignin1.0&wtrealm=urn%3afederation%3atrey+research&
wct=2006-07-13T07%3a32%3a27Z&wctx=https%3a%2f%2ftreyws-test%2fclaims%
2f%5chttps%3a%2f%2ftreyws-test%2fclaims%2fDefault.aspx&ttindex=1727
HTTP/1.1 302 Found: Cookie: WebSsoAuth=eNrNV1tz4rgS5qekmMcU4zvYVJI68hXbGGIbCLhO1
ZYv8iVg3ZBMF+/AkImYSe7c07Nw/pJarW7v/661ZIEsJfdDEFZwqJK8uzufaSKj90/QgY
nqJAIEj5DUZ2ahIME60Gm51F9zvWogA0DpnunlmUN1ays3Kx67JI43u/hgx5BzfDBkCKH5
MB50ykeu3WRDUMYwM190Rm6gVvVaffOcF/zYgGLEgkfufwQSNlnwSHdZOXwhPziJHfLpBx
mbgrLYeUPbWCMh8R3fOheA+g+PZxjE/IsSE6C8m6SVzwM8wJ+hRipTLNpAcLqhPWjDnvVe
bMK6iCBmQ8tWFZF4p/svzu6UXm6ibkqYHtXwBK6hR8/YJ91b+Y/N4/dBHZ1G0yTkWk3CEu
0mq8TlJUwH711/EbfaeVCX5r4RV7mYfUBWfdJiRf69ijqWkNjM8toFS5Wk51NPEvrwxwfsJ
3bf8b65fgNfxTCrEv9s1a7cCqZofvdZ/g8F81nZgFWCB38tn09KwybJgrwprzmya+8V+tX
bbIJKRQ1O2mECizs5L1IXeY+rajvEsNKPyeqW3xFPZe5uv+dFhPkbN01LbP486T6BIE2yB
CXDRfLiP5ey/e7n6Vv8n41fWXhGp0dL1fWKpRpr67g7cK/J4zPOO/eR6ffHrtS6iYbEAS
otMvu58Vy6/rwF0B2b30s3E0Nn9y/R32jfSv9J9xCnqZ5dhr/XtQGQowydKb89yBVirze/
16QwmkRjThNKgt23w0TbLdFvv911NiX+8pOogwVUQEvffEdV9M03xvqDAd1JRzDOQwpBGU
Sfete/oLBqfU9PQhulmdoJ2+S44cGdQc2UV4kVZx+YZLACPxksgcPfs8n6Oxb9w77gOcXz
XxCvPRur4xd4mzJgiESthfH3dxSH7vffu0If3qYFW5WogMxLT+M/zccMNvDTb6FQa+8hnO
G9OvmvmAH+whOTCJOJv4/PL1zdDFxqZopv4na7CgV/bDRJcCrebmFoppdUTg68T5qPmDv3
KLxxlp4z95FUYTRzKzEHQsG0j2jzI/zQtPmAkhVjAaI7T2/XbDxIZWJQCd9xvFxlI+EZ7
x+li2ltyMrSblviZn9mZpu6UGLGOW+xy2poX5QaY0XtZpZ7kScbPdqXuIi2BRD7Bm/erIu
B+mR3Rmb9KdUvnkggnwsugv8T4xE7eSv1lhTYjhfbx1j9kYBgfoP14C+QD+QYftJaolg3O

```

iW7mXkXC6VIWnews1N1UVCVEQQNFGoFF5EKm6QkWPbILTJx1mYMJH6128ThSuwXlgzUgA
sew/EY2V+LCNEWpYeausmkdmY/9zGLmKbcPRAD1Bm8mr4AyZmprzEBjiBP3LDteZdJZZvC
rgywCm48mCx4YM4GU16ulRXjLCE63OGWoSmgAXBHsnWKRHiWa0gkHALQyAaLAJ6bOR6Zor
3YpxsHJdEL1A9K2I7zeEdoLZey4qga6w7pt+ezEm6Pxst/pBMcQI/Eo8tV0rrnT+z2/AuJ
Sw45yNcNo/SBReuJsGPK5Rq3qaLFrHSD0ZbyK4wAUUOaFQWp7MXewly2IuLbRktX03mBzS
nFwsub3EERJKo5GWtgPhIOvEbulxRFHFujysVFFYAI+plU+PyLuHbYRI8S1hT8Dc4Sh4EQ
QzRpDMj/GrImNSRni/KSr6rmjxnt/gnImgEgABiJDEK66hoTyCCIRjPhoXoiRiFBHgEiKr
nkU0Tw48T0yaUmOzPnKt/DDpgrWNJwSsYZ5q7U3ONIEg6uN1JyxbXjQNMLJd7PieXM+Qr6
b0WXtleejRs7BXYKzfqQUshhQXcaeTVcerdg6U1DTebMwz7ohL9HNTOT1XEoiK3ckd2Thv
pjvx+Rkg+onDpZWPk6t2m+ZV4/SmnFKbFeUtvdlJLg0EJCpxvUku2kBa1H7K1Z7A/bX+kJ6
fNk3k5iIXujnVld9IPK8qElRPuiGpP+RHIMsvTh6MrGaasPvVybyiE4ES3fp5dZYavlou;
WebSsoAuth0=cFf+wmdmNhb4wZEBdGWqOftPmfHLeclv1R8xLqXzWPYybe8pB/+h3zhc
KPHWty1zhfzBBVqznBpxgN/uA/68D9C+NXlq19wfJjPM72Pp4hBQgXAkQLHeHjTKG61Fjo
3AYZANDgro73f7WUBzL4FZseFej6RRyPt7dZC6L7Bq5Dft+c8pNU8XhbrA7rW1kIyCKKSg
8JxsqX7pVbX9XBSKHDqyy7HY6yGnGyK/f1HkeWVVO61vY1GCjy0pMcaC71tOGt/PxwNtBL
WXEWwhtI53xwbZKBj4juassXVNuKQwVwdK+HygAz01BCXzcsNP11QSxoKR51b+Cu0EiL58
X7Lc2htVYbhs1VktBx5129y6qUm5BKXGjUYpOcX2BrtqxR1zWGRO+syU1lSncznLsWw3ZX
ByiS1Hq0/FWT/b5S8vBT33F+v1jpoFnGOjSun77SsM6wfstmNeJJuair132B+99/2ucn28
PhGdPzzSpXASvY04dOPT0dwa73mwz/RcxnVJos+6kPI6ndsXEXor7mGQFOjCnhdtB4iyvT
B64L+fbv4du0fOmB9U9J9fhqKf6L4LismDD4gqT5OcJ2Cpfs0yVL9QY/B8U6H6PzSu6vzk5dc
h+mAzudXxxvSh+98dbHvCJ2Pt+eO2JHS7SZvIURDw83cCBZ/AgPdteY=; _LSCleanup=
2006-07-13:07:32:27Zr0urn:federation:trey research; _TTPDest=urn:
federation:trey research; TTPData=eNrNWftzqsos/iuW6zGVBYg3rCRlhquAaAA
lyssphOEshVEGRpZle9BoTNYt59R62D4NPW3311/3dMD81DgFWHBXQJzb0CuyOK+maA1TC
+ItSjFsHJJNigde7bEZ5f12QFHYi2Di4u9kByN3+x11IVViqkXTHYpuUX1W4Lz59HBjGfo
fbbD89YDFE2DADGMMtj1DauK1V8bP436NAMGzD+vddps/ftFuzd91ewc79iu5y7Yv1+4HeaD
RXjAqopzt2UQCPOu/d0755hp3RvwLYGrZ7zppM9NossHQTQh51bOxm4vpsXSbNhuK8om8M
ME+FjkyGCOPOoOAdfoz0bQS608SB1E4gHuTewgTEaMN/pgXsJoPkWm4BSP64FuDFGOQ8D1
MFfoSQKk3SSgScvsd7q9C86F8YKP4apB0ly8iz2avtXR59Unj7FnGewamQQQzfzogfqo+6
n55+bpz4FdnHo7+PahOsHmOyidUyyEqChi6M3+ugdM31J7GU1oyC/QdZ8UgK5vj2Kulay0
tQyKoWLLHhnM8/S+vgB+ondK94312/g8wimeeydrNq5m8OEPDc+yv9QMB+VDZhHyP+xfD4
oDco49VGJLzmyi9Ur9PK3pzEpFdWtYMYZg0ZZYn7+5Pkbwd4wdTsedx8An4SpzFJhpuj7
D/nsv3ueQt/o/GL6kcAVC/4+XCWk4yvSpy+Hnj3xPGR5yN66r+22NTStx4A3yflDZuftz
EW9eDXwDZ/Oxj7m4K+OT+HvUn7c/SP+EWUJJKgtF7/XdQGQUwydKL87yBVMLRs/y5Iod4kC
U7inID9d8ME222G919HSf3yXNlxmJiIyt5m6hVXWZbfS/YEH3QlmlqI5iij40A6/Nc//gn7
d+p4eBDDfKtNjM/h406AaYBMiMlyj5BcmGYgha5P38ODde0w7/dZsUDd4vmjmA7IMu/c4c
pmTJQsGMKsnR2NmQY/Nb18b4U8P08xNMRmNcb5Z/284YLqHG7SF/j2+hHOC9HVzv2CHugU
nxiGZif8PTleOzibOVTPHN2GVHqWsG5S6BHgV4f3E0eycpcnEu9V8oK7ckvVtLVyzdlYUY
Tg1c1Gn/J5011Fmx1mmaTOBae+MEojVhb+d96NDIj09jLvrlopNJXwsPNN3kWwtuE2Vj4t
90Zram4XtYglY2jTyLrGUfMfYHmdV4WW87i6VIm9VO3UNABPOiR5XrV0emvSA5kpm9SXZK7
rXmHz/GWXdBd5mpuJX07ZiG4A4ru01X/mI6gf4De4zmQG/APQzOUS06NCE6uXteCfVLVVC
PLtLYVfVvKREEAWRWCUuVBqOoKGyayCeafdJiCMR+ud9E6VriS5oE5k4EIHMPyStlcinPTF
KWYm3OVTeXiFoS1VmeWcHtFBFAu6XL8ClhjqlbGFJSGOHZPsnNFJp1kBr88yCKw+XA854E
xZFVryermwmNViThsVzRqqEhiAVgr7p9jqihVNqcYBQFsZA1HgY1PnQ10017uE4uB4smhLv
ZZth3Sxx7Q1LthxeQF0p+9W+NmJNkfjZb/TGa7DDMWjyOeTmeZ07vb8EogLjTrK+ZRq6we
J1WNn02k9F6RVHa3+WqcZXRkto8gHGZR5oZfYq4g72IsKhFxFvFycmf0Eas4dKvg9xCeC
SIOh1rSDYWDpzG7tcUxxz7Q5WOpisAEPGqrPDos7p1+KYAES4t+BuaQISGJJIJyWhmTexqy
Jpcka4qzVWXXkqNHeG5OCSAUgEHIEISLriGRPIJQBEM+nGViaAj8ERCSwkseRfLs13wPz
bYkh+Zs6Vn0YZP76zacMfJGrZbrVe/YXvUjtZTKE7YND8pSqH2XS543Z0Piuxye9155Pix
1BGZyO6WHTaJc9Xt2NPJctVWbL2TsZNZOTdPugEvtcupyOtIikMLOZI7tGhPRPtRa7wh9
RP5CWuNEqvwqs7ritXKUcJs16y291pc4gtE1qL3Zmte+dK88BIOrwTux/ojel5L1qGDCC9
YqeuW; TTPData0=LyWeV5UxqSfdkNR3+REIOaud/KFVTuL+f1nbVmtGV8LPf16dhUYvF
3Pa1X/hmZVLc7xzZABdmSjOdqXMuqOK8xbKe4wL6bSWV6m2YxkH791vFMyVaOs1lrkk/uC
c7LNMQtigP58D/nQOyLk1eXYX3x3GU8rrUsn84LO+cGRAtt4eNHY1VDNEDcGhl/YOCujud
/up3+ZeFDpVB3s91IYB7+3VXuK+wLyUR23Le07YWTLP1D1lp62FeOiHAQuF53jLdrGTc+3
jPFPkwJfdrk+9H1C7ZNDdiYLPcivfSV2bCmN6ZEmxMRJ0z3KS6G426mldQLOMYcV1WJ3zz
JHR8pWvZolra10wbkuYqT0leD60fT1hBCVdIcNL1mwcRIKRIA9QjsGoiiff4T63Xg3zIFi
q6rSQ/RvR29y6LFoyBn2NHQ6T1oTaG/1lJe46h3nqJM8dbElFMpNTRKW7SYduLsJlaHXZK
02m0/TykrVn3ny93nGygPpVKClJt3qFQfFAfe6YZ28m5m1LXDvvee6/vKpFLKxH87tJe4u2
AtJhNDPEUXb8LbL/wXaBd8Qt2sRedfow4A6k1N+iOM3f3xwu9tyv2euTi+HppT9Owzebb
7eAP9x8bzXPTz9gOclvYv3Iy08/kzz9A6cx/kw=

<<<

HTTP/1.1 302 Found

Location: https://treysts-7/adfs/ls/?wa=wsignin1.0&ttpsize=2652&
ttpindex=1727&wctx=https%3a%2f%2ftreyws-test%2fclaims%2f%5chttps%
3a%2f%2ftreyws-test%2fclaims%2fDefault.aspx&wresult=kqNHeG5OCSAUgEH
IEISLriGRPIJQBEM%2bnGViaAj8ERCSwkseRfLs13wPzbYkh%2bZs6Vn0YZP76zacMJF
GrZbrVe%2fYXvUjtZTKE7YND8pSqH2XS543Z0Piuxye9155Pix1BGZyO6WHTaJc9Xt2

NPJctVWbL2TsZNZOTdPugEvtcupyOtIikMLOZI7tGhPRPtRa7wh9RP5CwuNEqvwwqs7rit
XKUcJsl6y291pc4gtElqL3Zmte%2bdK88BIOrwTux%2fojel5SlqGDCC9YgeuWLyWeV5U
xqSfdkNR3%2bREI0aud%2fKfVtUL%2bflnbVmTGV8LPf16dhUYvF3PalX%2fhMzVLC7xz
ZABdmSjOdqXMuqOK8xbKe4wL6bSWV6m2XykH791vFMYaOs1lrkk%2fuC7LnmQTigP58
D%2fnQOyLk1eXYX3x3GU8rrUsn84LO%2bcGRAtt4eNHylVDNEDcGhl%2fYOCuJud%2fup
3%2bZefDPvB3s91IYB7%2b3VXuK%2bwLyUR23Le07YWTLP1D1lp62FeOiHAQuF53jLdrG
Tc%2b3jPFPkwJFdrk%2b9HlC7ZNDdiYLPciVfSV2bCmN6ZEmxMRJ0z3KS6G426mLDqL0M
YcV1WJ3zzJHR8pWVozlral0wbkuYqT0leD60fT1hBCVdIcNL1mwcRIKRIAu9QjsGoiff4
T63Xg3zIFiq6rSQ%2fRvR29y6LFoyBn2NHQ6T1oTaG%2f11Je46h3nqJM8dbElFMpNTRK
W7SYduLSjlaHXZKO2m0%2fTykrVn3ny93nGygPpVKClJt3qFQfFAfe6YZ8m5m1LXDvvee
6%2fvKpFLKxH87tJe4u2AtJhNDPEUXb8LbL%2fwXaBdT8Qt2sRedfow4A6k1N%2biOM3f
3xwu9tyv2euTi%2bHppT9Owzebb7eAP9x8bzXPTz9gOclvYv3Iy08%2fkzz9A6cx%2fkW
%3d

>>> treysts-7 (Resource IP/STS)
GET /ads/ls/?wa=wsignin1.0&ttpsize=2652&ttindex=1727&wctx=https%3a%
2f%2ftreyws-test%2fclaims%2f%5chttps%3a%2f%2ftreyws-test%2fclaims%2fD
efault.aspx&wresult=kqNHEG50cCSAUgEHIEISLriGRPIJQBEM%2bnGviaAj8ERCSwk
serFLs13wPzbYkh%2bz56Vn0YZP76zacMJFGrZbrVe%2fYXvUjtZTKE7YND8pSqH2XS54
3Z0PiuXye9155PixlBGZyO6WHThaJc9Xt2NPJctVWbL2TsZNZOTdPugEvtcupyOtIikML
OZI7tGhPRPtRa7wh9RP5CwuNEqvwwqs7ritXKUcJsl6y291pc4gtElqL3Zmte%2bdK88B
IOrwTux%2fojel5SlqGDCC9YgeuWLyWeV5UxqSfdkNR3%2bREI0aud%2fKfVtUL%2bflnb
VmTGV8LPf16dhUYvF3PalX%2fhMzVLC7xzZABdmSjOdqXMuqOK8xbKe4wL6bSWV6m2Xyk
H791vFMYaOs1lrkk%2fuC7LnmQTigP58D%2fnQOyLk1eXYX3x3GU8rrUsn84LO%2bcG
RAtt4eNHylVDNEDcGhl%2fYOCuJud%2fup3%2bZefDPvB3s91IYB7%2b3VXuK%2bwLyUR
23Le07YWTLP1D1lp62FeOiHAQuF53jLdrGTc%2b3jPFPkwJFdrk%2b9HlC7ZNDdiYLPci
VfSV2bCmN6ZEmxMRJ0z3KS6G426mLDqL0MYcV1WJ3zzJHR8pWVozlral0wbkuYqT0leD6
0fT1hBCVdIcNL1mwcRIKRIAu9QjsGoiff4T63Xg3zIFiq6rSQ%2fRvR29y6LFoyBn2NHQ
6T1oTaG%2f11Je46h3nqJM8dbElFMpNTRKW7SYduLSjlaHXZKO2m0%2fTykrVn3ny93nG
ygPpVKClJt3qFQfFAfe6YZ8m5m1LXDvvee6%2fvKpFLKxH87tJe4u2AtJhNDPEUXb8LbL
%2fwXaBdT8Qt2sRedfow4A6k1N%2biOM3f3xwu9tyv2euTi%2bHppT9Owzebb7eAP9x8b
zXPTz9gOclvYv3Iy08%2fkzz9A6cx%2fkW%3d HTTP/1.1
Cookie: _TTPRealm=urn:federation:adatum; _TTPData=eNrNWFtzqsoS/iuW6zG
VBYG3rCRlhuAaaAlyssphOEShVEGRPzle9BoTNYt59R62D4NPW3311/3dDM81DgFWHBX
QJzb0CuyOK+maA1TC-ItSjFsHJUNigde7bEZ5fl2QFHYi2Di4u9kByN3+x11IVViqkXTH
YpuU1LW4Lz5HBjGfobfD89YDfZDADGMMtjlDauK1V8bP436NAMGZD+vddps/ftFuzd91
ewc79iuy5y7Yv1+4HeaDRXjAqopzt2UQCPOu/d0755hp3RvwLYGrZ7zppM9NossHQTQH51
bOxm4vpsXSbNhuK8om8MME+FjkyGCOP0oOAdfoz0bQS608SB1E4gHuTewgTEaMN/pgXsJ
oPkWm4BSP64FuDFGOQ8D1MFfoSQKk3SSgSCvsd7q9C86F8YKP4apB0ly8iz2avtXR59Unj
7FnGwamQQQzfzofqo+6n55+bpz4FdnHo7+PahOsHmOyidUyyEqChi6M3+uqdm31J7GUI
oyC/Qd8BUCkV5vj2Kulay0tQyKoW1LHhM8/S+vGB+ondK94312/g8wimeeydrNg5m80EPD
c+yv9QMB+VDZHzHyP+xFD4oDco49VGJLzmyi9Ur9PK3pzEpFdWvtYMYZg0ZZYn7+5Pkbw4
wdTsedx8An4SpzFJhpj7D/nsv3uoeQt/o/GL6xcAVC/4+XCWk4yvSpy+Hnj3xPGR5yN66
r+22NTStx4A3yflDZuftzEW9eDXwDZ/Oxj7m4K+OT+HvUn7c/SP+EWUJKgtF7/XdQGQUwy
dKL18bYbVM1Rs/y5Iod4kCU7inID9d8ME222G919HSf3yXN1xmJiiyt5m6hVXWZbfs/Yeh3
Q1mqI5iij40A6/Nc//gn7d+p4eBDDFKTnJm/h406AaYBMiMlyj5BcmGYqha5P38ODde0w7
/dzsUDD4vmjma7Imu/c4cpmTJQsGMKsnR2NmQY/Nb18b4U8P08xNMRmNCb5Z/284YLqHG7
SF/j2+hHOC9HVzv2CHugUnxiGZif8PT1eOzibOVTPhN2GVHqWgsG5S6BHgV4f3E0eypcne
u9V8oK7ckvLdLVyzd1YUYTg1c1Gn/J5011FmxlmmaTOBae+MEojVHb+d96NDIj09jLvrlo
pNjXwsPNN3kWwtuE2Vj4t90Zram4XtYglY2jTyLGrUFmYHmdV4WW87i6VIm9VO3UNaBPOi
R5XrV0emvSA5kpm9SXZK7rXmHZ/GWXdBd5mpuJX07ZiQ4Aru01X/mI6gf4De4zmQG/APOq
zOUS06Nce6uXteCfVLVVCPLtLYVfVfKREAAWRWCUvBqOoKGayCeQfdJiCMR+ud9E6Vris
5oE5k4EIHMPyStlcinPTFKWY3OVTeXifOSlVmeWcHtfBFAu6XL8ClhjqlbGFJSGOHZPsu
NFJp1kBr88yCKw+XA854ExFVryermwmNViThsVzRqqEhiAVgR7p9jqihVNqCYBQFSZAlHg
Y1PnQ10017uE4uB4smhLvZZth3SxY7QX1thxeQF0p+9W+NmJNkfjZb/TGa7DDMWjyOeTme
ZO7vb8EogLjTrK+ZRq6weJlWNn02k9F6RVHa3+WqcZXRkto8gHGZR5ozfYq4g72IsKhFxV
avFycmf0Eas4dKgc9xCeCSIOh1rSDYWDpzG7tcUxxz7Q5WOpisAEPGqrpDoS7p1+KYaES4
t+BuaQIsGJIJyWhmTexqyJpcka4qzVWXX

<<<
HTTP/1.1 302 Found
Date: Thu, 13 Jul 2006 07:32:28 GMT
Location: https://treyws-test/claims/?wa=wsignin1.0&ttpsize=2708&
ttindex=0&wctx=https%3a%2f%2ftreyws-test%2fclaims%2fDefault.aspx&
wresult=eNrNWFuTqjoW%2f1uW59HqDd7aS3V3TUBRUFAQFX2ZihAuKgsEPHXt9DW
bfc%2bu6dnaj8cnsLK41lvfumQ15CWjSddAhxTRZIBsNA6S3CQ7ha1EI4IpKp3CPaZd
pvZa9pMk6nIctX0UQvqDzVACox8k9riMcjWeb3J8jUvilCblt5cHZOR8wh57oTDcdw
G1KE4Cgkv3kdx7Lf%2b79VxD9eam%2f1Rttfinhms%2fP3U2z9WnTpu3EXSR7Wwa5Z
JMaYpkTBOIGTVm%2fPmJbz1V6ybf6tZr3Vp7%2fa4Tv5bTGHdd5KAYFka6SYzyUowo

grHt10sq3JJ4gWLK517LVSYI8EfBNQYF6SsWgTSgXQxDRLuJ3Z0Bddyt%2fuC78OZH
%2bd1FkWANkAS0pJFFQC6J0e%2fIMoUJnsTATQrKjzrtm84tcKkTIGwj1qMkDuwC%2
f27ok8pbkTPKklb4nNGnhH3C2XsYhJR74T6qfnr%2fe3Tuk183e84xKCCg49KuWODP
SBrbt%2fIpxNFqHYED0rc5CEh5bdPCYIOTNLwhfuMdsMnZBew3LtkCKn%2fTRM13Y
47sCe5R01PzWSen%2fydKC4c0uZfb5Y%2b4N7j8e7ae3ASH%2bEksC%2boswQmKGTv
pY%2fyL8qyxTL9UVlFiU%2bcX4oUf1DqZgF2SEZvJTBln1tkJ%2b9vGqtE2Sm03QDF
JYnEIfx6vb6XwHyqlD%2bAEwY4YmMGcYn%2fdY39D5uE7%2f5%2fBL9F5U6A%2byou
t6glrJI2aYI%2bT%2fxz3PjIs3QfFZ%2b9lvshDPbAcVjToOWPkzSCNvoGyfJnGwu4
T9Eb%2fJr1J%2b3P0v%2fGwYRhSHAX%2frOsVcaYZegS8j%2fDdBCtNPqzJiv6KYEO
iskRxf9slqwMwyBJvk%2bT%2b%2b26mgUeZkUUv7fe07Esy35k9Qsf1pV4ju9wTMGh
gfdX%2bfoVcorW9%2fYiQkwwW8n74PzQoEpg7xG2hfvhbyCrXJUvIJ%2fQyX6yqw38
V7nEPfD5JswHHzjGFT9SH1QuSgVwUFztTaW7Ir%2bW%2fvndQeHsxY4gp23lD%2bjD%
2b33ggfER7EiHnid7cuVD6PtxvosM9kusFHttz%2f5843WN0hbhwjYmb1qkyIaitDM
fgMM09sVbFyT6d2mzHe9R84e6xZePHWrhn76qTe4MFMB5xsE6nDdirEdh5gyHJ%2f
uUq67WbrbgcpKl%2fDpYLMKV2YjsYfYU3XYpt%2bnIUoY7SjupI7encWwukl045HaN
TFLtTSiOecXPZ33SYFT0e2LKRu53oulhut6G2%2fPIMTLenu%2fCkwqja9aUsbPORq
cnX98odatB%2bePkLEQVK5QtZapMturr1ZEh8i8j1f%2b9spp8pwcTeB2JxZnNLbYu
1thkucdvRRHEBw9ksqA8eTRoSerdA8XTP52BJni7g78LBP2MF4A%2b10APrFXDziR9
1Vvoeq%2bfNedwsM%2fxkuDb2GjOw87R6QEkZxymbUfDNeVcnfuZ21PgrXA%2byeSL
TBXU038LdMHTFgJQT9Ey8rW1PztDNDH0oB%2f4dCXAjgIWx1k3moneqs%2bSGP5Gc
9FEHhDX6%2fk00yY7kS0MqU5oBNOqsg5TiZnrjoeEB7OipXDQPjaIittxro5Ned0Zzd
FxrEqLSRue832FHV2DyUi11%2bPFRLGGk4TD2dLeWs2phG1nq00lUqkNfTWWN66Rzt
Ozu%2bHTsamPDkdRrX3rEQjAPPKST%2f0nzdN15zshZU%2ba4JyZCSzPgBwIgKoZp
43cFXQGDq3OllfYHL9D7LQ7%2fdE%2fqPPot6u8%2fiVi10Rf
Set-Cookie: _TTPData=; expires=Wed, 12-Jul-2006 07:32:28 GMT; path=
/adfs/ls/Set-Cookie: _WebSsoAuth=eNrNVluXqrgS9qe4nEeXG/COq7vXCSAKC
gre0JdZacJFJSgXEX/9BG3dds/uffqcNQ/DU6hUqr76qlJJXmIY7HsgjlGU+CEuP0a
S8F5s90u0bLhNSYToeuNR2rXWPNn1Nju7SFOIMs22xWylIcp0jCcQJx8lqp03S7R
ndqTGN0d3qNeq/e3bzrRK+VNMI9B9kogoWTXhKhvByhGMHI8iplBW7DaImimMy9Vhg
i8PFHwTnY47hXgL7ZCmHsxz0MAxT3Eqs3A8q4x/yge/AeR+Xt5RoIH2LbLwRxWQ0TD
jlhhL4CSxQmeBIBJykgP+t07zrvVfQ+whbSEdxEv1WYf/h6JPKm5ckh7hHUUXMWVx
LyBLK2km/iKkXGqPqp/9fw6c+xxX3Z5/8wgS0nbjHF/ZnYRp26J26QnyjLvCtKiXDJ
3lKSOXtU4KgDZM0Eke+W7vbD8OdT3LvHEMYe990UdesiIWC6JxULZ0ni/zs7Xh+aYd
d+vXu6YPdBx/vob2Tk3gIJ751tTpLYIIC81/+KP9NWXZIpj8qKyjxQvtvRQo/KPUyH
9thFt9LYJaaW2Ql1738qQUtJLrQdH0VlMYwCSLwXUSeppjy0PBTd+QXIKQ3j4EUbuVQQ
WU7XyBuzAxx5JNcY6D837n9YfYae/Ofjd1YeAKjF8XJnLSGVZKYJ+jzx7wnjI87yY
1Qse630A+jvgW2TphFXPk7GB2ihb4CsfpaxhPsUvcHfo/6k/Vn633DzYRCeUbj/s6g
Vgphk6Er5P4N0EIXp4Z8FwdrPGRwOUXhC0b8bJSnDwE+S780kvtXm9/FpIii99b7A
JZ12Y+scvDuhJN0SxkFFOZyd/+o3FYhu2h9by88xCeM03nvX54aVBns3TDyEy/4wiR
dMXRhsobOVs1imviPsp16wvNNMx+QRTGsxR5krpZ05KCoJnK116rfzxvYvC28s8g
jgmJ28QP43/NxwIn9a+PCC7Ft/DuUL6vrkv2KGewQm+S87c/4enB0c3E7eqsXHTOFc
nIerKwzE4zrAwVhmc7N0pRU68Z80X6sEtGT/XwiN7N0UsDJahjucX7G+CRTPC2iHI7
OHwbJ1zxVG7rQ5cTbU3vjlZbCeNw/WikLKbruSuvHIKic7GLdSW+pOo2i+TM7BitO
1m1FJNzXpWUvn/XJKkx0L+Fc0nOPPWyP08022DZGsp7R1mIXnBV4mAoNQ5yxphafP
c2UG0YzpkT3ceoGoFkyFN5slVeb4E8gx8ZofwWldGiWQEm8DbiiuzubUxxdpLFJkKb
ver5ERxdKEgdcaTRoinLDBcXXP1+Aymr7o7fzB2xGc0BbiEAG0W3M1FbC0tNE/pZa
wEH+3wjcp6F9dYiYe+2AJCY0Zm6BQ11LuXKvJ8pggyvsstdJl1lCqed+lugca665IA
y5w093xj7iz1UEvXcc/hvGPoiABNuq4HMXe94d90HaSS18YIHvjv0tGo+zbjpkfRu
bgA8YQSQ1KOk8mFYobgOK+qKRx0T81Qwd0oV8ZzacOOFug0VsTlPAsv+b5Krq7+ZKR
Yq/FyIhVDSULhbGVtjdZUxJY91PBKrtahnhJJpqOni/TimHQ6nmuj4wkZa9sZy4cRg
Hn1rB37bbPlhGdraaRt1ZNpBGTWbWBoeACVzHUHjgKaA8AsbCHrc1Sm9Uke+12B6z/
HzGvdPuGNKXR5/jiYSWZD0BSe5Au4AiHdDe+6AgckHkgg47K1OMqaAnCtjJAKjcKN5
J0sFWRWNixyp9MTjlV3RXwoJ3ZibqpOW7Qm
jnLiKdZlOknesBROuWKTNU1RQEH8x4OiBrhM4RWue52Tmm2tcBCIHJJZrg6bGjMbpI
vdJRHuU22fQvayV8TFVXeY6ROln2WzIyeFftjQBx5tD0F7nLPbjSHtA2NJQ761M7GS
rutsbvGtBK7U3Gyopw3WC1kypT9uqTPRxpLJHGSn5cA7WIGurcl6tCSyBtFryLEiDS
DhJeLcvshpFmbGbed; path=/adfs/ls/; secure; HttpOnly
Set-Cookie: _WebSsoAuth0=85y/5fgneQbCdR0OddoSwto4frXt2YYefvazri9ze
yAy9sBzvvBJK33tJ0cXwHtbL4Cr837id0+GeP4Zo3gWirG+auVm/RwbG59+DUYPoSH
vFgmxt4f7mVln6XGg5mRP/X0vAlJHPGSMZGxhR7Ufyhlj0S7Enc5yKpzmlBFE4V4Jt
44h64Y9SEORD7DuL6n5eileZiK/pAWW96Z43jRpZxox5DHkAdobY09K/FXSddWmq3H
M1Btgme3sZT1Yq055TBqMqtSbx6HcMHWns0eSYUHA4TEt8U6nzfeNxZoz3E4w97VLJ
7In5FPCU6GeLQKkWh30moibyProFdN52L7YLZi80A7dLfLqjwFKp/vJvokX515/uQ
OAm6ky1W4pGT2kieYhkyX0xlm0ZkLqhOzwAr96hxt92cRbzKn2ozmxuG8M7Akr7ACr
H57ZWH43oozOXp2mofvWrrH8yiPWWufezvtgtV23e5oOB268x2rZ3BqHraM+0qa6ue
OeZPcuinl6LA/e+/jrnJ/G78xpT+7dNNqs6ZYZ2iTnNzNFqpBx2rVGp02w3QcFjWad
qn0y1dhiXm6VuPpdVnl+1VgmUvnoqlLjSV/fxEl96vvSWnNKH6+X9/3GR+ww53Ngf
; path=/adfs/ls/; secure; HttpOnly
Set-Cookie: _LSRealm=urn:federation:adatum; expires=Sat, 12-Aug-2006

```
07:32:28 GMT; path=/adfs/ls/; secure; HttpOnly
Set-Cookie: _LSCleanup=2006-07-13:07:32:28ahttps://treyws-test/
claims/; path=/adfs/ls/; secure; HttpOnly
Set-Cookie: _TTPDest=https://treyws-test/claims/; path=/adfs/ls/;
secure; HttpOnly
Set-Cookie: TTPData=eNrNWfuTqjoW/iuW59HqDd7aS3V3TUBRUFAQFX2ZihAuK
gkSEPHXT9DWbfc+u6dnaj8cnsLK4lvfumQ15CWjSddAhxTRZlBsNA6S3CQ7hA1EI4I
pKp3CPaZdpvZa9pMk6nIctX0UQvgDzVACox8k9riMcjWeb3J8jUvilCblt5cHZOR8w
H57oTDcdwG1KE4Cgkv3kdx7Lf+79Vxd9eam/1Rttfinhms/P3U2z9WnTpu3EXSR7Ww
a5ZJMaYpkTBOIGTVm/PmJbz1V6ybf6tZr3Vp7/a4Tv5bTGHdd5KAYFka6SYzyUowog
rHtl0sq3JJ4gWLK517LVSYI8EfBNQYF6SsWgTSgXQxDRLuJ3Z0Bddyt/uC78OZH+d1
FkWANkAS0pJFEQC6J0e/YiMoUJnsTATQrKjzrtm84tcKkTIGwj1qMkDuwC/27ok8pbk
TPKklb4nNgNhH3C2XsYhJR74T6qfnr/e3Tuk183e84xKCCg49KuWODPSBrbt/IpxNf
QhYEdE0rc5CEh5bdPCYIOTNLwhfuMdsMnZBew3LtkCKn/TRM13Y47sCe5R01PzWSen
/ydKC4c0uZfb5Y+4N7j8e7ae3ASH+EksC+oswQmKGTvpY/yL8qyxTL9UV1FiU+cX4o
UflDqZgF2SEZvJTBLN1tkJ+9vGqtE2Sm03QDFJYnEIxf6vb6XwHyqld+AEwY4YmGC
Yn/dY39D5uE7/5/BL9F5U6A+yout6glrJI2aYI+T/xz3PjIs3QfFZ+9lvshDPbAcVj
ToOWPkzSCNvoGyfJnGwu4T9Eb/Jr1J+3P0v/GWYRhSHAX/rOsVcaYZegS8j/DdBCTN
PqzJiv6KYEOiskRxf9slqWmwyBJvk+T++26mgUeZkUUv7fe07Esy35k9Qsf1pV4ju9
wTMGhgfdX+foVcorW9/YiQkwwW8n74PzQoEpg7xG2hfVhbyCrXJUvIJ/QtyX6yqw38V
7nEPfD5JswH2jGFT9SH1QuSgVwUFztTaW7Ir+W/vndQeHsxY4gp23lD+jD+33ggfER
7E1Hnid7cuVD6PtxvosM9kusFHttz/5843WN0hbhwjYmb1qkyIaitDMfGMMO9sVbFy
T6d2mzHe9R84e6xZePHWrhn76qIe4MFMB5xsE6nDdirEdh5gyHJ/uUq67WbrbgcPK
l/DpYLMKv2YjsYfU3XYpt+nIuOy7SjupI7encWwukl045HaNTFLtSi0ecXPZ332F
Wa6Z2LKRu53oulhut6G2/pIMTLenu/Ckwqjaa9uSbPORqcnX98odatB+ePkLEQVK5Q
tZapMturrlZEh8i8j1F+9spp8pwcTeB2JxZnNLbYulthkucdvRRHEBw9ksgA8eTRoS
ErdA8XTP52BJni7g78LBp2MF4A+l0APrFXDzi9lVvoeq+fNedwsM/XkuDb2GjOw87
R6QEkZXYmbUFDeNeVcuZ2lPgRxa+yeSLTBX0U38LdMHTFgJQTdEY8rWlPztDNDHoo
oB/4dCXAjGiwX1k3moneqs+SGP5Gc9FEHhDX6/k00yY7ks0MqU5oBNOqsg5TiZnrjo
EB70ipXDQPjaIittxr05Ned0ZzdFxrEqLSRue832FHV2DyUill+PFRLGGk4TD2dLeW
s2phG1nq00lUqkNfTWWN66RztOzu+HTsamPDkdRrX3rEQjAPPKST/0nzdnL5zshZU
+a4JyZCSzPgBwIqKoZp43cFXQGIQ30llfYHL9D7LQ7/dE/qPPot6u8/iVi10RfEwm
Mmbek9XRZYv4PVYMDzvptsTgCwCGWRCTpJGwMaMHPDtjQZJHZC37RlsDmZ0Ni9wZ/EQ
QVn1JGxqJk2zWffdzsieuGuVib1dbSV63VUG9cFN0XVUBYbbpoKgBIVNFVWhf5uRMX
6kCBJKA1I5Qgw290huk8905HFSEXZDCznmvSvOL7jAzJmo/y2YHQZS9ODAGPu8MwfM
472zXlsKvrAUPxeZug9V0Vevkttm4FLLN3XtuMZ6; path=/adfs/ls/; secure;
HttpOnly
Set-Cookie: _TTPData0=IUvGvLZndSZtsHLcDLLjYuBHdmjoK/Y9WjBZnenVFark
A8jiEgteXxJm9XNTNiaohBsBfFBngGyqsGhwds9chzXLti+Yxnks51VbZE7A6nqDH
z3NzZ5ta//jNEziP7WD+HytJ8E7aMlnX76KJ16xdhYvNN7USTwU+7VtUg0FJ284GU
08P9bFP8ONQ9y9ma+nUtAlZHRGSsZGxjv7kfKilrrs2lndERNdJNuQOIYv4lW9dSDM
sZpEQSQ2wEC85cLaTzTBIXfk8j+1NsNja80j1byhiKAO2tsS8nwTJpelrD04Xq1B9g
pdPaK0Y417zTmDUYTao1DkOlvjF0WRnJlG2BgMe8LLgtZ7FvzVeC5bVCM9DPrdiZze
cB13MzJKl1QZLTSiuJso01l1823LMTgNmyk4d61N4uKsoUaGK+mxITfhkSxam3CIWR
oVTgg1M65wBQqptuahW5y2zp7m0A2wSVEy03Z8kvM7cSiM2rei0s7CfLLEK7P7zzq
bRYTXsmcp0ZdejpUGLIsppx97n/k4/Y+255rR0nA49c9cxMjjdRNUq98qa6ueOeZVc
uy1377A/e+/9rHL7N2aCr64GMhp12aFtHyBqkvvtQ/SN24dGsSNGZB/Y+eX6AXb72I
lIgJOfJ4cbHvweXpvg9F40/QH23jHf/wK+/rF+VLY+/ULlIn9w9WNY/vYu5u0//Xcu
dA==; path=/adfs/ls/; secure; HttpOnly

>>> treyws-test (WS Resource)
GET /claims/?wa=wsignin1.0&httpsize=2708&ttpindex=0&wctx=https%3a%2f%
2ftreyws-test%2fclaims%2fDefault.aspx&wresult=eNrNWfuTqjoW%2fuiW59Hq
Dd7aS3V3TUBRUFAQFX2ZihAuKgkSEPHXT9DWbfc%2bu6dnaj8cnsLK4lvfumQ15CWjSd
dAhxTRZlBsNA6S3CQ7hA1EI4IpKp3CPaZdpvZa9pMk6nIctX0UQvgDzVACox8k9riMcj
Web3J8jUvilCblt5cHZOR8wH57oTDcdwG1KE4Cgkv3kdx7Lf%2b79Vxd9eam%2flRttf
inhms%2fP3U2z9WnTpu3EXSR7Wwa5ZJMaYpkTBOIGTVm%2fPmJbz1V6ybf6tZr3Vp7%2
fa4Tv5bTGHdd5KAYFka6SYzyUowogrHtl0sq3JJ4gWLK517LVSYI8EfBNQYF6SsWgTSg
XQxDRLuJ3Z0Bddyt%2fuC78OZH%2bd1FkWANkAS0pJFEQC6J0e%2fIMoUJnsTATQrKjz
rtm84tcKkTIGwj1qMkDuwC%2f27ok8pbkTPKklb4nNgNhH3C2XsYhJR74T6qfnr%2fe3
Tuk183e84xKCCg49KuWODPSBrbt%2fIpxNfQhYEdE0rc5CEh5bdPCYIOTNLwhfuMdsMn
ZBew3LtkCKn%2fTRM13Y47sCe5R01PzWSen%2fydKC4c0uZfb5Y%2b4N7j8e7ae3ASH%
2bEksC%2boswQmKGTvpY%2fyL8qyxTL9UV1FiU%2bcX4oUflDqZgF2SEZvJTBLN1tkJ%
2b9vGqtE2Sm03QDFJYnEIxf6vb6XwHyqld%2bAEwY4YmGCYn%2fdY39D5uE7%2f5%2f
BL9F5U6A%2byout6glrJI2aYI%2bT%2fxz3PjIs3QfFZ%2b9lvshDPbAcVjToOWPkzSC
NvoGyfJnGwu4T9Eb%2fJr1J%2b3P0v%2fGWYRhSHAX%2frOsVcaYZegS8j%2fDdBCTNP
qzJiv6KYEOiskRxf9slqWmwyBJvk%2bT%2b26mgUeZkUUv7fe07Esy35k9Qsf1pV4
ju9wTMGhgfdX%2bfoVcorW9%2fyiQkwwW8n74PzQoEpg7xG2hfVhbyCrXJUvIJ%2fQyX
6yqw38V7nEPfD5JswH2jGFT9SH1QuSgVwUFztTaW7Ir%2bW%2fvndQeHsxY4gp23lD%2
```

bjD%2b33ggfER7EiHnid7cuVD6PtxvosM9kusFHttz%2f5843WN0hbhwjYMb1qkyIait
DMfgMMO9sVbFyT6d2mzHe9R84e6xZePHWrhn76qIe4MFMB5xsE6nDdirEdh5gyHJ%2f
uUq67WbrbgcpKl%2fDpYLMKV2YjsJYfU3XYpt%2bnIUoY7SjupI7encWwukl045HaNTF
LTtSi0ecXPZ332Fwa6Z2LKRu53oulhut6G2%2fpIMTLenu%2fCkwqjaa9uSbPORqcnX9
8odatB%2bePkLEQVK5QtZapMturrlZEh8i8j1f%2b9spp8pwcTeB2JxZnNLbYulthkuc
dvRRHEBw9ksgA8eTRoSerdA8XTP52BJni7g78LBp2MF4A%2b10APrFXDziR91Vvoeq%2
bfNedwsM%2fXkuDb2GjOw87R6QEckZxymbUFdNeVcNfuZ21PgRXa%2byeSLTBX0U38LdM
HTFgJQTdEy8rW1PztDNdH0ooB%2f4dCXAjGIWx1k3moneqs%2bSGP5Gc9FEHhDX6%2fk
00yY7kSOMqU5oBNOqsg5TiZnrjoEB7OipXDQPjaIittxro5Ned0ZzdFxrEqLSRue832F
HV2DyUi11%2bPFRLLGGk4TD2dLeWs2phGlnqOOLUqkNfTWWN66RztOzu%2bHTsamPDkdk
rRx3rEQjAPPKST%2f0nzdN15zshZU%2ba4JyZCSzPgBwIgKoZp43cFXQGIDq3OllfYHL
9D7LQ7%2fde%2fqPPot6u8%2fiVi10Rf HTTP/1.1

<<<
HTTP/1.1 302 Found
Cache-Control: private
Content-Length: 316
Content-Type: text/html; charset=utf-8
Location: https://treysts-7/adfs/ls/?wa=wsignin1.0&wreply=
https%3a%2f%2ftreyws-test%2fclaims%2f&wct=2006-07-13T07%3a32%3a28Z&
wctx=https%3a%2f%2ftreyws-test%2fclaims%2fDefault.aspx&ttindex=1758
Set-Cookie: _TTPData=eNrNWFTqjoW/iuW59HqDd7aS3V3TUBRUFAQFX2ZihAuKqk
SEPHXT9DWbfc+u6dnaj7cnsLK41vfumQ15CWjSddAhxTRZIBsNA6S3CQ7hAlEI4IpKp
3CPAdpzyZa9pMK6nIctX0UQvqDzVACox8k9riMcjWeb3J8jUvilCblt5cHZORhW57o
TDcdwGLKE4Cgkv3kdx7Lf+79VxD9eam/1Rttfinhms/P3U2z9WnTpu3EXSR7Wwa5ZJM
aYpKtBOIGTVm/PmJbz1V6ybf6tZr3Vp7/a4Tv5bTGHdd5KAYFka6SYzyUowogrHt10sq
3JJ4gWLK517LVSyi8EFBNQYF6SsWgTSgXqXDRLuJ3Z0Bddyt/uc78OZH+d1FkWANKAS0
pJFEQC6J0e/IMoUJnsTATQrKjzrtm84tcKkTIgwlqMkDuwC/27ok8pbkTPKklb4nNGn
hH3C2XsYhJR74T6qfnr/e3Tuk183e84xKCCg49KuWODPSBrbt/IpxNfQhYEdE0rc5CEh
5bdPCYiOTNLwhFuMdsMnZBew3LtkCKn/TRM13Y47sCeS0R1PzWSen/ydK4C0uZfb5Y+
4N7j8e7ae3ASH+Eksc+oswQmKGTvpY/yL8qyxTL9UVlFiU+cX4oUf1DqZgF2SEZvJTBL
N1tkJ+9vGqtE2Sm03QDFJYnEIfx6vb6XwHyqlD+AEWY4YmMGcYn/dY39D5uE7/5/BL9F
5U6A+yout6glrJi2aYI+T/xz3PjIs3QfFZ+9lvshDPbAcVjToOWPkzSCNvoGyfJnGwu4
T9Eb/Jr1J+3P0v/GWYRhSHAX/rOsVcaYZegS8j/DdBCtNPqzJiv6KYEoisKRx9slqWm
wyBjvk+T++26mgUeZkUUV7fe07Esy35k9Qsf1pV4ju9wTMGhgfdx+foVcorW9/YiQkww
W8n74PzQEpPg7xG2fhvhyCrXJUvIJ/QtyX6yqw38V7nEPfD5JswH2jGFT9SH1QuSgVwU
FztTaW7r+W/vndQeHsxY4gp231D+jD+33ggfER7EiHnid7cuVD6PtxvosM9kusFHttz
/5843WN0hbhwjYMb1qkyIaitDMfgMMO9sVbFyT6d2mzHe9R84e6xZePHWrhn76qIe4MF
MbB5xsE6nDdirEdh5gyHJ/uUq67WbrbgcpKl/DpYLMKV2YjsJYfU3XYpt+nIUoY7Sjup
I7encWwukl045HaNTFLtSi0ecXPZ332Fwa6Z2LKRu53oulhut6G2/pIMTLenu/Ckwqj
aa9uSbPORqcnX98odatB+ePkLEQVK5QtZapMturrlZEh8i8j1f+9spp8pwcTeB2JxZnN
LbYulthkucdvRRHEBw9ksgA8eTRoSerdA8XTP52BJni7g78LBp2MF4A+10APrFXDziR9
1Vvoeq+fNedwsM/XkuDb2GjOw87R6QEckZxymbUFdNeVcNfuZ21PgRXa+yeSLTBX0U38L
dMHTFgJQTdEy8rW1PztDNdH0ooB/4dCXAjGIWx1k3moneqs+SGP5Gc9FEHhDX6/k00yY
7kSOMqU5oBNOqsg5TiZnrjoEB7OipXDQPjaIittxro5Ned0ZzdFxrEqLSRue832FHV2D
yUi11+PFRLLGGk4TD2dLeWs2phGlnqOOLUqkNfTWWN66RztOzu+HTsamPDkdkrRx3rEQj
APPKST/OnzdN15zshZU+a4JyZCSzPgBwIgKoZp43cFXQGIDq3OllfYHL9D7LQ7/de/qP
Pot6u8/iVi10Rf; path=/claims; secure; HttpOnly

>>> treysts-7 (Resource IP/STS)
GET /adfs/ls/?wa=wsignin1.0&wreply=https%3a%2f%2ftreyws-test%2fclaims
%2f&wct=2006-07-13T07%3a32%3a28Z&wctx=https%3a%2f%2ftreyws-test%
2fclaims%2fDefault.aspx&ttindex=1758 HTTP/1.1
Cookie: TTPRealm=urn:federation:adatum; WebSsoAuth=eNrNVluXqrgS9qe4
nEeXG/Coq7vXCSAKCgre0JdZAcJfJSgXEX/9BG3dds/uffqcNQ/DU6hUqr76q1JJXmIY7
HsgjlGU+CEuP0aS8Fr5s9Ouo0bLbNSYToeuNR2rXWPNN1nju7SfoIMs22xWylIcp0jCcQ
Jx8lqp03S7RndqTGNod3qNeq/e3bzzrRK+VNMI9B9kogoWTXhKhvByhGMHI8iplBW7DaIm
imMy9Vhgi8PFHwTny47hXgL7ZCmHsxz0MAxT3Eqs3A8q4x/yge/AeR+Xt5RoiH2LbLwRxW
Q0TDj1hhL4CSxQmeBIBJykgP+T07zrvVkfGq+whbSedxEvlWYf/h6JPKm5ckh7hHUUXMWVx
LyBLK2km/ikKx6qPqp/9fW6c+xxX3Z5/8wgS0nbjHF/ZnYRpZ6J26QnyjLvCtKiXDJ31KS
OxtU4Kgd2M0eKE+W7vbD8OdT3LvhEMYe990UdesiIWC6JxULZ0ni/zs7Xh+aYdd+vXu6YP
dBx/vob2Tk3giJ751tTpLYIIC81/+KP9NWXZIpj8qKyjxQvtvRQo/KPUYH9thFt9LYJaaW
2Q1738qqUTJLrQdH0VLMYwCSLwXuSepjy0PBTd+QXiKQ3j4EUbuVQQWU7XyBuzAxz5JNkz
C6D837n9YyfAe/0fjd1YeAKjf8XJnLSGVZKYJ+jzx7wnji87yY1Qse630A+jvgW2TphFXP
k7GB2ihh4CsfPaxhPsUvcHfo/6k/Vn633DzYRCEuBj/s6gVgphk6Er5P4N0EIXp4Z8FwdR
PGRwOUXhC0b8bJsnDwE+S780kvtxXM9/FpIii99b7AJZl2Y+scCVduhJN0SxFFOzYd/+o3
FYhu2h9by88xCeM03nvX54aVBns3TDyEy/4wiRDMXRhsobOVs1imviPSp16wvNNMx+QRTG
sxR5krpZ05KCoOJnKC116rfzxvYvC28s8gJgmJ28QP43/NxwIn9A+PCC7Ft/DuUL6vrkv2
KGewQm+S87c/4enB0c3E7eqsXHTOfcnIerKwzE4zrAwVhmc7NOpRU68Z80X6sEtGT/XwiN

7N0UsdJahjucX7G+CRTPC2iHI7OHwbJ1zxVG7rQ5cTbKU3vjLZbCeNw/WikLKbruSuvHIk
Ic7GLdSW+p0o2i+TM7Bit0lM1FJNzzXpWUvn/XJKkx0L+Fc0nOPPyP08022DZGsp7R1mI
XnBV4mAoNQ5yxphafPc2UG0Yzpk+TC3eoGoFkyFN5slVeb4E8gX8ZofwWldGiWQEm8Dbii
zubUxxdPLfJkKbVer5ERxdkEgdcaTRoiLDBcXPl+Ayrnm7o7fzB2xGc0BbiEAAg0W3M1F
bc0tNE/pZawEH+3wjcp6F9dYiYE+2AJCY0Zm6BQ1lLuXKvJ8pggyvsstdJl1lCqed+lugc
a665IAy5w093xj7izIUevXCc/hvGPoiABNuq4HMXe94d90HaSS18YIHvjv0tGo+zbjpkf
rubgA8YQsq1K0K8mfYobgOK+qKRx0T81Qwd0oV8ZzacOOFug0VsTlpAsv+b5Krq7+ZKRYq
/FyIhVDSULhbGVtjdZUxJY91PBKrtAHnhJJpqOni/TimHQ6nmuj4wkZa9sZy4cRgHn1rB3
7bbPlhGdraaRtLZNpBGTWBwBoeACVzHUHjgKaA8AsbCHrc1Sm9Uke+12B6z/HzGvdPuGNK
XR5/jiYSWZD0BSe5Au4AiHDde+6AgckHkgg47K1OMqaAnCtJJAkjcKN5J0sFWRWNixyp9M
Tjlv3RXWoJ3ZibqPOW7QmjnlI
kdZlOknesBRouWKTNUlRQEh8x4OiBrhM4RWue52Tmm2tcBCIHJJZrg6bGjMbpIvdJRhUu
Z2fQvayV8TFVXeY6ROln2WzIyEffTjQBx5td0F7nLPbjSHTa2NJQ761M7GSrutsbvGtBK7
U3Gyopw3WC1kyptU9gTPRxpLJHGSn5ca7WIGurcl6tCSyBtFryLEiDSdhJeLcvshpFqmbG
bed;_WebSsoAuth0=85y/5fgneQbCdR0OdoSwt04frXt2Yyefvazri9zeyAy9sBzvvBJ
K3tJ0cXwHtb4Cr87Id0+GeP4zo3gWirG+auVm/RwbG59+DUYPosHvFgmxt47fmVln6X
Gg5mRP/X0vAlJHPGSMZGxh7Ufyhlj0s7Enc5yKpzmlBFE4V4Jt44h64Y9SEORD7DuL6n5
eileZiK/pAWW96Z43jRpZxox5DHkAdobY09K/FXSddWmq3HM1Btgme3sZT1YqO55TBqMqt
Sbx6HcMHWns0eSYUHA4Tet8U6nzfeNxxZoz3E4w97VLJ7Ini5FPCU6GeLQKkWh30moibyPR
oFdn52L7YlZLi87dLFLqjwFKp/vJvokX515/uQOAm6ky1W4pGT2kieYhkyX0xlm0ZkLqh
OzWAr96hxt92cRbzKn2ozmxuG8M7Akr7ACrH57Z8WH43oozOXp2mocWrrH8yiPWWufezvt
gtV23e5oOB268x2rZ3BqHraM+0qa6ueOeZPcuin16LA/e+/jrnJ/G78xpT+7dNNqs6ZYZ2
iTnNZNFqBx2rVGp02w3QcFjWadqn0y1dhiXm6VuPpdIvnl+lvgmUvnoqlJJSV/fxEl96
vvSWWhNKHb+X9/3GR+vw53Ngf;_LSRealM=urn:federation:adatum;_LSCleanup=
2006-07-13:07:32:28Zahttps://treyws-test/claims/;_TTPDest=
https://treyws-test/claims/;_TTPData=eNrNWfuTqjoW/iuW59HqDd7aS3V3TUB
RUFAQFX2ZihAukgSEPHXT9Dwbfc+u6dnaj8cnsLK41vfumQ15CWjSddAhxTRZIBsNA6S
3CQ7h1AE14IpKp3CPaZdpvZa9pMk6nIctX0UQvgDzVAc0x8k9riMcjWeb3J8jUvilCblt
5cHZOR8wH57oTdcdwGLKE4Cgkv3kdx7Lf+79VxD9eam/lRttfinhms/P3U2z9WnTpu3EX
SR7Wwa5ZJMaYpkTBOIGTVm/PmJbZ1V6ybf6tZr3Vp7/a4Tv5bTGHdd5KAYFka6SYzyUow
ogrHt10sq3JJ4gWLK517LVSYI8EFBNQYF6SsWgTSgXQXDRLuJ3Z0Bddyt/uC780ZH+d1F
kWANKASOpJFEQc6J0e/IMoUJnsTATQrKjzrtm84tcKkTIGwjlqMkDuwC/27ok8pbkTPKk
lb4nNGnhH3C2XsYhJR74T6qfnr/e3Tuk183e84xKCCg49KuWODPSBrbt/IpxNfQhYEdE0
rc5CEh5bdPCYIOTNLwhfuMdsMnZBew3LtkCKn/TRM13Y47sCe5R01PzWsen/ydKC4c0uZ
fb5Y+4N7j8e7ae3ASH+Eksc+oswQmKGTvpY/yL8qyxTL9UVlFiU+CX4oUf1DqZGF2SEZv
JTBln1tkJ+9vG8te2Sm03QDFJYnEIfx6vb6XwHyql+dAEwY4YmMGCYn/dY39D5uE7/5/B
L9F5U6A+Yogt6glrJ12aYI+T/xz3PjIs3QfFZ+9lvshDPbAcVjToOWPpkzSCNvoGyfJnGw
u4T9Eb/Jr1J+3P0v/GWYRhSHAX/rOsVcaYZegS8j/DdBCTNPqzJiv6KYEoisKRx9slqw
MwyBJvk+T++26mgUeZkUUV7fe07Esy35k9Qsf1pV4ju9wTMGhgfdX+foVcorW9/YiQkww
w8n74PzQoEpg7xG2hfvhbyCrXJUvIJ/QyX6yqw38V7nEPfD5JswH2jGFT9SH1QuSgVwUF
ZtTaW7Iz+w/vndQEHsX4gpp23ld+jD+33ggfER7EiHnid7cuVd6PtxvosM9kusFHttz/
5843WN0hbhwjYmb1qkyIaitDMfgMMO9sVbFyT6d2mzHe9R84e6xZePHWrh76qIe4MFMb
B5xsE6nDirEdh5gyHJ/uUq67WbrbgcpK1/DpYLMKV2YjsJYfU3XYpt+nIUoY7SjupI7e
ncWwukl045HaNTFLTtSi0ecXPZ332Fwa6Z2LKRu53oulhut6G2/pIMTLenu/Ckwqjaa9u
SbPORqnX98odatB+ePKLEQVK5QZapMturr1ZEh8i8jlf+9spp8pwcTeB2JxZnNLbYul
thkucdvRRHEBw9ksgA8eTRoSErdA8XTP52BJni7g78LBp2MF4A+10APrFXDziR91Vvoeq
+fNedwsM/Xkudb2GjOw87R6QEkdXymbUFdNeVcNfuZ21PgRXa+yeSLTBX0U38LdMHTFgJ
QTDey8rW1PztDNdH0oob/4dCXAjgIWx1k3moneqs+SGP5Gc9FEHhDX6/k00yY7kS0MqU5
oBNOqsg5TiZnrj0E7oipXDQPjaIittxro5Ned0ZzdFxrEqLSRue832FHV2DyU11+PFR
LGGk4TD2dLeWs2phG1nqOOLUqkNfTWWN66RztOzu+HTsamPdkdkrRx3reQjAPPKST/Onz
dN15zshZU+a4JyZCSzPgBwIgKoZp43cFXQGIDq3O11fYHL9D7LQ7/dE/qPPot6u8/iVi1
0RfEwmMmbek9XRZYv4PVYMDzvptsTgCwCGWRCTpJGwAMHPDjtjQZJH2C37R1sDmZ0Ni9wZ
/EQQVn1UGxqJk2zWFFdZsieuGuVi1dbSV63VUG9cFN0XVUBYbbpKGBIVNFVWhf5uRMX
6kCBJKALi5Qgw29Ohuk8905HFSEXZDCznmvSvOL7jAzJmo/y2YHQSZ9ODAGPu8Mwfm472
zX1sKvrAUPxeZug9VOVevktthM4FLN3XtuMZ6;_TTPData0=IUvGvLZndSZtsHLcDLL
jYUBHdmjok/Y9WjBZnenVFARKA8jiEgteXxJ0m9XNTNiaohBsBfFBngGyqsGhws9chzX
Lti+Yxnks51VbZE7A6nqDH33Nz5ta//jNEZiP7WD+HytJ8E7aMlnX76KJ16xdhYNvNN7
UStwU+7VtUg0FJ284GU08P9bFP8ONQY9ma+nUtAlZHRGSsZGxjv7kfKilrrs2lnderND
jNuQOIyV41W9dSDMSzPESQ2wEC85cLaTzTBIXFK8j+1NsNja80j1byhiKAO2tsS8nwTJ
pelrD04Xq1B9gpdPaK0Y417zTmDUYTao1DkOlVjF0wRnJlg2BgMe8LqtZ7FvzVeC5bVC
M9DPrdiZzEcB13mJK1QZLTsiuJso011823LMTgNmyk4d61N4uKsoUaGK+mxITfhkSx
aM3CIWROvtgglM65wxBQqptwahW5y2zp7m0A2wSVEy03Z8kvM7cSiM2rei0
s7CsLLEK7P7zzqbRYTXsmcp0ZdejpuGLIssppx97n/k4/Y+255rR0nA49c9cxMjjdRNuq9
8qa6ueOeZvcuy1377A/e+/9rHL7N2aCr64GMhp12aFtHyBqkvvtQ/SN24dGsSNGZB/Y+e
X6AXb721lIgjOfJ4cbHvweXpv9GF40/QH23jHf/wK+/rF+VLy+/ULlIn9w9WNY/vYu5u0
//XcudA==

<<<
HTTP/1.1 302 Found

Location: https://treyws-test/claims/?wa=wsigninl.0&ttpsize=2708&ttpindex=1758&wctx=https%3a%2f%2ftreyws-test%2fclaims%2fDefault.aspx&wresult=EwmMmbek9XRZYv4PVYMDzvpTsTgCwCGWRCTpJGwAMHPDtjQZJHZC37R1sDmZ0N19wZ%2fEQQVn1JGxqJk2zWFFdZsieuG uV1b1dbSV63VUG9cFN0XVUBYbbpoKgBIVNFVWhf5uRMX6kCBJKAL15Qgw290huk8905HFSEXZDCznmvSvOL7jAzJmo%2fy2YHQSZ90DAGPu8MwFM472zX1sKvrAUPXeZug9V0VevktthM4FLLN3XtuM26IUvGvLZndSZtsHLcD LLjYuBHdmjok%2fy9WjBZnenVFarKA8jiEgteXxJ0m9XNTNiaohBsBfFBngGyqsGhwds9chzXLti%2bYxnks51VbZE7A6nqDH3NzZ5ta%2f%2fjNEziP7WD%2bHytJ8E7aMlnX76KJ16xdhYNvNN7USTuW%2b7VtUg0FJ284GU08P9bFPr8ONQy9ma%2bnUtAlZHRGSsZGxjv7kfKi1rrs21ndERNdjNuQOIyV41W9dSDMsZpEQSQ2wEC85cLaTzTBIXfK8j%2blNsNja80j1byhiKAO2tsS8nwTJpelrD04Xq1B9gpdPaK0Y417zTmDUYT a01DkOl1vJf0wRnJlq2BgMe8LLqtZ7FvzVeC5bVCM9DPrdiZzEcB13MzJKI1LQZLTSiuJs o01i1823LMTgNmyk4d61N4uKsoUaGK%2bmxiTfHkSxaM3CIWROVTgg1M65wxBQqptwah W5y2zp7m0A2wSVey0328kvM7cSiM2rei0s7CsLLEK7P7zzqbRYTXsmcp0ZdejpuGLIsp px97n%2fk4%2fy%2b255rR0NA49c9cxMjjdRNUq98qa6ueOeZVcuy1377A%2fe%2b%2f9rHL7N2aCr64GMhp12aFtHyBqkvvtQ%2fSN24dGGSNGZB%2fy%2beX6AXb72I1IlgJ0fJ4cbHvwExpv9GF40%2fQH23jHf%2fwK%2b%2frF%2bVLY%2b%2FUL1In9w9WNY%2fvYu5u0%2f%2fXcudA%3d%3d

>>> treyws-test (WS Resource)

GET /claims/?wa=wsigninl.0&ttpsize=2708&ttpindex=1758&wctx=https%3a%2f%2ftreyws-test%2fclaims%2fDefault.aspx&wresult=EwmMmbek9XRZYv4PVYMDzvpTsTgCwCGWRCTpJGwAMHPDtjQZJHZC37R1sDmZ0N19wZ%2fEQQVn1JGxqJk2zWFFdZsieuG uV1b1dbSV63VUG9cFN0XVUBYbbpoKgBIVNFVWhf5uRMX6kCBJKAL15Qgw290huk8905HFSEXZDCznmvSvOL7jAzJmo%2fy2YHQSZ90DAGPu8MwFM472zX1sKvrAUPXeZug9V0VevktthM4FLLN3XtuM26IUvGvLZndSZtsHLcD LLjYuBHdmjok%2fy9WjBZnenVFarKA8jiEgteXxJ0m9XNTNiaohBsBfFBngGyqsGhwds9chzXLti%2bYxnks51VbZE7A6nqDH3NzZ5ta%2f%2fjNEziP7WD%2bHytJ8E7aMlnX76KJ16xdhYNvNN7USTuW%2b7VtUg0FJ284GU08P9bFPr8ONQy9ma%2bnUtAlZHRGSsZGxjv7kfKi1rrs21ndERNdjNuQOIyV41W9dSDMsZpEQSQ2wEC85cLaTzTBIXfK8j%2blNsNja80j1byhiKAO2tsS8nwTJpelrD04Xq1B9gpdPaK0Y417zTmDUYT a01DkOl1vJf0wRnJlq2BgMe8LLqtZ7FvzVeC5bVCM9DPrdiZzEcB13MzJKI1LQZLTSiuJs o01i1823LMTgNmyk4d61N4uKsoUaGK%2bmxiTfHkSxaM3CIWROVTgg1M65wxBQqptwah W5y2zp7m0A2wSVey0328kvM7cSiM2rei0s7CsLLEK7P7zzqbRYTXsmcp0ZdejpuGLIsp px97n%2fk4%2fy%2b255rR0NA49c9cxMjjdRNUq98qa6ueOeZVcuy1377A%2fe%2b%2f9rHL7N2aCr64GMhp12aFtHyBqkvvtQ%2fSN24dGGSNGZB%2fy%2beX6AXb72I1IlgJ0fJ4cbHvwExpv9GF40%2fQH23jHf%2fwK%2b%2frF%2bVLY%2b%2FUL1In9w9WNY%2fvYu5u0%2f%2fXcudA%3d%3d HTTP/1.1
Cookie: _TTPData=eNrNWFuTqjow/iuW59HqDd7aS3V3TUBRUFaQFX2ZihAuKgkSEPHXT9DWbfc+u6dnaj8cnsLK41lvumQ15CWjSddAhxTRZIBsaN6S3CQ7hA1EI4IpKp3CPazdpvZa9pMkdn1IctX0UQvqZVACox8k9riMcjWeb3J8jUvilCblt5cHZOR8wH57oTDcdwG1KE4Cgkv3kdx7Lfl+79VxD9eam/lRtftinhms/P3U2z9WnTpu3EXSR7Wwa5ZJMaYpkTBOIGTVm/PmJbz1V6yb6f6tZr3Vp7/a4Tv5bTGHdd5KAYFka6SYzyUowogrHt10sq3JJ4gWLK517LVSYI8EfBNQYF6SsWgTSGXQXDRLuJ3Z0Bddyt/uc780ZHD1FkWANKAS0pJFEQC6J0e/IMOUJnsTATQrKjzrtm84tcKkTIGwj1qMkDuwC/27ok8pbkTPKklb4nNGnhH3C2XsYhJR74T6qfnr/e3Tuk183e844KCCG49KuWODPSBrbt/IpxNfQhYEdE0rc5CEh5bdPCYIOTNLwhfuMdsMnZBeW3LtkCKn/TRM13Y47sCe5R01PzWSen/ydKC4c0uZfb5Y+4N7j8e7ae3ASH+EksC+oswQmKGtvpY/yL8qyx7L9UV1FiU+cX4oUf1DqZGF2SEzvJTBLN1tkJ+9vGgtE2Sm03QDFJYnEIfx6vb6XwHyqld+AEWY4YmMGcYn/dY39D5ue7/5/BL9F5U6A+yout6glrJI2aYI+T/xz3PjIs3QfFZ+91vshDPbAcVjToOWPKzSCNvoGyFJnGwu4T9Eb/Jr1J+3P0v/GWYRhSHAX/rOsVcaYZegS8j/DdBCNTpqzJiv6KYEOiskRxf9slqwmwyBJvk+T++26mgUeZkUUv7fe07Esy35k9Qsf1pV4ju9wTMghgfdX+foVcorW9/YiQkww8n74PzQoEpg7xG2hfvhbyCrXJUvIJ/QyX6yqw38V7nEPfD5JswH2jGFT9SH1QuSGvWUFztTaW7Ir+W/vndQeHsxY4gp231D+jD+33ggfERYE1i8id7cuVD6PtvxvosM9kusFHttz/5843WN0hbhwjYmb1qkyIaitDMfgMMO9sVbFyT6d2mzHeR984e6xZePHWRhn76qIe4MFMbB5xsE6nDdirEdh5gyHJ/uUq67WbrbgcpK1/DpYL MKV2YjsYfU3XYpt+nIUoY7SJupI7encWwuk1045HaNTFLTtSi0ecXPZ332FWa6Z2LKRu53oulhut6G2/pIMTLenu/Ckwqjaa9uSbPORqcnX98odatB+ePkLEQVK5QtZapMturr1ZEH8i8jlF+9spp8pwcTeB2JxZnLbYulthkucdvRRHEBw9ksgA8eTRoSerdA8XTP52BJni7g78LBp2MF4A+l0APrFXDziR91Vvoeq+fNedwsM/XkuDb2GjOw87R6QEekZXymbUFdNeVcnfuZ21PgRXa+yeSLTBX0U38LdMHTFgJQTDey8rW1PztDNdHOooB/4dCXAjGIWx1k3moneqs+SGP5Gc9FEHhDX6/k00yY7kS0MqU5oBN0qsg5TiZnrjoeB7OipXDQpJaIittxro5Ned0ZzdFxrEqLSRue832FHV2DyU111+PFRLGgk4TD2dLews2phG1nq00LUqkNfTWWN66RztOzu+HTsamPDkdkrRx3REqJAPPKST/0nzdN15zshZU+a4JyZCSzPgBwIqKoZp43cFXQGIDq3O1lfYHL9D7LQ7/dE/qPPot6u8/iVi10Rf

<<<

HTTP/1.1 302 Found

Location: https://treyws-test/claims/Default.aspx

Set-Cookie: _TTPData=; expires=Wed, 12-Jul-2006 07:32:28 GMT;

path=/claims

Set-Cookie: WebSsoAuth=eNrNV9uSosoS/RXDeTR6wLsY3R2nAEFQUBAVfdlRQnFRqV
VIuIn79LrL1unvOzOmzYx62T0WYKnPlyiSrfE5gtO+DJEfXGhJceawU8aX6V7fTQM32pv
lU73bZp5bndJ64Taf+xPVYB0EPOe6mVa0oSziHBScpXOlLtcGynSe2+1RvWmy332z0G73
lGyZ+qWYx7nvIRTEsg/TTGBWVGUCIxxk5QrWhws+IFihP67qVap4YQfzScoz10+iXpmy8C
kzDpYxihpJ86/RnQxv36d7YP731UX5+vKQoEu2FpSCo6SXnkkRj9iifFTPakB15aUn6P6
d0xb15B5oYIO8hESRqHTun/EegT5DVI00PSZ5gy5zx5SukWxtndMEqYZ+Yj9NPzf/fOeM
rrHs89haUL6HpJXyJ9z0gWO+hNutJ8ky4KnZgkxEvfFaT6+q1A0IVpFj0zn73d/ROyC2n
tPTKESfDfEA3DiTkoSt5JNzIrnRfnYCCIC5f02Jd7pA9+H3q8pfYmThognIb0lessHSmK
6HPlo/03bdlm1f4I1lAaEPenJoUfQP08xC7Jk3sLzLLNFjnp25NOO1FxS7QXorgikTiCN
HpZelr6xAlQBJPvVKEEWMN3Evv3FphP9eorcKMqH7TYMCXxf27af3dI9Jb/R+d3VR4EmN
/pclctPz20yVL0+cW/J42PPCuPvbnTPtqIYLgHrkuHRL9+DI5QAd9gWT1c4wF3GfoFF6
e9Sf0Z+v/4i2QKC4XP9Z1hplTCT01fzPMJVjkh3+LMmyfyrgcIjJCcX/bpa0DaMwTb9O
k/nldzULfUybKH4bVq9ieZ5/z5tXpNqqsQzLMRTgJqH/rXrbhdxy9L0+CxATTl/kfXh5N
6AqYO+TOEyD6Bcu60ydlV0+obPz5NRb+Fu1wrzj80U3H5jFCXxKali/eyJkRh+LyZKrmTe
Wl+ulrF4XXZyuGOKEnb5S8W/9/PBA+oT05IPcpuadzpfRld79Qh3lPTgx9eub+E50eGt1
c3LrGxS37XJsQ1FOHY3CcYXGs13G6z6YOPfHeI5+Zh7Z0/b4XhtW7AbEoL4iJRqsO19G8
FWPjEOXucHh2zoXm6b12Fy4necauw8UiWlmtg7NkkLbbLp
VeMrLV4Q4m7cxVetM4thbpOVoyulYuadla4HusGhSzaD2FKfZCLMUuS6wPU7X22jbHk1
mzjrzXXTW4GEqNmlpxm2M5BwYG7VptxL2NLnwh5odKbY6VSDb7eWWyDvyzyNU3LKy2ywn
whTEvKJ5Z/PK04sONkUR2a0ggPjog1zhga+M5JakNn1Q/gbnC9B5f3cMdgHM5SwPjLkER
LDWTCeXjJW4MAxxkLfnUN4Xa4kPHGy25xF3ckWApJzn9S1oapZSaNYg10QVxm2Xu0252j
TeOA+2wOB9fcEDzRjss1jb+4s71FL9IvD4Jw4DCYAjvzVA7q92gr8agCwWongugNAfBka
tmOb8dCeg1SXNQJTJhpJpS4
HRyYepDcLRqegbl3qlFNNyLC21sKWtuNEensSYtJj14KfY1enUNJyPNWY4XE9UeTlIG50
tna7enEnbcoYGXaq0xDLRY2XhmNs8u3obNxpYxOp6QvXK9sXoYAVjUzsZx0Nm0PXJ2Fnb
W0Xn1REnmAwDgRABQy31f9jTQkkf97or5gGdyY0DrMOiJ/OB9zoLRG1Dd6iVWEI7yTNk0
RUMTaL2AL1IxfP+OFXmgCEABOZ+vpFHeoHv5FQkZUTWsnBydJA7+bCsncl0eH41kPShm
brpZ13zOpIz8bRDgYxevZsWTUfjtSs31TA0DRaa05HLHuBzTdD43vWdkhsrjYdA4pHK8Q
3YMuoz0ZvvLpFc43dhBrnLXpPmV+wwNyfaIM9nR14hAyibcsC6Q9AZF9x2bavsy16wUGj
vNljLVg2ucIR2Cpd6sWnqpzU2Sls6ZvU97TNpg9XTRs5PCzk40JFprOh+tKC2JsU11URT
ZEhl1x1/IPGGQ/tmxm8tgQ+3; path=/claims; secure; HttpOnly
Set-Cookie: WebSsoAuth0=vPDongOyasChyToiOY0bV9+Ba5vkc5xVY1G4s1R35cD7
RUxWGxg/NLoAIdgGEVye95Owd7Kl848cpbNYrs1lu9g0zokt/4hr100CbXU316XChE5nm
wbHjiO9n/Uz98ioH1EW8ZOxw402vuh2rXn+lzamRyvW2nBHEFM9hrZerZq2q6cEUmIsB
kuGGu1kC4zSV1wIicEU2y1NqW2vdjgQAOA7elxoKThMu35ess3+Po0kLHKdfegGc11/zy
mA0bXGq3jUgluTIN3R4rtQMDjMasIXrcjDOz5irf9bmSFxqUbu5P5KGREL0cCWhIkud2s
lqrbWLLZzCu7uCGYLBkiMg697aKmTofLuJOSmWZ0E4+XLEj0y1BheMy1lyBAmp93izX
p93LVH3Eg44JkxZaLs/S3ide7VWbNmH887GirrEGnAGnZ2THI6roWip05XTPLTNQBBQkX
DOvgh2xgXrnYbbNXA29K0dZ+Zwujs6/4LHaqfJ+bNcpumzGPC/pi9j7vK/b/x69+bsI8T
; path=/claims; secure; HttpOnly

```
>>> treysw-test (WS Resource)
GET /claims/Default.aspx HTTP/1.1
Cookie: _WebSsoAuth=eNrNV9uSosoS/RXDeTR6wLsY3R2nAEFQUBAVfdlRQnFRqVVIuIn
79LrL1unvOzOmzYx62T0WYKnPlyiSrfE5gtO+DJEfXGhJceawU8aX6V7fTQM32pv1U73bZ
p5bndJ64Taf+xPVYB0EPOe6mVa0oSziHBScpXOlLtcGynSe2+1RvWmy332z0G73lGyZ+qW
Yx7nvIRTEsg/TTGBWVGUCIxxk5QrWhws+IFihP67qVap4YQfzScoz10+iXpmy8CkzDpYxih
pJ86/RnQxv36d7YP731UX5+vKQoEu2FpSCo6SXnkkRj9iifFTPakB15aUn6P6d0xb15B5o
YIO8hESRqHTun/EegT5DVI00PSZ5gy5zx5SukWxtndMEqYZ+Yj9NPzf/fOeMrrHs89haUL
6HpJXyJ9z0gWO+hNutJ8ky4KnZgkxEvfFaT6+q1A0IVpFj0zn73d/ROyC2ntPTKESfDfEA
3DiTkoSt5JNzIrnRfnYCCIC5f02Jd7pA9+H3q8pfYmThognIb0lessHSmK6HPlo/03bdlm1
lf4I1lAaEPenJoUfQP08xC7Jk3sLzLLNFjnp25NOO1FxS7QXorgikTiCNHpZelr6xAlQBJ
PvVKEEWMN3Evv3FphP9eorcKMqH7TYMCXxf27af3dI9Jb/R+d3VR4EmN/pclctPz20yVL0
+cW/J42PPCuPvbnTPtqIYLgHrkuHRL9+DI5QAd9gWT1c4wF3GfoFF6e9Sf0Z+v/4i2QKC
K4XP9Z1hplTCT01fzPMJVjkh3+LMmyfyrgcIjJCcX/bpa0DaMwTb9Ok/nldzULfUybKH4b
vQ9ieZ5/z5tXpNqqsQzLMRTgJqH/rXrbhdxy9L0+CxATTl/kfXh5N6AqYO+TOEyD6Bcu60
ydlV0+obPz5NRb+Fu1wrzj80U3H5jFCXxKali/eyJkRh+LyZKrmTeWl+ulrF4XXZyuGOKEn
b5S8W/9/PBA+oT05IPcpuadzpfRld79Qh3lPTgx9eub+E50eGt1c3LrGxS37XJsQ1FOHY3
CcYXGs13G6z6YOPfHeI5+Zh7Z0/b4XhtW7AbEoL4iJRqsO19G8FWPjEOXucHh2zoXm6b12
Fy4necauw8UiWlmtg7NkkLbbLpVeMrLV4Q4m7cxVetM4thbpOVoyulYuadla4HusGhSzaD
2FKfZCLMUuS6wPU7X22jbHk1mzjrzXXTW4GEqNmlpxm2M5BwYG7VptxL2NLnwh5odKbY6
VSDb7eWWyDvyzyNU3LKy2ywnwhTEvKJ5Z/PK04sONkUR2a0ggPjog1zhga+M5JakNn1Q/g
bnC9B5f3cMdgHM5SwPjLkERLDWTCeXjJW4MAxxkLfnUN4Xa4kPHGy25xF3ckWApJzn9S1o
apZSaNYg10QVxm2Xu0252jTeOA+2wOB9fcEDzRjss1jb+4s71FL9IvD4Jw4DCYAjvzVA7q
92gr8agCwWongugNAfBkatmOb8dCeg1SXNQJTJhpJpS4HRyYepDcLRqegbl3qlFNNyLC21s
KWtuNEensSYtJj14KfY1enUNJyPNWY4XE9UeTlIG50tna7enEnbcoYGXaq0xDLRY2XhmNs
8u3obNxpYxOp6QvXK9sXoYAVjUzsZx0Nm0PXJ2FnbW0Xn1REnmAwDgRABQy31f9jTQkkf9
7or5gGdyY0DrMOiJ/OB9zoLRG1Dd6iVWEI7yTNk0RUMTaL2AL1IxfP+OFXmgCEABOZ+vpF
HeoHv5FQkZUTWsnBydJA7+bCsncl0eH41kPShmbrpZ13zOpIz8bRDgYxevZsWTUfjtSs3
```

```
1TA0DRAaO5HLHuBzTdD43vWdkhsrjYdA4pHK8Q3YMuozOZvvLpFc43dhBrnLXpPmV+wwNy
faIM9nR14hAyibcsC6Q9AZF9x2bavsy16wUGjvN1jLVg2ucIR2Cpd6sWngpzU2S1s6ZvU9
7TNpg9XTRs5PCzk40JFprOh+tKC2JsU11URTZEh1iXl/IPGGQ/tmxm8tgQ+3;_WebSsoA
uth0=vPDongOyasChyToiOY0bv9+Ba5vkc5xVY1G4s1R35cd7RUxWGxg/NLoAIdgGEVye9
5Owd7K1848cpbNYrsl1u9g0zokt/4hr100CbXU316XCHe5nmwbHjiO9oN/Uz98ioH1EW8Z
Oxw4O2vuh2rXn+lzamRyvW2nBHEFM9hrZerZq2q6cEUmIsBkuGGulKc4zSViwiIcEU2y1N
qw2vdjqGAoA7elxoKThMu35ess3+Po0kLHKdfeqGc11/zymA0bXGq3jUGluTIN3R4rtQMD
jMasIXrcjDOz5irf9bmSFxqUbu5P5KGREL0cCWhIkud2slqrbWLLZzc7uCGYLBkiMg697
aKmToEuFJuJOSmWZ0E4+XLEj0y1BheMyllYBamp93izXp93LVH3Eg44JKxZaLs/S3ide7V
WbNmH887GirrEGnAGnZ2THI6roWip05XTPLTNQBQkXDovgh2xgXrnYbbNXA29K0dZ+Zwu
jls6/4LHaqfJ+bNcpumzGPC/pi9j7vK/b/x69+bsI8T
```

```
<<<
```

```
HTTP/1.1 200 OK
```

```
[Application specific content]
```

4.2 SAML 1.1 Assertion Extension

Following is a **SAML assertion** fragment that illustrates the message syntax of the SAML 1.1 Assertion Extension elements in the advice element, as specified in section [2.2.3](#).

```
<saml:Advice xmlns:adsf="urn:microsoft:federation">
  <adsf:WindowsIdentifiers>
    AAAAAAEAAAABAAAAAABRUAAAVU+0xvWJxlc9CDm4GAAAA9AEAAAYCAAHAgaACAIaA
    AECAAAAAGAA
  </adsf:WindowsIdentifiers>
  <adsf:CookieInfoHash>
    K6GNTL15/jljype53+PFRAiOfek=
  </adsf:CookieInfoHash>
  <adsf:WindowsUserIdentifier>
    S-1-5-21-837636885-2507236029-1846428367-500
  </adsf:WindowsUserIdentifier>
  <adsf:WindowsUserName>
    ADFSVM-A\Administrator
  </adsf:WindowsUserName>
</saml:Advice>
```

The raw octets of the [WindowsIdentifiers \(section 3.1.5.2.1.5\)](#) binary structure, after **base64** decoding are as follows.

```
00 00 00 00 01 00 00 00 01 04 00 00 00 00 05 15 00 00 00 15 53 ED
31 BD 62 71 95 CF 42 0E 6E 06 00 00 00 F4 01 00 00 06 02 00 00 07 02
00 00 08 02 00 00 01 02 00 00 00 02 00 00
```

The octet stream is structured as follows (see section [2.2.3.2](#)).

```
00 00 00 00  WindowsIdentifierFlags = 0
                TryLocalAccount = 0
                NoUserSid = 0
01 00 00 00  PackedSidsCount = 1 (0x00000001)
                PackedSids1
                    DomainSid
01                Revision = 1 (0x01)
    04                SubAuthorityCount = 4 (0x04)
                IdentifierAuthority[0..1] = {0, 0, ...
00 00 00 05                IdentifierAuthority[2..5] = {0, 0, 0, 5
                (0x05)}
15 00 00 00                SubAuthority1 = 21 (0x00000015)
15 53 ED 31                SubAuthority2 = 837636885 (0x31ED5315)
BD 62 71 95                SubAuthority3 = 2507236029 (0x957162BD)
```

CF 42 0E 6E	SubAuthority4 = 1846428367 (0x6E0E42CF)
06 00 00 00	RidCount = 6 (0x00000006)
F4 01 00 00	Rid1 = 500 (0x000001F4)
06 02 00 00	Rid2 = 518 (0x00000206)
07 02 00 00	Rid3 = 519 (0x00000207)
08 02 00 00	Rid4 = 520 (0x00000208)
01 02 00 00	Rid5 = 513 (0x00000201)
00 02 00 00	Rid6 = 512 (0x00000200)

The **SIDs** encoded in the structure are as follows:

- S-1-5-21-837636885-2507236029-1846428367-500
- S-1-5-21-837636885-2507236029-1846428367-518
- S-1-5-21-837636885-2507236029-1846428367-519
- S-1-5-21-837636885-2507236029-1846428367-520
- S-1-5-21-837636885-2507236029-1846428367-513
- S-1-5-21-837636885-2507236029-1846428367-512

5 Security

5.1 Security Considerations for Implementers

Security considerations for the Microsoft Web Browser Federated Sign-On Protocol Extensions are specified in the following subsections. Additionally, the security considerations outlined in [\[MS-MWBF\]](#) section 5 apply to the Microsoft Web Browser Federated Sign-On Protocol Extensions.

5.1.1 Data Integrity

Data integrity concerns, as described in [\[MS-MWBF\]](#) section 5.1.1, apply to the extensions specified in this document. Of particular concern are the **SAML advice** elements specified by the SAML 1.1 Assertion Extension. These elements are included in the **SAML assertion**, which is signed to prevent tampering (see [\[MS-MWBF\]](#) section 2.2.4.2.2).

5.1.2 Privacy

The privacy considerations in [\[MS-MWBF\]](#) section 5.1.5 apply to the extensions in this document. The extensions also introduce new privacy concerns.

The Query String Response Transfer Protocol is used to communicate **security tokens** from one party to another by using the query string of the HTTP request URL. The use of Secure Sockets Layer/Transport Layer Security (SSL/TLS) prevents the exposure of **user** information outside the services participating in the protocol (see [\[MS-MWBF\]](#) section 5.1.3); however, a GET message might provide a lesser degree of confidentiality than a POST message due to URL tracking concerns. For example, **web browser requestor** implementations might track URL history, or web proxy servers might log URLs.

The SAML 1.1 Assertion Extension provides a method for including **SIDs** in a **SAML assertion**. These SIDs might identify user identity, capabilities, or affiliations. For this reason, SIDs should not be included indiscriminately; rather, their distribution should be limited to specific **relying parties**.[<24>](#)

5.1.3 Authorization Validation and Filtering

When processing **SIDs** from an **IP/STS**, **relying parties** must ensure that the IP/STS is authorized to issue SIDs that fall under a particular set of subauthorities. This is similar to namespace collision concerns with UPN and EmailAddress **claims** (as specified in [\[MS-MWBF\]](#) section 5.1.6).[<25>](#)

5.2 Index of Security Parameters

Because the Microsoft Web Browser Federated Sign-On Protocol Extensions is an authentication protocol, the security details are in the message processing rules section.

Security parameter	Section
WindowsUserIdentifier	2.2.3.1
WindowsUserName	2.2.3.1
WindowsIdentifiers	2.2.3.1

6 Appendix A: Product Behavior

The information in this specification is applicable to the following Microsoft products or supplemental software. References to product versions include updates to those products.

- Windows Server 2003 R2 operating system
- Windows Server 2008 operating system
- Windows Server 2008 R2 operating system
- Windows Server 2012 operating system

Exceptions, if any, are noted in this section. If an update version, service pack or Knowledge Base (KB) number appears with a product name, the behavior changed in that update. The new behavior also applies to subsequent updates unless otherwise specified. If a product edition appears with the product version, behavior is different in that product edition.

Unless otherwise specified, any statement of optional behavior in this specification that is prescribed using the terms "SHOULD" or "SHOULD NOT" implies product behavior in accordance with the SHOULD or SHOULD NOT prescription. Unless otherwise specified, the term "MAY" implies that the product does not follow the prescription.

[<1> Section 1.5](#): The Windows implementation of the SAML 1.1 Assertion Extension is integrated with Windows **Active Directory**. All **SIDs** that are transmitted in the **SAML assertion** correspond to Active Directory **accounts**. In order to transmit SIDs between a **requestor IP/STS** and a **resource IP/STS** that reside in different **forests**, the requestor IP/STS forest must be a **trusted forest** of the resource IP/STS forest.

[<2> Section 1.5](#): Local configuration is used to modify several of the Windows behaviors described in this document. In all cases, the local configuration must exist before the protocol is initiated. Specific instances where local configuration is used are listed below.

By default, an **IP/STS** does not issue **security tokens** with SIDs (see section [3.1.5.2](#)). The issuance of SIDs (that is, including the WindowsIdentifier (section [3.1.5.2.1.3](#)), WindowsUserName (section [3.1.5.2.1.4](#)), and WindowsIdentifiers (section [3.1.5.2.1.5](#)) elements in issued SAML assertions) can be enabled for specific **relying parties** by using local configuration.

By default, when a **user** authenticates to a resource IP/STS by using a security token from a requestor IP/STS (see section 3.1.5.2), any SIDs in the SAML assertion are ignored (that is, the WindowsIdentifier (section 3.1.5.2.1.3), WindowsUserName (section 3.1.5.2.1.4), and WindowsIdentifiers (section 3.1.5.2.1.5) elements). Processing these SIDs (as described in section 3.1.5.2) can be enabled for specified requestor IP/STSs using local configuration. In this case, the local configuration also specifies a Windows **domain** associated with the IP/STS, so that SID filtering can be performed (as specified in section [5.1.3](#)).

When a user authenticates to a resource IP/STS by using a security token from a requestor IP/STS (as specified in section 3.1.5.2), **claims** can be mapped to SIDs based on local configuration. In this case, the local configuration also specifies the desired value of the **TryLocalAccount** flag (as specified in WindowsIdentifierFlags Structure (section [2.2.3.2.1](#))).

A special algorithm is used to determine when the Query String Response Protocol is used (as specified in section [3.2.5.1.1](#)). This algorithm uses local configuration to force the use of the protocol and for the **FileExtensionBypassList** flag.

[<3> Section 1.6](#): The Query String Response Transfer Protocol is supported only in Windows Server 2003 R2, Windows Server 2008, Windows Server 2008 R2, and Windows Server 2012.

<4> [Section 1.6](#): SAML1.1 Assertion Extension is supported only in Windows Server 2003 R2, Windows Server 2008, and Windows Server 2008 R2.

<5> [Section 1.9](#): The following URIs are used as local assignments in fields specified by the SAML 1.1 Assertion Extension: "urn:federation:activedirectory" and "urn:microsoft:federation".

<6> [Section 2.2.3.1](#): Support for this field is conditional, as specified in [CookieInfoHash Element \(section 3.1.5.2.1.2\)](#).

<7> [Section 2.2.3.1](#): WindowsUserName Element (section 3.1.5.2.1.4) specifies how Windows constructs the WindowsUserName value.

<8> [Section 3.1.1.1.1](#): The **pending result** is maintained in secure session cookies written using Set-Cookie headers (as specified in [\[RFC2965\]](#)), which are added to the HTTP 302 response to the **web browser requestor**.

<9> [Section 3.1.1.1.2](#): The recommended value is used.

<10> [Section 3.1.2](#): There are no new timers. The pending result is stored by using secure session cookies (see section [3.1.1.1.1](#)).

<11> [Section 3.1.5.2.1.1](#): The IP/STS only includes this element when it is a resource IP/STS. The value identifies Active Directory, a **Lightweight Directory Access Protocol (LDAP)** service, or a **security realm**. If the user authenticated by using Active Directory, the [ClaimSource \(section 3.1.5.2.1.1\)](#) value is "urn:federation:activedirectory". If the user authenticated by using an LDAP service, the ClaimSource (section 3.1.5.2.1.1) value is a URI associated with the LDAP service, for example "LDAP://ldap.example.com". If the user authenticated by using a SAML assertion from a Requestor IP/STS, the ClaimSource (section 3.1.5.2.1.1) value is the value of the Security Assertion Markup Language (SAML) issuer. For more information about user authentication, see [\[MS-MWBF\]](#) section 3.1.5.4.3.

<12> [Section 3.1.5.2.1.2](#): The IP/STS includes the CookieInfoHash (section 3.1.5.2.1.2) element when an authentication cookie is issued at the same time as a SAML assertion. The authentication cookie is a **base64-encoded** binary structure. Windows includes the base64-encoded SHA-1 hash value (as described in [\[FIPS180\]](#)) of the raw octets.

<13> [Section 3.1.5.2.1.3](#): By default, the IP/STS does not include the WindowsUserIdentifier (section 3.1.5.2.1.3) element. It can be enabled for specific relying parties (see section [1.5](#)). A requestor IP/STS populates this field only if the user authenticated by using Active Directory. A resource IP/STS populates this field only if the user authenticated by using a security token from a requestor IP/STS, in which case the value, if present, is copied from that security token.

<14> [Section 3.1.5.2.1.4](#): By default, the IP/STS does not include the WindowsUserName (section 3.1.5.2.1.4) element. It can be enabled for specific relying parties (see section [1.5](#)). Windows constructs the user name by including the **NetBIOS** domain name (as it is always available in this context) and Active Directory account name, separated by a backslash (for example, "DOMAIN\user name"). A requestor IP/STS populates this field only if the user authenticated by using Active Directory. A resource IP/STS populates this field only if the user authenticated by using a security token from a requestor IP/STS, in which case the value, if present, is copied from that security token.

<15> [Section 3.1.5.2.1.5](#): By default, the IP/STS does not include the WindowsIdentifiers (section 3.1.5.2.1.5) element. It can be enabled for specific relying parties (see section [1.5](#)).

A requestor IP/STS populates this field only if the user authenticated by using Active Directory. In this case, a WindowsIdentifiers (section 3.1.5.2.1.5) binary structure is constructed that contains a subset of the security identifiers (SIDs) associated with the Active Directory account. The subset consists of the user SIDs, the **global group** SIDs, and the **universal group** SIDs. That structure is base64-encoded and included in this field.

A resource IP/STS populates this field only if the user authenticated by using a security token from a requestor IP/STS. If the SAML assertion issued by the requestor IP/STS contains a **WindowsIdentifiers** field, its value is copied to this field. Otherwise, the resource IP/STS maps claims from the SAML assertion of the requestor IP/STS to SIDs based on local configuration (see section 1.5). If any claims are mapped to SIDs, those SIDs are encoded in a **WindowsIdentifiers** (section 3.1.5.2.1.5) binary structure with the **NoUserSid** flag set to 1 and the **TryLocalAccount** flag set based on local configuration (see section 1.5). The structure is base64-encoded and included in the **WindowsIdentifiers** field.

<16> [Section 3.1.6](#): There are not any new timer events. The pending result is stored by using secure session cookies (see section 3.1.1.1.1).

<17> [Section 3.2.1.1.1](#): The **aggregated result** is maintained in secure session cookies written using Set-Cookie headers (as specified in [RFC2965]), which are added to the HTTP 302 response to the web browser requestor.

<18> [Section 3.2.2](#): There are no new timers. The aggregated result is stored using secure session cookies (see section [3.2.1.1.1](#)).

<19> [Section 3.2.5.1.1](#): A special algorithm (described later) determines whether to use the Query String Response Transfer Protocol or not.

The algorithm is presented in pseudocode.

Let KnownClients be a constant set composed of the following strings, which specify **user agent** string fragments that are known to identify user agents that do not support scripting:

- "Microsoft FrontPage"
- "Microsoft Office"
- "Test for Web Form Existence"
- "Microsoft Data Access Internet Publishing Provider"
- "Microsoft-WebDAV"

Let FileExtensionBypassList be a set of strings retrieved from local configuration (see section 1.5) that identify file name extensions.

The algorithm is as follows.

```

IF local configuration forces Query String Response Transfer
Protocol
OR User-Agent is empty
OR User-Agent is in KnownClients
OR NOT User-Agent contains "Mozilla"
OR NOT Http-Verb is in (GET,POST) THEN
    Use Query String Response Transfer Protocol
ELSE
    IF local machine state indicates Windows Sharepoint Services
    AND NOT Request-URL's file extension is in
FileExtensionBypassList THEN
        Use Query String Response Transfer Protocol
    ELSE
        Use the base protocol specified in [MS-MWBF]
    END IF
END IF

```

<20> [Section 3.2.5.2](#): By default, the resource IP/STS ignores these fields and processes the security token as though the extension elements are absent. Their processing can be enabled for specific requestor IP/STSs (as specified in section 1.5). Section [3.1.5.2.1](#) specifies how the resource IP/STS

includes the fields. The **web service resource** uses the fields to provide authorization services to a web-based application on the same machine by using methods that are outside the scope of this protocol.

[<21> Section 3.2.6](#): There are no new timer events. The aggregated result is stored by using secure session cookies (as specified in section 3.1.1.1.1).

[<22> Section 3.3.1](#): Windows Internet Explorer supports the use of session and persistent HTTP cookies (for more information, see [RFC2965]). The Windows implementation of this protocol requires that web browser requestor support at least session cookies. It uses persistent cookies to preserve security realm identifiers if they are supported by the web browser requestor.

[<23> Section 3.3.5](#): The RMS 2.0 client in Windows Vista operating system with Service Pack 1 (SP1), Windows Server 2008, Windows 7 operating system, Windows Server 2008 R2, Windows 8 operating system, and Windows Server 2012 adds a *whr* parameter to the wsignin 1.0 Request Message (section [2.2.2](#)) if the wsignin 1.0 Request Message does not already contain a *whr* parameter.

[<24> Section 5.1.2](#): The inclusion of SIDs can be enabled for specific relying parties (as specified in section 1.5).

[<25> Section 5.1.3](#): A relying party that accepts security identifiers from an IP/STS in another security realm performs SID filtering based on local configuration that associates the IP/STS with a Windows domain (as specified in section 1.5). For more information about SID filtering, see [\[MS-PAC\]](#) section 4.1.2.2.

7 Change Tracking

No table of changes is available. The document is either new or has had no changes since its last release.

8 Index

A

Abstract data model
 [IP/STS](#) 18
 [relying party](#) 21
[Aggregated result](#) 21
[Applicability](#) 12
[Assertion extension example](#) 42
[Assertion Extension syntax](#) 15
[Authorization validation security](#) 44

C

[Capability negotiation](#) 12
[Change tracking](#) 49
[ClaimSource element](#) 20
[CookieInfoHash element](#) 20

D

[Data - security integrity](#) 44
Data model - abstract
 [IP/STS](#) 18
 [relying party](#) 21
[Directory service schema elements](#) 17

E

[Elements - directory service schema](#) 17
Examples
 [full network trace](#) 28
 [Query String Response Transfer Protocol](#) 25
 [Query String Response Transfer protocol example](#) 25
 [Security Assertion Markup Language \(SAML\) v1.1 assertion extension example](#) 42

F

[Fields - vendor-extensible](#) 13
[Filtering - security](#) 44

G

[Glossary](#) 7

H

Higher-layer triggered events
 [IP/STS](#) 19
 [relying party](#) 22

I

[Implementer - security considerations](#) 44
[Implementer security considerations](#) 44
[Index of security parameters](#) 44
[Index security parameters](#) 44
[Informative references](#) 11
Initialization
 [IP/STS](#) 19
 [relying party](#) 22

[Integrity - data](#) 44
[Introduction](#) 7
IP/STS
 [abstract data model](#) 18
 [higher-layer triggered events](#) 19
 [initialization](#) 19
 [local events](#) 21
 [message processing](#) 19
 [overview](#) 18
 [sequencing rules](#) 19
 [timer events](#) 21
 [timers](#) 19

L

Local events
 [IP/STS](#) 21
 [relying party](#) 23

M

[Maximum query string response message length](#) 18
Message processing
 [IP/STS](#) 19
 [relying party](#) 22
Messages
 [overview](#) 14
 Query String Response Transfer protocol ([section 2.1.1](#) 14, [section 2.2.2](#) 14)
 [SAML 1.1 Assertion Extension](#) 15
 [syntax](#) 14
 [transport](#) 14
 [XML namespace](#) 14
 [XML Namespace References](#) 14

N

[Normative references](#) 10

O

[Overview \(synopsis\)](#) 11

P

[PACKED_SIDS packet](#) 16
[Parameters - security](#) 44
[Parameters - security index](#) 44
[Pending result](#) 18
[Preconditions](#) 12
[Prerequisites](#) 12
[Privacy - security considerations](#) 44
[Processing - complete aggregated result](#) 23
[Product behavior](#) 45
Protocol Details
 [overview](#) 18

Q

Query String Response Transfer Protocol ([section 1.3.1](#) 11, [section 3.1.1.1](#) 18, [section 3.1.5.1](#) 19, [section 3.2.1.1](#) 21, [section 3.2.5.1](#) 22)

[syntax](#) 14
[transport](#) 14
[Query String Response Transfer protocol example](#) 25
[Query String Response Transfer Protocol message](#) 14

R

[References](#) 10
 [informative](#) 11
 [normative](#) 10
[Relationship to other protocols](#) 12
Relying Party
 [abstract data model](#) 21
 [higher-layer triggered events](#) 22
 [initialization](#) 22
 [local events](#) 23
 [message processing](#) 22
 [overview](#) 21
 [sequencing rules](#) 22
 [timer events](#) 23
 [timers](#) 21

S

[SAML 1.1 Assertion Extension message](#) 15
[SAML advice elements](#) 15
[SAML v1.1](#) 12
 [assertion extension example](#) 42
 [Assertion Extension syntax](#) 15
[Schema elements - directory service](#) 17
Security
 [authorization validation and filtering](#) 44
 [data integrity](#) 44
 [implementer considerations](#) 44
 [parameter index](#) 44
 [privacy](#) 44
[Security Assertion Markup Language \(SAML\) V1.1 -
 \[assertion extension\]\(#\) 23](#)
Security Assertion Markup Language (SAML) v1.1
 Assertion Extension ([section 1.3.2](#) 12, [section
3.1.5.2](#) 20)
 [syntax](#) 15
[Security Assertion Markup Language \(SAML\) v1.1
 \[assertion extension example\]\(#\) 42](#)
Sequencing rules
 [IP/STS](#) 19
 [relying party](#) 22
[Standards assignments](#) 13
Syntax
 [messages](#) 14
 [Query String Response Transfer protocol](#) 14
 [Security Assertion Markup Language \(SAML\) V1.1
 \[Assertion Extension\]\(#\) 15](#)

T

Timer events
 [IP/STS](#) 21
 [relying party](#) 23
Timers
 [IP/STS](#) 19
 [relying party](#) 21
[Tracking changes](#) 49
[Transport](#) 14
Triggered events - higher-layer

[IP/STS](#) 19
[relying party](#) 22

V

[Vendor-extensible fields](#) 13
[Versioning](#) 12

W

[WindowsIdentifierFlags packet](#) 16
[WindowsIdentifiers element](#) 21
[WindowsIdentifiers packet](#) 15
[WindowsUserIdentifier element](#) 21
[WindowsUserName element](#) 21
[wsignin1.0 message](#) 14
 [common parameters](#) 14
 [response](#) 14
wsignin1.0 request
 receiving - ttpindex not specified ([section 3.1.5.1.1](#)
 19, [section 3.2.5.1.2](#) 22)
 [receiving - ttpindex of 0 specified](#) 19
 [receiving - ttpindex other than 0 specified](#) 19
 [receiving - ttpindex specified](#) 22
 [responding - ttpindex specified](#) 20
[wsignin1.0 request - sending](#) 22
[wsignin1.0 request response](#) 20

X

[XML namespace message](#) 14
[XML Namespace References message](#) 14